ĐẠI HỌC QUỐC GIA THÀNH PHỐ HỒ CHÍ MINH
TRƯỜNG ĐẠI HỌC KINH TẾ - LUẬT
KHOA HỆ THỐNG THÔNG TIN

BỆNH VIỆN ĐA KHOA
KHU VỰC THỦ ĐỨC

Project Name: Building Mobile Application for Appointment Scheduling and Records Tracking System - Thu Duc Regional General Hospital

Information System Analysis and Design Course

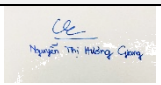
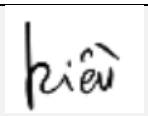
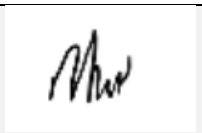
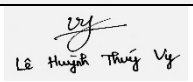
Group 2

Final Version

10/11/2024

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Acknowledgements

We would like to express our deepest gratitude to Assoc. Prof. Ho Trung Thanh for his invaluable guidance, mentorship, and support throughout our journey in the Analyzing and Development of Information Systems (ADIS) subject. His expertise, encouragement, and insightful feedback have been instrumental in helping us complete this project. His dedication to teaching and commitment to our success have inspired us greatly, and we are profoundly grateful for his contribution to our academic and professional growth.

Thank you, Assoc. Prof. Ho Trung Thanh, for your unwavering support and guidance.

Ho Chi Minh city, 10/11/2024

Group 5

Commitments

Our team would like to commit that the project "Mobile Appointment Scheduling System for Thu Duc Regional General Hospital" was carried out by our team under the guidance of Assoc. Prof. Ho Trung Thanh and refer to relevant sources for research purposes. Our team would like to make sure that the entire content of the report is truthful, unique and there is no fraud. At the same time, all data and information contained in the report are the results of the practice process. If there is any issue of fraud or error, our team accepts all responsibility.

Ho Chi Minh city, 10/11/2024

Group 5

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Executive Summary

This project aims to design and implement a Mobile Appointment Scheduling System for Thu Duc Regional General Hospital to enhance the hospital's operational efficiency and improve patient experience. Established in 1978, Thu Duc Regional General Hospital now faces challenges in managing increasing patient volumes due to outdated systems. This new system will address these challenges by streamlining the appointment process, reducing patient wait times, and optimizing staff workload.

Key components of the system include appointment scheduling, real-time updates, secure payment processing, and integration with the hospital’s existing management system. Development will use secure, scalable cloud-based infrastructure, with compliance to healthcare data protection laws. The system will also support both iOS and Android platforms for broad accessibility.

Major milestones include the planning and requirements phase by January 2025, prototype development by May 2025, testing and user training by August 2025, and full system deployment by September 2025. Upon completion, the system will enhance patient engagement, increase operational efficiency, and support the hospital's commitment to high-quality healthcare services.

List of Abbreviations

No.	Abbreviations	Meanings
1	KPI	Key performance indicators
2	FDD	Function Decomposition Diagrams
3	BPM	Business Process Modeling
4	BPMN	Business Process Modeling Notation
5	DFD	Data Flow Diagram
6	UML	Unified Modeling Language
7	BRD	Business Requirements Document
8	SRD	Systems Requirements Document
9	2FA	Two-Factor Authentication
10	ID	Identification
11	HR	Human resources
12	BLOB	Binary large object

Pert chart

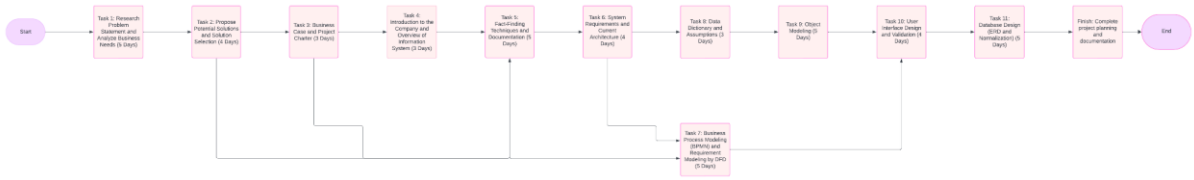
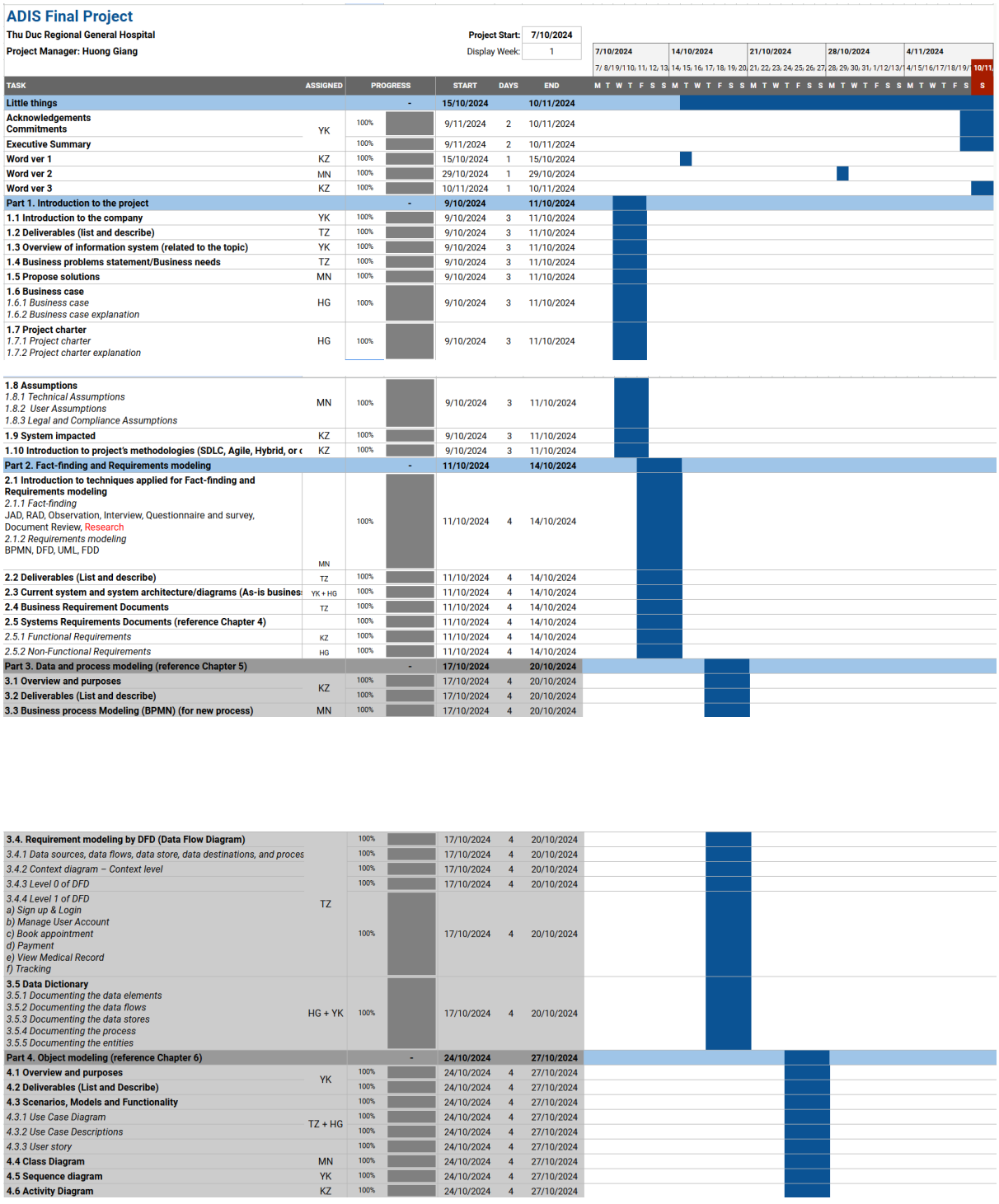


Figure 0. 1. Pert Chart of the project (Source: Author)

Gantt chart



ADIS Final Project

Thu Duc Regional General Hospital

Project Manager: Huong Giang

Project Start: 7/10/2024

Display Week: 1

TASK	ASSIGNED	PROGRESS	START	DAYS	END
Little things		-	**15/10/2024**		**10/11/2024**
Acknowledgements **Commitments**	YK	100%	9/11/2024	2	10/11/2024
Executive Summary		100%	9/11/2024	2	10/11/2024
Word ver 1	KZ	100%	15/10/2024	1	15/10/2024
Word ver 2	MN	100%	29/10/2024	1	29/10/2024
Word ver 3	KZ	100%	10/11/2024	1	10/11/2024
Part 1. Introduction to the project		-	**9/10/2024**		**11/10/2024**
1.1 Introduction to the company	YK	100%	9/10/2024	3	11/10/2024
1.2 Deliverables (list and describe)	TZ	100%	9/10/2024	3	11/10/2024
1.3 Overview of information system (related to the topic)	YK	100%	9/10/2024	3	11/10/2024
1.4 Business problems statement/Business needs	TZ	100%	9/10/2024	3	11/10/2024
1.5 Propose solutions	MN	100%	9/10/2024	3	11/10/2024
1.6 Business case *1.6.1 Business case* *1.6.2 Business case explanation*	HG	100%	9/10/2024	3	11/10/2024
1.7 Project charter *1.7.1 Project charter* *1.7.2 Project charter explanation*	HG	100%	9/10/2024	3	11/10/2024
1.8 Assumptions *1.8.1 Technical Assumptions* *1.8.2 User Assumptions* *1.8.3 Legal and Compliance Assumptions*	MN	100%	9/10/2024	3	11/10/2024
1.9 System impacted	KZ	100%	9/10/2024	3	11/10/2024
1.10 Introduction to project's methodologies (SDLC, Agile, Hybrid, or	KZ	100%	9/10/2024	3	11/10/2024
Part 2. Fact-finding and Requirements modeling		-	**11/10/2024**		**14/10/2024**
2.1 Introduction to techniques applied for Fact-finding and Requirements modeling *2.1.1 Fact-finding* JAD, RAD, Observation, Interview, Questionnaire and survey, Document Review, Research *2.1.2 Requirements modeling* BPMN, DFD, UML, FDD	MN	100%	11/10/2024	4	14/10/2024
2.2 Deliverables (List and describe)	TZ	100%	11/10/2024	4	14/10/2024
2.3 Current system and system architecture/diagrams (As-is busines	YK + HG	100%	11/10/2024	4	14/10/2024
2.4 Business Requirement Documents	TZ	100%	11/10/2024	4	14/10/2024
2.5 Systems Requirements Documents (reference Chapter 4)		100%	11/10/2024	4	14/10/2024
2.5.1 Functional Requirements	KZ	100%	11/10/2024	4	14/10/2024
2.5.2 Non-Functional Requirements	HG	100%	11/10/2024	4	14/10/2024
Part 3. Data and process modeling (reference Chapter 5)		-	**17/10/2024**		**20/10/2024**
3.1 Overview and purposes	KZ	100%	17/10/2024	4	20/10/2024
3.2 Deliverables (List and describe)		100%	17/10/2024	4	20/10/2024
3.3 Business process Modeling (BPMN) (for new process)	MN	100%	17/10/2024	4	20/10/2024
3.4. Requirement modeling by DFD (Data Flow Diagram)		100%	17/10/2024	4	20/10/2024
3.4.1 Data sources, data flows, data store, data destinations, and proces		100%	17/10/2024	4	20/10/2024
3.4.2 Context diagram – Context level		100%	17/10/2024	4	20/10/2024
3.4.3 Level 0 of DFD		100%	17/10/2024	4	20/10/2024
3.4.4 Level 1 of DFD *a) Sign up & Login* *b) Manage User Account* *c) Book appointment* *d) Payment* *e) View Medical Record* *f) Tracking*	TZ	100%	17/10/2024	4	20/10/2024
3.5 Data Dictionary *3.5.1 Documenting the data elements* *3.5.2 Documenting the data flows* *3.5.3 Documenting the data stores* *3.5.4 Documenting the process* *3.5.5 Documenting the entities*	HG + YK	100%	17/10/2024	4	20/10/2024
Part 4. Object modeling (reference Chapter 6)		-	**24/10/2024**		**27/10/2024**
4.1 Overview and purposes	YK	100%	24/10/2024	4	27/10/2024
4.2 Deliverables (List and Describe)		100%	24/10/2024	4	27/10/2024
4.3 Scenarios, Models and Functionality		100%	24/10/2024	4	27/10/2024
4.3.1 Use Case Diagram	TZ + HG	100%	24/10/2024	4	27/10/2024
4.3.2 Use Case Descriptions		100%	24/10/2024	4	27/10/2024
4.3.3 User story		100%	24/10/2024	4	27/10/2024
4.4 Class Diagram	MN	100%	24/10/2024	4	27/10/2024
4.5 Sequence diagram	YK	100%	24/10/2024	4	27/10/2024
4.6 Activity Diagram	KZ	100%	24/10/2024	4	27/10/2024

Figure 0. 2. Gantt Chart of the project (Source: Author)

Part 1: Introduction to the project

This part provides a general overview of Thu Duc Regional General Hospital, including its history, mission, vision, and structure. It highlights the hospital's development, current challenges, and the motivation behind the new project. The chapter introduces the need for a mobile appointment scheduling system to address operational inefficiencies and improve patient care. Additionally, it mentions the hospital's business model and the key project deliverables that aim to enhance healthcare services through digital transformation.

1.1. Introduction of the hospital

1.1.1. The process of establishment and development

The hospital was established in 1978 on the basis of taking over the monastery located in Tan Phu commune, Thu Duc district (now Linh Trung ward, Thu Duc district). In 1994, it became Thu Duc District Medical Center with a scale of 250 inpatient beds according to Decision No. 725/QD-UB-NC of the People's Committee of Ho Chi Minh City (People's Committee of Ho Chi Minh City, 1994).

In 1999, Thu Duc District Medical Center was upgraded to Thu Duc Regional General Hospital under the Department of Health of Ho Chi Minh City according to Decision No. 6784/QĐ-UB (People's Committee of Ho Chi Minh City, 1994).



Figure 1. 1. Thu Duc Regional General Hospital (Source: https://thuonghieucongluan.com.vn/)

Currently, Thu Duc Regional General Hospital sees more than 2,000 outpatient visits each day and treats over 700 inpatients. It has 8 functional rooms and 25 clinical and subclinical departments. The hospital has a team of more than 600 staff, including experienced and caring doctors. It also works with big hospitals in the city and is a training place for medical students. However, the facilities have become old even after many repairs. Therefore, the Ho Chi Minh City People's Committee approved the new hospital project with a total investment of VND 1,915 billion in 2019, starting construction from the end of 2021 (Loc, 2021).



Figure 1. 2. Representatives of the Ministry of Health and the Ho Chi Minh City health sector at the groundbreaking ceremony for the construction of Thu Duc Regional General Hospital in 2021 (Source: https://tuoitre.vn/)

Accordingly, the hospital project has 1 basement and 10 floors, built on a land area of over 58,000m2. The project has a technical infrastructure scale of 1,000 beds and a scale of inpatient treatment and treatment of 500 modern, high-quality beds. The project will create specialized departments with modern, synchronous medical equipment to reduce the load on end-line hospitals, inner-city areas, and improve professional qualifications for health care for people in Thu Duc District and neighboring areas. The construction of the hospital is expected to undergo inspection in January 2025, with operations starting in April 2025 (Lien, 2024). The new hospital will be located at the northeastern entrance of Ho Chi Minh City, meeting the healthcare needs of the area with modern facilities, elevators, and high-end examination rooms, while also creating a green and spacious environment for patients.

In addition, the unit has achieved many excellent accomplishments during its operation, notably receiving the Second-Class Labor Medal. In addition, the unit has also been awarded the Flag of Excellence by the People's Committee of the city, along with many certificates of merit from the People's Committee of the city and the Ministry of Health. This reflects recognition and appreciation for the contributions and efforts of the team in serving the community.

1.1.2. Mission, Vision

a) Mission

The hospital develops with 3 main missions. *First,* to become a prestigious general hospital in the region, with good professional quality and comprehensive service. *Second,* to provide a variety of specialized medical services. *Finally,* the hospital always has a team of good and experienced staff, best meeting the needs of screening, treatment and health care of everyone.

b) Vision

The hospital develops in each investment phase, gradually expanding its scope of operations, becoming a center for training and scientific research, improving the quality of medical examination and treatment, providing high-quality and professional medical care services; always innovating to aim for perfection in all activities to sustainably develop the hospital brand.

In addition, the hospital always creates the best working environment and conditions for all employees; the hospital is always a gathering place for talented, progressive and experienced employees; actively invests in and applies modern diagnostic and treatment equipment. All are aimed at bringing patients the best quality of medical examination, treatment and health care services.

Moreover, the hospital upholds the spirit of responsibility to the community; takes the interests and health of patients as the goal in all activities; always listens and shares to achieve the satisfaction and trust of patients and their relatives. The satisfaction and trust of patients and their relatives is the measure of success of Thu Duc Regional General Hospital.

1.1.3. Structure Manager

The organizational chart of Thu Duc Regional General Hospital is organized by function, as departments and faculties are divided based on specific tasks and areas of expertise. Specifically, there are functional departments such as the General Planning Department, the Finance and Accounting Department, and the Information Technology Department, which are responsible for supporting management, finance, and technology. At the same time, clinical and paraclinical departments are divided according to medical specialties such as internal medicine, surgery, laboratory testing, and diagnostic imaging, in order to provide medical examination and treatment services in each specific area. This model helps optimize work efficiency by assigning tasks clearly and appropriately to each functional department.

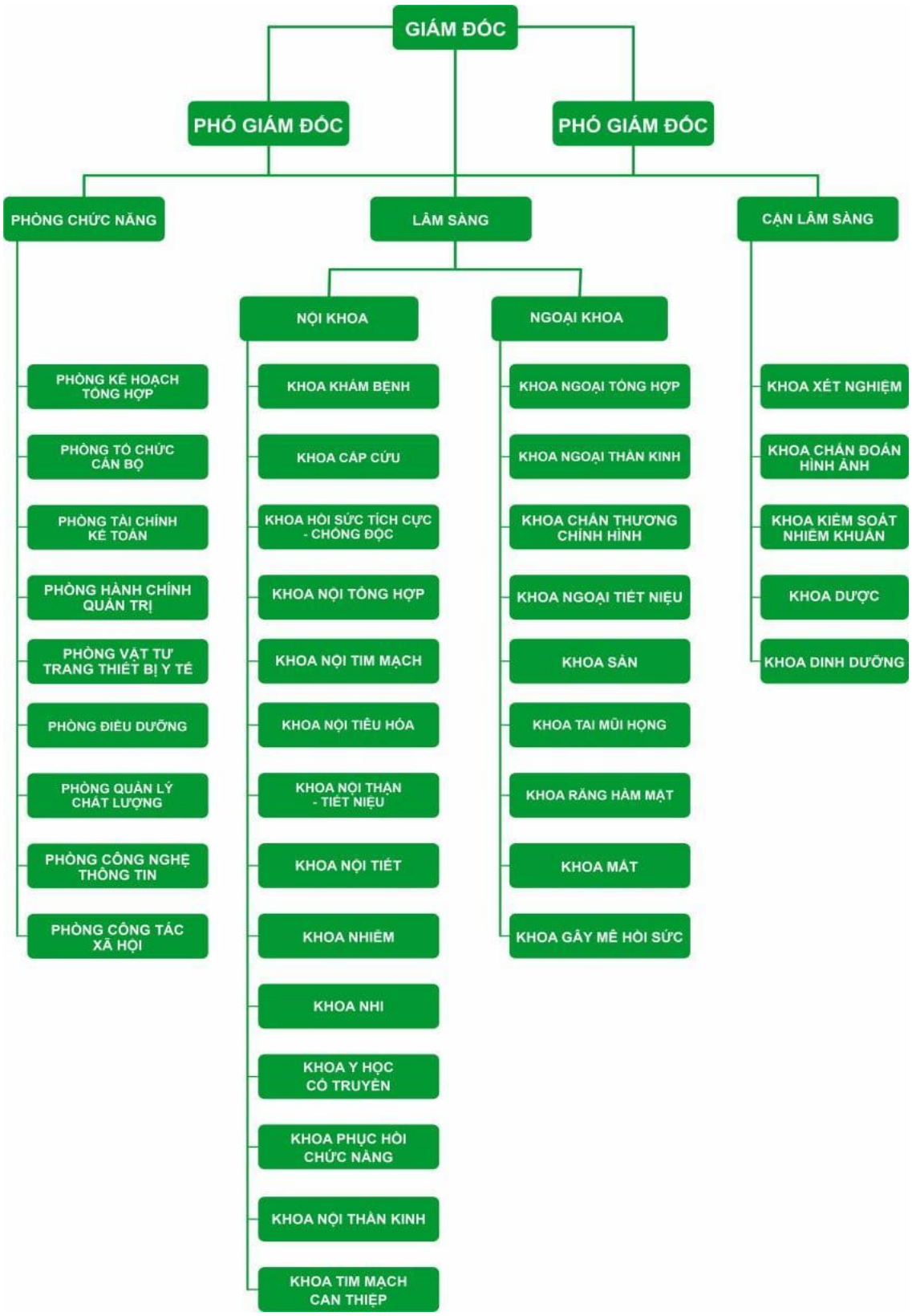


Figure 1. 3. Organization chart of hospital (Source: http://www.benhvienkhuvucthuduc.vn/gioi-thieu/)

1.1.4. Business model

Table 1. 1. Business model canvas of Hospital (Source: Authors)

Component	Details
Market Segment	Target Customers: Residents in the Thu Duc area and nearby regions, including adults, children, and the elderly. Services: Providing a wide range of medical services from basic to specialized, such as internal medicine, surgery, obstetrics and gynecology, pediatrics, chronic disease treatment, and supporting medical services like diagnostic imaging (X-ray, ultrasound, CT scan). Medical services for both insured and uninsured patients, meeting the needs of various customer groups.
Core Values	Dedication: Our hospital is always devoted to patients. The health care services we provide to the community always ensure quality and safety. Responsibility: Our hospital is always responsible to the community and society. Each of our employees is always responsible for their own profession. Sharing: Every employee in our hospital strives to learn, listen to meet customer needs, empathize, and share difficulties with patients and colleagues.
Distribution Channels	Direct: Patients can visit the hospital directly to receive services. Online: Integration of IT platforms allows patients to schedule appointments via the website or hotline and receive online consultation from doctors.

Customer Relationships	Customer Service: Providing consultation and support to patients during examination and treatment, including answering questions and providing detailed guidance. Post-Treatment Care: Monitoring the health status of patients after discharge, providing information on nutrition, exercise, and symptoms that need attention.
Revenue Sources	Periodic Health Checkups: Offering health checkup packages for individuals and businesses, including comprehensive testing and health checks. Private Services: Offering non-insurance health packages with better service quality, meeting the need for fast medical care. Additional Services: Developing other services such as mental health care, nutrition consulting, and community health training programs.
Operating Costs	Personnel Costs: Costs for doctors, nurses, and support staff, including salaries, bonuses, and benefits. Medical Supplies Costs: Costs for purchasing medicines, medical equipment, and consumables required for daily hospital operations. Facility Maintenance Costs: Costs for maintaining and upgrading medical equipment, as well as ensuring a clean and safe hospital environment. Marketing Costs: Investment in communication and advertising activities to enhance brand awareness and attract patients.

Other Factors	Competitors: Other hospitals in the area, private clinics, and other medical facilities that may offer similar services. Policies and Regulations: Adherence to government regulations regarding healthcare, health insurance, and national and international medical standards. Patient Feedback: Surveys and feedback collection from patients to improve service quality and better meet customer needs.

1.2. Deliverables

This chapter outlines the key deliverables for the project, each of which serves a critical function in ensuring the success of the system development for Thu Duc Regional General Hospital. These documents are essential for identifying and understanding the business issues faced by the hospital. They provide the foundation for decision-making and ensure that the system to be implemented will meet the hospital's operational requirements.

Preliminary investigation report: This report provides a comprehensive analysis of the hospital's current systems and operational processes. It identifies key inefficiencies, gaps, and challenges related to appointment scheduling, data management, and patient interaction. The report includes a detailed examination of the manual appointment scheduling process, its impact on hospital operations, and patient experience.

The purpose of this deliverable is to gather sufficient data to understand the business problems, define the project's scope, and form the foundation for decision-making during the project. This early-stage investigation is critical in determining the feasibility and guiding the next phases of the project.

Business case: The business case outlines the rationale behind the project. It includes the justification for implementing the new mobile appointment scheduling and records tracking system, along with an analysis of the expected benefits, costs, and risks.

The business case serves to align the project with the hospital's strategic goals. It ensures that the project will solve identified business problems and provides a formal proposal to gain approval from key stakeholders. The case includes options considered, projected benefits such as reduced patient wait times, and potential risks, ensuring that decision-makers are well-informed.

Project charter: The project charter defines the objectives, scope, and stakeholders of the project. It formally authorizes the project and outlines key components such as the project manager, deliverables, timelines, budget, and risk considerations.

The project charter is essential as it officially launches the project, clarifying the roles and responsibilities of all involved parties. It ensures that the project team and stakeholders have a clear understanding of the project's scope, objectives, and constraints. This document also serves as a reference point throughout the project to maintain alignment with initial goals.

Business profile document*:* This document provides detailed information about Thu Duc Regional General Hospital, its mission, vision, structure, and core values. It highlights the hospital's current business model, patient demographics, and the healthcare services provided. The profile is integral to understanding the hospital's operational context and is critical for aligning the project goals with the hospital's long-term strategic objectives.

Each of these deliverables is integral to the planning and implementation of the new appointment scheduling system. They help guide the project from its initial stages through development, ensuring that all stakeholders are aligned and that the project remains on track to meet its goals.

1.3. Overview of the information system(related to the topic)



Figure 1. 4. Overview of making an appointment and medical examination process (Sources: Authors)

To understand why having a clear hospital visit process is important, it helps to first look at the steps patients go through now. Figure … shows the process, from when the patient arrives until they leave the hospital. First, the patient chooses how to register—either by going straight to the hospital or booking an appointment online or through the hotline. If the patient goes to the hospital, they take a queue number, complete the registration, and make a payment. Afterward, they wait in the waiting area for their check-up. If the patient booked an appointment, they would have already paid the registration fee and simply arrived at the scheduled time. After the check-up, if the doctor gives a prescription, the patient pays for the medicine and collects it. If there's no prescription, they can skip this step. Finally, once all steps are done, the patient can leave the hospital.

Understanding the current system helps find problems that need fixing, like long waiting times or complicated procedures. By learning about these issues, we can design a new system that makes everything run more smoothly. This will reduce stress on hospital staff and improve the patient experience, making the whole process easier and more efficient for everyone.

1.4. Business problems statement

Business problems statement: The hospital's reliance on manual processes for managing patient appointments and tracking records results in operational inefficiencies, long wait times, and administrative burden. Although an online appointment system exists, it is underutilized and lacks effectiveness due to limited functionality and awareness. Patients face challenges, such as the absence of real-time updates and inadequate digital record management, causing frustration and burdening both patients and staff. With the hospital's expansion in 2024, there is an urgent need to streamline these processes to meet growing demand and alleviate pressure on central and inner-city hospitals.

Customer feedback insight: According to numerous reviews on Google, a significant number of patients have expressed dissatisfaction with the hospital's slow administrative procedures and long waiting times. Common complaints include delays in scheduling appointments and receiving treatment. However, several patients mentioned that the hospital's website offers quick appointment booking, suggesting a growing demand for online scheduling options. These reviews highlight both the inefficiency of the current system and the potential for improving the patient experience through digital solutions.

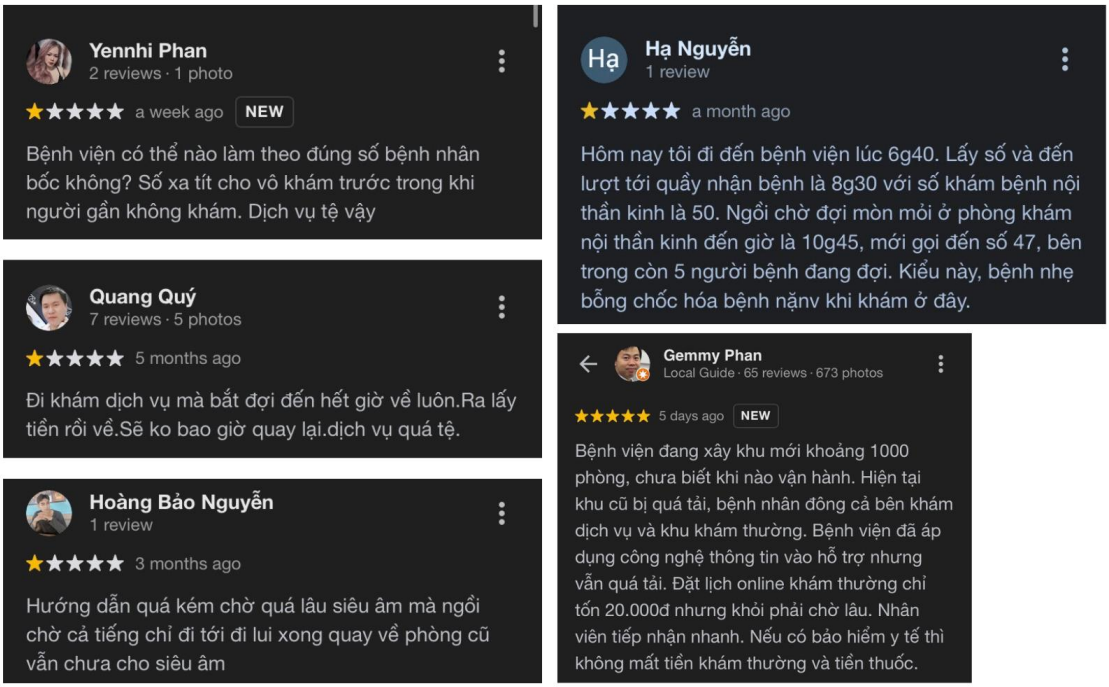


Figure 1. 5. Patient reviews highlighting long wait times and administrative delays (Source: Google).

Key Issues:

- Inefficient appointment scheduling: The current manual system is prone to delays and errors. Patients often experience confusion due to miscommunications with staff, and administrative tasks such as rescheduling or cancellations are time-consuming.

- High administrative workload: Staff must manually handle all appointment bookings and updates, which increases their workload and exposes the hospital to a higher risk of human error during data entry. Staff productivity is reduced, as valuable time is spent on routine tasks that could be automated.

- Disjointed data management: A lack of integration between the appointment system and the hospital's database complicates the management of patient records. This disjointed process increases the likelihood of errors in patient data, which can lead to mismanagement of appointments or medical information.

- Absence of real-time updates: Patients are not provided with timely updates about their appointments, transitions between services, or payments. This lack of communication leads to confusion, missed appointments, and longer wait times as patients navigate the hospital's services.

- Resource misallocation: Without an efficient system to manage bookings and resources (e.g., doctors, medical equipment), the hospital struggles to optimize its capacity. As a result, patients may encounter delays, and healthcare resources are not utilized effectively.

- Limited patient access to medical records: Patients have no easy way to access their appointment history or medical records digitally, which contributes to a lack of transparency. This disconnect prevents patients from actively engaging with their healthcare and reduces their ability to track treatments or follow-up visits.

Problem Analysis:

- Operational inefficiencies: The manual processes slow down the hospital's operations, leading to increased labor costs and making it difficult for staff to handle the rising patient volumes. The hospital risks losing its competitive edge as patient dissatisfaction grows due to prolonged wait times.

- Patient experience: Patients are dissatisfied with the hospital's complicated and lengthy appointment procedures. The lack of access to real-time updates and medical records limits transparency and negatively affects patient engagement with their healthcare.

- Data inaccuracy and management: Manual data entry increases the risk of errors in patient records, which could result in missed appointments or incorrect medical information. This undermines the reliability of the hospital's healthcare services.

- Real-time communication gap: Without real-time communication systems, patients remain uninformed about their appointment status and next steps. This often results in missed appointments, delayed check-ups, and longer waiting periods, further exacerbating the hospital's inefficiencies.

- Limited digital transformation: The hospital’s dependency on manual processes limits its ability to adopt modern digital solutions, such as automated scheduling and digital record management. This shortfall restricts the hospital’s capacity to improve its service quality and efficiency in response to the 2024 expansion.

1.5. Propose Solutions

To solve problems as mentioned above, there are numerous solutions with many benefits and drawbacks. The report will focus on the most straightforward and easy-to-deploy solution which is building a mobile application to book appointments and view medical history.

The mobile application will help patients easily schedule, manage, and track appointments. The system will help to enhance the patient's experience when going to the hospital by providing a quick and convenient way to book appointments and view their medical record. This helps solve problems mentioned in section 1.4 for example data inaccuracy and management, operational inefficiencies, and high administrative workload, and paves the way for digital transformation in the healthcare industry.

1.5.1. Solution Purposes

The traditional process of registering for medical examinations at hospitals is often cumbersome and inefficient, creating a need for a more streamlined, user-friendly system. Optimizing this process through a simple and accessible application can offer numerous benefits, not only to individual patients but to the healthcare system in general:

- *Improved Service Quality:* Patients will experience faster, more accurate, and more efficient registration and medical examinations. Additionally, the application allows patients to track their check-up progress, providing them with greater transparency and convenience. It allows patients to schedule, manage, and track appointments efficiently, with real-time updates.

- *Reduced Operating Costs:* With the option for online payments, the hospital can decrease the need for cashiers and administrative staff to handle patient scheduling. This reduces overhead costs, including staff time, paperwork, and manual data entry typically associated with traditional registration processes. Automating the appointment booking

process reduces manual work for hospital staff, leading to improved operational efficiency.

- *Time Savings for patients:* Digital record-keeping makes it easier for patients to access their records, ensuring faster access and seamless continuity when patients return for future visits. Enhancing the overall patient experience by providing convenient, online access to appointments, medical records, and billing information, reducing wait times and errors.

- *Integration with hospital management system*: Seamless integration between the appointment system and the hospital's existing patient management system to ensure accurate and timely updates of medical records.

These benefits affect not only patients but also hospital managers. In a business sense, improved service will attract more customers, the same can also be applied to the healthcare system attracting more patients.

1.5.2. Solution Plans

The proposed solutions will be carried out according to the project charter below and monitored by deliverables in the above section. The project charter helps the development process by marking out the scope, timeline, KPI, and so on so everyone on the team can keep track of the process. Deliverables will help the team with the goals of each stage, ensuring the project stays on track and does not deviate from the goals. By following the mentioned documents, the project can be ensured to stay on track and be completed in a timely manner.

1.6. Business case

1.6.1. Business case

Table 1. 2. Project Business Case (Source: Author)

PROJECT BUSINESS CASE	
Project Name	Mobile Appointment Scheduling System

Project Sponsor	Head of Customer Service Department	Project Manager	Huong Giang
Date of Project Approval	19th Sep 2024	Last Revision Date	19th Sep 2024
Contribution to Business Strategy	The hospital was established to serve the local community and reduce pressure on central hospitals. With rising demand, the current appointment and consultation process is inefficient. This system will streamline scheduling, reduce wait times, and improve overall efficiency, aligning with the hospital's goal of delivering timely and accessible healthcare.		
Options Considered	1. Use an Third-Party software to manage patient appointments and records. 2. Building an In-house Mobile Appointment and Medical Record Management System (selected)		
Benefits	1.Reduced patient wait time - expected reduction in average patient wait time to no more than 20 minutes 2. Higher appointment attendance - expected 15% reduction in no-show rates compared bookings made on the website 3. Increased operational efficiency - anticipated 30% decrease in administrative time spent manually handling appointment scheduling and follow-ups,		
Timescales	The system development is estimated to take approximately 10 months from inception to deployment.		
Risks	Two key risks of the project include ensuring compliance with healthcare regulations to protect sensitive patient information and avoid legal repercussions. Additionally, integration challenges may occur when connecting the new system to the existing hospital management system.		

1.6.2. Business case explanation

Contribution to Business Strategy

As mentioned in section 1.1.1 the Thu Duc Regional General Hospital commenced construction in 2021, with a huge investment. The healthcare sector in Ho Chi Minh City envisions that upon its completion, the hospital will develop specialized departments equipped with advanced medical technology to provide comprehensive healthcare services for local residents (Loc, 2021). *This initiative aims to alleviate the patient load on tertiary hospitals and those in the city center.*

Consequently, the implementation of an effective appointment scheduling system will optimize the process of scheduling consultations, streamline patient interactions, reduce waiting times, and enhance overall operational efficiency. This improvement will not only *increase the hospital's capacity to serve a larger patient population but also improve the quality of care delivered to the community.*

Options Considered

Two main approaches were evaluated for managing patient appointments and tracking medical records: utilizing third-party software and developing an in-house mobile appointment and medical record management system. Although third-party software offers quick implementation, it presents challenges such as higher costs and potential data privacy risks. Given the need to integrate with the hospital's existing database for retrieving patient information, outsourcing this task could expose sensitive healthcare data to external entities, increasing the risk of breaches. Moreover, reliance on an external provider could create dependency, limiting the hospital’s ability to make timely improvements or adapt to new requirements. Developing an in-house system, on the other hand, not only ensures data security but also helps reduce long-term costs by eliminating the need for ongoing licensing fees and external support. This approach also allows the hospital to promptly modify the application’s features and facilitates quicker, more efficient enhancements or reengineering in response to future demands.

Benefits

The proposed system offers several significant benefits aimed at enhancing patient care and operational efficiency. First, it is expected to *reduce patient wait times*, with a target of ensuring that the average wait time does not exceed 20 minutes. This improvement is crucial for enhancing the overall patient experience, as shorter wait times can lead to higher satisfaction levels and more effective patient management. Additionally, the system aims to *increase appointment attendance* by achieving a 15% reduction in no-show rates compared to bookings made on the hospital's current website. By implementing easy rescheduling options, the system is designed to engage patients more effectively, thereby reducing the number of missed appointments and optimizing resource allocation. Lastly, the anticipated *increase in operational efficiency* is noteworthy, with an expected 30% decrease in administrative time spent on manually handling appointment scheduling and follow-ups. This reduction will allow staff to focus on more critical tasks and improve the overall workflow within the hospital.

Timescales

The development of the system is projected to take approximately 10 months from inception to deployment. This timeline encompasses several critical stages, including requirement gathering, design, development, testing, and implementation.

Planning & Requirements Gathering: December 2024 – January 2025 (2 months)
+ Sprints: 2 Sprints (1 month each)
+ Define objectives, gather user stories from stakeholders, and establish the Agile framework.

Design Phase: February 2025 (1 month)
+ Sprint: 1 Sprint
+ Develop wireframes and UI/UX designs, incorporating stakeholder feedback.

Development Phase: March 2025 – May 2025 (3 months)
+ Sprints: 6 Sprints (2 sprints per month)
+ Develop a working prototype with core functionalities. Each sprint will focus on specific user stories, integrating feedback and adding features iteratively.

Testing Phase: June 2025 – July 2025 (2 months)

+ Sprints: 4 Sprints (2 sprints per month)

+ Perform thorough testing (functional, usability, security) and conduct user acceptance testing (UAT). Adjustments based on feedback will be incorporated in each sprint.

Training & Rollout Preparation: August 2025 (1 month)

+ Sprint: 1 Sprint

+ Conduct training for staff and finalize the deployment strategy.

Launch & Post-Launch Support: September 2025 (1 month)

+ Activities: Launch the system and provide ongoing support, monitoring for issues and user feedback.

Risk

The project involves several risks that must be addressed to ensure its success. A key risk is ensuring compliance with healthcare regulations to protect sensitive patient information. To mitigate this, a robust data protection strategy will be implemented throughout development and operational phases. Another significant risk is the potential for integration challenges with the existing hospital management system, which could affect data flow and operational efficiency. To address this, the project team will analyze current systems and develop a clear integration plan, ensuring the new system enhances existing infrastructure without causing disruptions.

1.7. Project charter

1.7.1. Project charter

Table 1. 3. Project Charter (Source: Author)

PROJECT CHARTER			
Project Name	Mobile Appointment Scheduling System		
Project Sponsor	Hospital Director	Project Manager	Huong Giang
Date of Project	19th Sep 2024	Last Revision Date	19th Sep 2024

Approval			
Project Description	Build a Mobile Appointment Scheduling and Records Management System to offer a seamless solution for patients to book, manage medical appointments, and track medical records efficiently.		
Scope	The Mobile Appointment Scheduling and Records Management System will enable patients to efficiently book medical appointments (khám bệnh) and view their medical records retrieved from the hospital database. However, it will not manage or modify these medical records, focusing solely on appointment scheduling and access to existing patient information.		
Business Case	Enhance customer engagement by providing a personalized learning experience. Increase enrollment rates by 15% through improved customer support. Boost student satisfaction by 20 points through tailored course recommendations and career advice.		
Constraints (in priority order)	Time	Full system deployment by May 2025.	
	Budget	4 Developers, 1 Healthcare Consultant, 1 BA, 1 QA Tester, 1 Project Manager	
	Scope	Providing a fully functional, secure mobile appointment scheduling system integrated with the hospital management system.	
	Qualiy	The system must meet high standards of usability, security, and performance	
Project deliverables	- The mobile application (iOS and Android) features user registration, appointment scheduling, secure payment integration, real-time tracking, appointment history, and full integration with		

<table>
<tr><td></td><td colspan="3">the hospital management system.
- User training and documentation.</td></tr>
<tr><td rowspan="4">Benefits (measurable results)</td><td colspan="3">See KPI belows + business case above</td></tr>
<tr><td>KPI</td><td>Baseline</td><td>Goal</td></tr>
<tr><td>Patient wait times</td><td>60 minutes</td><td>15 minutes</td></tr>
<tr><td>No-show rate</td><td>20%</td><td>5%</td></tr>
<tr><td rowspan="4">Steering Committee</td><td>Patient satisfaction score</td><td>70</td><td>90</td></tr>
<tr><td>CIO</td><td rowspan="4">Project team</td><td>4 Developers</td></tr>
<tr><td>Lead Developer</td><td>1 BA + 1 QA Tester</td></tr>
<tr><td>Project Manager</td><td>1 Healthcare Consultant</td></tr>
<tr><td>Head of Patient Services</td><td>1 Project Manager</td></tr>
<tr><td>Key Stakeholders</td><td colspan="3">Hospital Executive Board
Hospital IT Department
Medical Staff (Doctors, Nurses, Administrators)
Patients
Finance Department
Compliance and Legal Teams</td></tr>
<tr><td>Risks</td><td colspan="3">Two key risks of the project include ensuring compliance with healthcare regulations to protect sensitive patient information and avoid legal repercussions. Additionally, integration challenges may occur when connecting the new system to the existing hospital management system.</td></tr>
</table>

1.7.2. Project charter explanation

Project Description

The project description specifies the development of a system that *enables patients to efficiently book medical appointments and access medical records.* This feature addresses a growing need for healthcare institutions to modernize their patient management systems. As hospitals increasingly focus on improving patient experiences, this system plays a key role in providing a user-friendly interface for booking appointments and accessing medical information, which not only improves convenience for patients but also reduces administrative burden on staff.

Scope

The scope is intentionally limited to *appointment booking and viewing medical records* pulled from the hospital database, without altering or managing the medical records themselves. This decision is driven by the need to mitigate legal and operational risks associated with managing patient data. Handling sensitive information such as medical records comes with significant regulatory requirements, and limiting the system's role to accessing records, rather than managing them, simplifies compliance. This also ensures that the project remains focused on improving patient experience without overcomplicating the system's development.

Business Case

The business case for this project revolves around the hospital's need to improve operational efficiency, patient satisfaction, and overall care quality. By digitizing the appointment scheduling process and allowing patients to access their medical records, the hospital aims to *reduce administrative workload and enhance the patient experience.* These improvements are expected to directly impact key performance indicators (KPIs) like patient wait times and no-show rates, which have a significant influence on both hospital operations and patient satisfaction.

Constraints (in priority order)

Time: With a goal to fully deploy by May 2025, the project operates under a tight timeline due to the hospital's operational needs and rising patient demand for digital services. Delays could frustrate patients, disrupt hospital efficiency, and lead to higher costs. Timely delivery is crucial to meet expectations and ensure smooth operations.

Budget: The project team consists of 4 developers, 1 healthcare consultant, 1 business analyst, 1 QA tester, and 1 project manager. This team size is designed to balance expertise and cost-effectiveness.

Scope: The scope is carefully defined to ensure the project focuses on building a secure, functional system that meets the hospital's current needs. The integration with the existing hospital management system is a critical requirement, as it allows for seamless data exchange without disrupting existing workflows.

Quality: The system must protect patient data and comply with healthcare regulations while offering an intuitive, user-friendly interface for both patients and staff.

Project Deliverables

A *fully functional mobile application* that offers essential features like appointment scheduling, secure payments, real-time tracking, and historical appointment views. The inclusion of *iOS and Android versions* ensures accessibility across the most commonly used mobile platforms.

User training and documentation will be provided to ensure hospital staff and patients can make full use of the system. Training is critical to ensuring that both medical staff and patients adopt the system smoothly, preventing miscommunication and potential errors in its early stages.

Benefits (Measurable Results)

To measure the success of the project, KPIs have been set to ensure tangible improvements. These KPIs reflect the hospital's focus on *improving patient experiences, reducing unnecessary delays, and optimizing the scheduling process.*

Achieving these goals will result in greater patient loyalty, higher hospital throughput, and a more efficient use of hospital resources.

Steering Committee

The steering committee includes critical stakeholders such as the Chief Information Officer (CIO), who will oversee the project's IT integration, and other key members like lead developers and healthcare consultants. *Their combined expertise ensures the project stays on track and aligns with both technical and medical requirements.*

Key Stakeholders

Hospital Executive Board:
+ Role: Provide overall strategic direction and project approval.
+ Interaction Method: Monthly progress meetings to present project updates, discuss challenges, and gather feedback.
+ Feedback Collection: Utilize formal reports and presentations during meetings to capture board insights and decisions.

IT Department:
+ Role: Responsible for ensuring technical integration with existing hospital systems.
+ Interaction Method: Bi-weekly technical review meetings to assess integration progress and address any technical issues.
+ Feedback Collection: Continuous collaboration through shared project management tools to document integration challenges and solutions, with ongoing discussions during sprints.

Medical Staff:
+ Role: Ensure the system meets their scheduling and patient data access needs.
+ Interaction Method: Weekly focus group meetings during the Design and Development phases to provide input on usability and features.
+ Feedback Collection: Gather direct feedback through surveys and usability testing sessions to ensure that the system aligns with their workflow.

Patients:

+ Role: End-users whose satisfaction is crucial for the system's success.
+ Interaction Method: Conduct user testing sessions and surveys at key milestones, particularly at the end of the Testing Phase.
+ Feedback Collection: Use feedback forms and interviews to gather insights on patient experience and preferences, integrating this feedback into subsequent sprints.

Compliance and Legal Teams:

+ Role: Ensure adherence to healthcare regulations and legal standards.
+ Interaction Method: Monthly compliance check-ins to review project progress and compliance with legal requirements.
+ Feedback Collection: Collect documentation and reports from these meetings to ensure that any compliance issues are identified and addressed promptly.

1.8. Assumptions

To build the application, the analyst and the development team will need to make some assumptions about users to focus on the functionalities that are most important. Cases other than the assumptions will not be considered when designing the application due to reasons such as out-of-control variables and common functionalities that are widely used.

1.8.1. Technical Assumptions

- The hospital infrastructure must be able to provide the necessary services like servers and data storage for the application to be running 24/7, handling loads up to 500 users, and be managed by a competent IT department.
- The external services that the application uses such as APIs must be reliable and have little downtime

1.8.2. User Assumptions

- The user's device must be capable enough processing power-wise to run the application without any problems.
- Users will provide accurate information when registering, and scheduling appointments such as name, email address,..

- Users will enable notifications about upcoming appointments and other reminders.
- Users will use all of the features that are available in the application and grant all necessary permissions (location, camera, …) to enable full functionality of the application.
- Users will use the application in the way the developers intended, following specific workflows or behaviors that match the application's design.
- Users are familiar with mobile applications and can quickly learn how to use the features without needing detailed guidance.

1.8.3. Legal and Compliance Assumptions

- The application complies with data protection laws of The Socialist Republic of Vietnam under the decree for the protection of personal data 13/2023/NĐ-CP (Government of Vietnam, 2023).
- Users will agree with the terms and conditions of the application and consent to the use of their information.
- The application development team automatically owns all code, content, and assets created during the development process.
- It is permissible to use third-party libraries, open-source software, and APIs without any legal implications as long as the code works within the application.

1.9. System impacted

The development of the new mobile application will have a significant impact on the hospital's existing systems, particularly concerning data management and system integration. The key areas of impact include database architecture, system workflows, data privacy, and user experience.

Integration with the Current Hospital System

The mobile application will not function independently but will be closely integrated with the hospital's existing management system. This integration is essential to ensure that patients have seamless access to medical records, appointment details, and other relevant healthcare information. To achieve this, the mobile application must communicate effectively with the hospital's central database to enable real-time data synchronization, ensuring that updates made to patient records and appointment

schedules are immediately reflected across both platforms. The integration strategy will prioritize maintaining consistency and accuracy in patient information, minimizing the risks of data fragmentation or discrepancies.

Real-Time Data Synchronization and System Workflow Changes

The success of the mobile application depends on its ability to synchronize data in real-time between the hospital's central database and the mobile interface. This will enable patients to receive timely updates on appointment schedules, medical records, and notifications. Implementing real-time data synchronization will require robust system architecture to manage high transaction volumes without compromising system performance. Additionally, the shift to automated data updates will reduce the administrative workload associated with manual data entry and patient record management, leading to optimized hospital workflows and more efficient use of staff resources.

Data Privacy and Security Compliance

Ensuring data security and compliance with healthcare regulations is paramount when integrating the new mobile application with the existing hospital systems. The mobile application and backend infrastructure must be designed to protect sensitive patient information through encryption, access controls, and secure data transmission protocols. Compliance with local data protection laws, such as the decree on personal data protection in Vietnam, will be a critical factor in the project's design and implementation phases. Regular security audits and system monitoring will be essential to identify and mitigate potential vulnerabilities.

Operational Implications and Workflow Adaptation

The introduction of the mobile application will streamline the hospital’s operations, altering existing workflows for patient information management and appointment scheduling. With automated processes and digital access to records, staff will experience a reduced administrative burden, allowing them to focus more on direct patient care. This change will require staff training and potential role adjustments to adapt to the new digital workflows. Additionally, the enhanced data accessibility may drive further

innovation in patient engagement, such as personalized health recommendations and follow-up care reminders based on patient history.

Database Management and Infrastructure Enhancement

Cloud Migration Strategy: Given the increasing demand for digital services and the need for accessibility, migrating the hospital's database to a cloud-based solution is a key consideration. A cloud infrastructure would not only provide scalability to accommodate growing data needs but also ensure that patients can access their medical records remotely at any time. This migration would involve selecting a cloud provider that meets the hospital's data security requirements and planning for a seamless transition to avoid downtime.

Integration of Legacy Data: If the existing database remains in use, there must be a plan to integrate legacy data into the new system architecture. This may involve upgrading the current database or employing middleware to facilitate communication between the mobile application and the hospital's existing management system. The choice of integration approach should align with the database's structure and the hospital's data management policies.

By modernizing the hospital's information system through the integration of a mobile application, the institution can significantly enhance patient experience, improve data management, and achieve greater operational efficiency while ensuring compliance with regulatory standards. This initiative represents a key step toward the hospital's digital transformation and long-term strategic goals.

1.10. Introduction to project's methodologies

The Agile methodology has been chosen as the most suitable approach for developing the new mobile application. Agile is ideal for projects where requirements may evolve over time, offering flexibility, collaborative development, and iterative progress. The methodology aligns well with the complexities of integrating the mobile application with existing hospital systems and the need for ongoing feedback from users to refine

the features and functionalities. The following step-by-step approach aligns Agile phases with the provided project phases to guide the development process.

Planning Phase

Tasks: Define the objectives, constraints, and deliverables for the mobile application project. This phase involves gathering requirements from stakeholders, such as hospital administrators, doctors, and patients. The objectives include providing functionalities like viewing medical records, booking appointments, and sending notifications. Constraints could include data privacy regulations and integration challenges with the existing system.

Agile Approach: Requirements are documented as user stories and stored in the product backlog. These stories help clarify the application's functionalities and guide the development process. The planning phase also involves setting up the Agile framework, such as organizing the project into sprints and defining sprint goals.

Risk Analysis Phase

Tasks: Identify potential risks, such as data security issues, integration challenges, or user adoption concerns. Develop strategies to mitigate these risks, including adopting secure data transmission protocols, planning integration with the existing hospital database, and providing user training.

Agile Approach: Risk assessment is carried out at the beginning of each sprint, and mitigation strategies are integrated into the sprint plan. The Agile process allows the team to revisit and adjust risk management strategies as the project progresses, ensuring that new risks are addressed promptly.

Engineering Phase

Tasks: Develop a working prototype that includes the planned deliverables. The prototype should implement core features such as accessing patient records, booking appointments, and notifying users. This phase focuses on incremental development, where features are added iteratively.

Agile Approach: Each sprint involves developing specific features identified in the planning phase. The development team works on user stories and integrates them into the prototype. Continuous integration is applied, with frequent code commits and testing to ensure the system's stability and functionality.

Evaluation Phase

Tasks: Perform thorough assessments and testing of the prototype to identify areas for improvement. This includes functionality tests, usability tests with hospital staff, and security checks. Feedback gathered during this phase is used to refine objectives and plan for the next iteration.

Agile Approach: At the end of each sprint, a sprint review is conducted to demonstrate the prototype to stakeholders and gather feedback. The team also holds a retrospective to evaluate the sprint's performance, discuss what worked well, and identify areas for improvement. The findings guide the objectives and adjustments for the upcoming sprint.

Part 2: Fact-Finding And Requirements Modeling

This part mentions the methodologies and approaches applied for gathering information and modeling requirements for the new system. It includes techniques such as observation, document review, interviews, and research to understand the current state of the hospital's appointment processes. This part also details the use of Business Process Modeling, Data Flow Diagrams, and Unified Modeling Language to effectively represent system requirements. Additionally, it introduces the Agile methodology used for iterative development to ensure flexibility and continuous improvement throughout the project. This foundation will guide the development of the Mobile Appointment Scheduling System to address the operational needs of Thu Duc Regional General Hospital.

2.1. Introduction to techniques applied for Fact-finding and Requirements modeling

2.1.1. Fact-finding

Observation

Direct Observation of Existing Processes: Analysts would spend time in Thu Duc Regional General Hospital's outpatient clinics, reception areas, and registration counters to observe the current patient flow for appointments. This includes:

- Mapping the patient journey from arrival to consultation, noting waiting times at each stage.
- Observing how patients interact with any existing appointment systems (kiosk, website, phone calls).
- Analyzing staff workflows related to appointment scheduling and patient registration.
- Identifying bottlenecks and areas of inefficiency in the current process.

Environmental Observation:

- Assessing the physical layout of the hospital to understand how patients navigate the space.

- Evaluating the availability and accessibility of information related to appointments (signage, brochures, etc.).

Document Review

Analyzing Existing System Documentation: Analysts would examine any existing documentation related to Thủ Đức hospital's current appointment system, including:

- Appointment logs and records
- Patient medical records and history
- Internal procedures and guidelines for appointment scheduling
- Statistical reports on patient volumes and appointment trends

Reviewing Hospital Policies:

- Understanding hospital policies regarding appointment scheduling, cancellations, and rescheduling.
- Examining any service level agreements or performance targets related to patient wait times.

Interview

Patient Interviews: Gathering diverse perspectives from various patient groups at hospital:

- Outpatients with different types of appointments (consultations, follow-ups, procedures)
- Inpatients who require follow-up appointments after discharge
- Emergency room patients who may need follow-up care
- Patients with chronic conditions who require regular appointments
- Gathering feedback on their experiences with the current appointment process.
- Identifying pain points and challenges they face (long wait times, difficulty reaching the hospital, lack of clarity on doctor's availability).
- Understanding their needs and expectations for a new appointment system (online booking, mobile app, appointment reminders, doctor information).

Staff Interviews:

- Interviewing doctors, nurses, receptionists, and administrators to gather insights on the current system and their needs for a new system.
- Understanding staff challenges and how a new system can improve their workflow.

Research

- **Benchmarking:** Investigating best practices and innovative solutions for hospital appointment scheduling systems in Vietnam and internationally.
- **Literature Review:**
    - Reviewing healthcare publications and research on patient satisfaction and efficient appointment scheduling.
    - Staying updated on relevant government regulations and guidelines related to healthcare data management in Vietnam.
- **Market Research:** Analyzing trends in patient expectations and technology adoption in healthcare.

By meticulously applying these fact-finding techniques, the project team can gain a thorough understanding of the current situation at Thu Đuc Regional General Hospital, identify areas for improvement, and develop a new appointment scheduling system that meets the needs of both patients and staff.

2.1.2. Requirements Modeling

Business Process Modeling (BPM)

BPM is a tool that can describe business processes into diagrams that show events, processes, and workflow. During modeling, analysts use a standard language called Business Process Modeling Notation or BPMN. In a BPMN the diagram is called a pool and each customer process is called swim lanes. Using BPM helps identify business processes faster leading to reduced cost and fewer errors in development.

Data Flow Diagram (DFD)

DFD is a diagram that is created from FDD. Analysts use DFD to create data flow of each function in an FDD to show how the system stores, processes, and transforms data. DFD shows how data moves through a system but does not show how that system works. DFD is mainly used during the system analyzing phase so focusing on the implementation problem is not a main point.

DFD uses basic symbols that represent processes, data stores, entities, and data flow. DFD also has multiple levels starting from context diagram, diagram 0 to diagram 3 DFD. Each shows a more detailed data flow with the context diagram showing the simplest data flow.

Unified Modeling Language (UML)

UML is used to visualize and document software systems and uses the concept of object-oriented design. Analysts use UML to represent the system from the perspective of the user. UML can be used in use case diagrams, and sequence diagrams that show the steps in a specific business process and show how classes or objects interact with each other.

2.1.3. System Analysis

Agile Method

Agile is a new methodology that focuses on developing a system incrementally by building prototypes and adjusting them to the user's needs and requirements. This methodology emphasizes continuous feedback and improvements are affected by what was learned in the previous iteration. One approach to Agile development is called Scrum where in a session, each member of the team plays a specific role and has a specific guideline that focuses on time blocks and interactions that deliver systems as a result (Beck et al., 2001).

The advantage of agile is it is very flexible and adaptable to changes, agile also brings deliverables consistently, reducing risk and validating the project to the stakeholders. However, due to the nature of agile, it lacks structure and documentation, bringing

significant risks. Agile also requires a high level of technical and interpersonal skills of each team member.

2.2. Deliverables

Following chapter 2, the project will produce the following key deliverables, each of which is crucial to the successful development and implementation of the Mobile Appointment Scheduling System for Thu Duc Regional General Hospital:

Fact-Finding Reports

These reports will document the outcomes of observation techniques. The primary purpose of these reports is to provide a comprehensive understanding of the current appointment scheduling processes, identifying inefficiencies and gaps.

They will serve as a foundation for informing the design and development of the new system, ensuring alignment with the operational needs of the hospital and enhancing the patient experience.

BPMN

Utilizing BPM and BPMN, these diagrams will visually represent the existing (as-is) business processes related to appointment scheduling and patient interactions.

By illustrating the current workflows, these models will enable stakeholders to identify areas for improvement and streamline operations. BPMN diagrams are essential for ensuring that the project objectives are in harmony with the hospital's operational framework.

Business Requirements Document (BRD)

The BRD will outline the functional and nonfunctional requirements of the proposed system, derived from insights gained during the fact-finding phase.

This document will specify the critical capabilities that the system must deliver to fulfill the hospital's requirements, serving as a reference for all stakeholders to ensure mutual

understanding and agreement before the development phase. The BRD is instrumental in minimizing the risk of scope creep and ensuring project alignment.

Systems Requirements Document (SRD)

The SRD will detail the technical specifications required for the system, encompassing hardware, software, and network infrastructure requirements.

This document will guide the development team in implementing a system that not only meets the business needs but also ensures compatibility with the existing hospital management framework. The SRD is vital for facilitating a clear technical roadmap for the implementation process.

Each of these deliverables will significantly contribute to the project's success, ensuring that the Mobile Appointment Scheduling System is effectively designed and implemented to address the identified challenges and meet the operational requirements of Thu Duc Regional General Hospital.

2.3. Current system and system architecture/diagrams (as-is business)

2.3.1. Making a medical appointment

The medical appointment booking process at Thu Duc General Hospital can be divided into two main methods: in-person and online/hotline booking.

a) In-person appointment booking process

Step 1: Go to hospital

The patient prepares personal documents and goes to Thu Duc General Hospital.

Step 2: Get a ticket automatic number

After entering the hospital, the patient goes to the area to get an automatic number. There are machines where the patient presses a button and receives a paper with a number. This paper includes information about the queue number and the medical department the patient registered for.

Step 3: Complete registration

After getting the number, the patient waits in the waiting area near the registration desk. The electronic screen shows the queue number currently being served. When it's the patient’s turn, they will be called to the registration desk to complete the process. At this point, the medical staff will ask for personal information, choose the examination area, and provide necessary instructions.

Step 4: Pay the examination fee

Afterward, the patient completes the payment for the examination and receives an appointment slip along with directions to the examination room.

b) Online and Hotline Appointment Booking Process

Step 1: Book an appointment

The patient can book an appointment via the hospital's hotline or official website. When booking, the patient needs to provide personal information, choose the medical department, and preferred time.

Step 2: Pay the examination fee and confirm the appointment

After booking, the patient pays the fee. Once completed, the hospital confirms the appointment via text message with details about the time and place.

Step 3: Arrive at the hospital

On the appointment day, the patient goes directly to the assigned examination area without needing to register or get a number at the hospital.

The diagram illustrates this process visually, detailing the sequential steps for both in-person and online/hotline booking procedures.

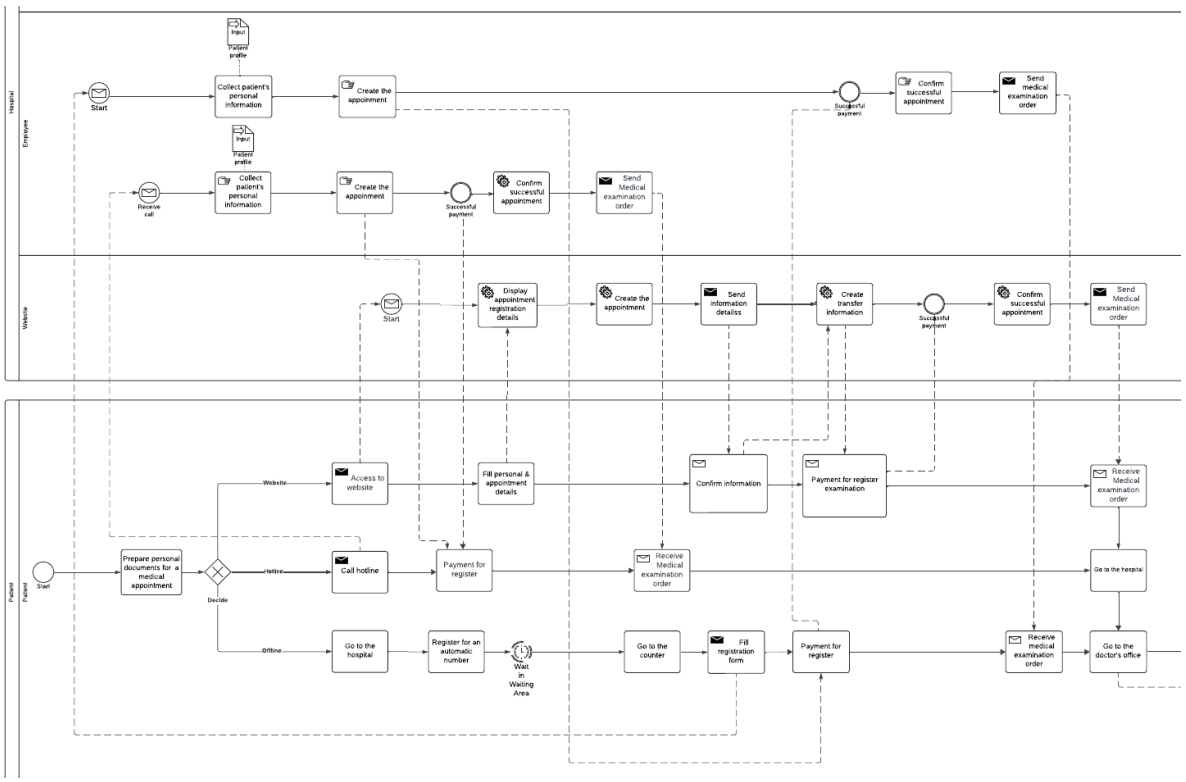


Figure 2. 1. BPMN of as-is business appointment booking process (Source: Authors)

The BPMN diagram illustrates the current business process for booking medical appointments. It includes three main participants: **Employee**, **Website**, and **Patient**.

1. **Employee**: Begins by collecting the patient's information and creating the appointment, confirming successful scheduling, and sending the medical examination order.
2. **Website**: Supports the booking process by displaying available appointment slots, collecting appointment details, confirming appointments, and sending necessary examination orders.
3. **Patient**: Initiates the process by preparing necessary documents, choosing between online (website) and offline (phone or in-person) registration. They proceed with payment and receive confirmation, eventually attending the medical examination.

The process flows through multiple stages, including information collection, appointment confirmation, and order processing, with interactions across both online and offline channels.

2.3.2. Medical Examination Process

Step 1: Undergo the initial examination

When arriving at the examination area, the patient waits in the waiting area. The electronic screen displays the queue number, and the patient's name is called when it's their turn. The doctor will examine the patient, discuss symptoms, and provide an initial diagnosis. If tests are required, the doctor will issue test orders.
If no tests are required, skip steps 2 and 3.

Step 2: Pay for the tests

The patient takes the test order and goes to the cashier's desk to pay for the necessary tests. After payment, the patient will receive instructions on where to go for the testing.

Step 3: Perform the tests

After paying, the patient goes to the designated testing area (e.g., blood tests, X-rays, ultrasounds, etc.). The patient will undergo the tests according to the doctor's instructions. Once completed, the patient waits to receive the test results.

Step 4: Return to the doctor with test results

After receiving the test results, the patient returns to the doctor for further consultation. The doctor will review the results, provide a final diagnosis, and decide if medication or additional treatment is needed.
If no prescription is needed, skip steps 5 and 6.

Step 5: Pay for the medication

If the doctor prescribes medication, the patient takes the prescription to the cashier's desk and pays for the prescribed medication.

Step 6: Receive the medication

After payment, the patient goes to the pharmacy to collect the medication. The pharmacy staff will prepare the medication based on the doctor's prescription.

Step 7: Leave the hospital

Once all procedures are complete, the patient can leave the hospital. If a follow-up is required, the medical staff will assist in scheduling the next appointment.

The BPMN diagram illustrates this process visually, detailing the sequential steps involved in the patient's journey. Each phase of the process is clearly outlined, providing a comprehensive overview of the medical examination workflow.

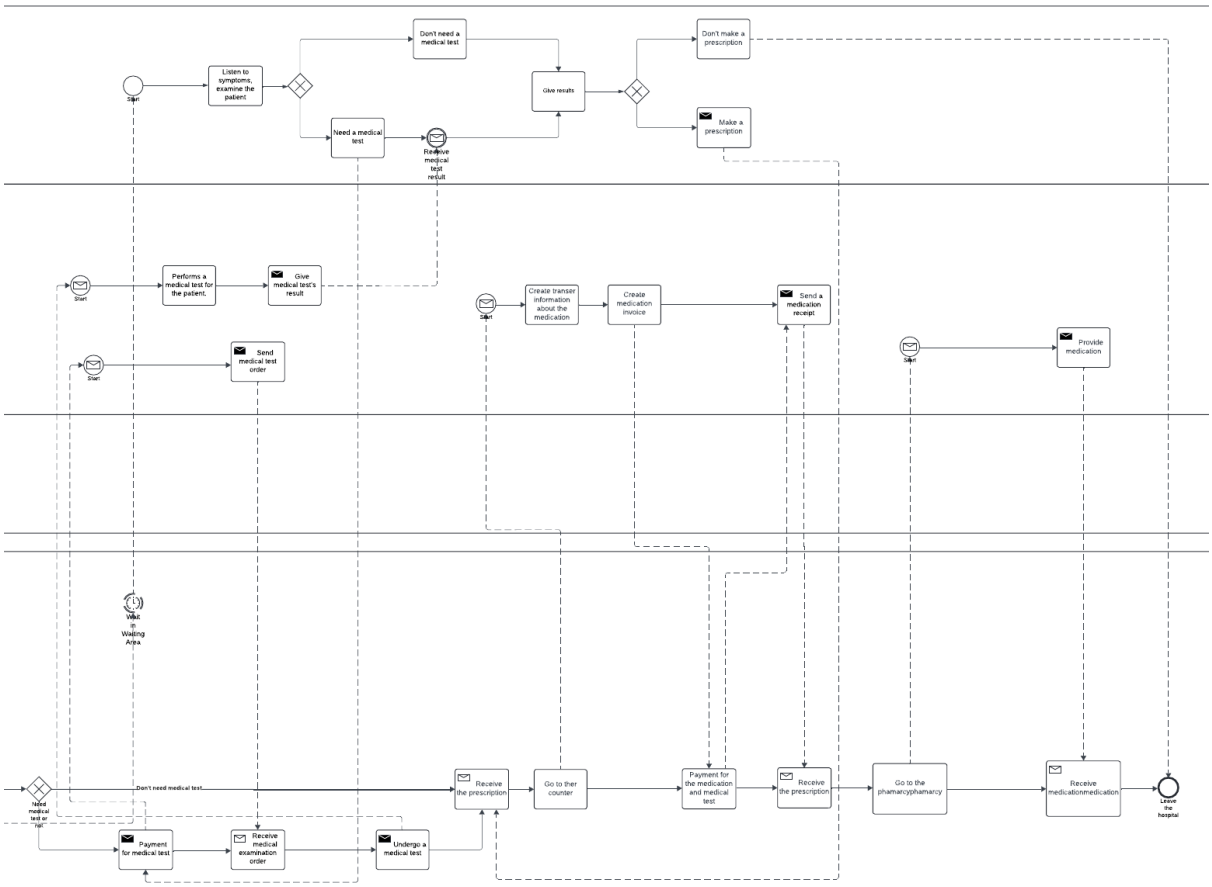


Figure 2. 2. BPMN of as-is business medical examination process (Source: Authors)

The BPMN diagram depicts the current process for conducting a medical examination, involving three main participants: **Doctor**, **Pharmacy**, and **Patient**.

Doctor: Begins by examining the patient to determine if a medical test or prescription is needed. If a test is required, the doctor performs it, provides the result, and sends a medical test order. Based on the examination, the doctor may

prescribe medication, create an invoice, and send a medication request to the pharmacy.

Pharmacy: Receives the medication request and prepares the medication for the patient.

Patient: Initiates the process by consulting the doctor, undergoing any required tests, and receiving prescriptions. They proceed with payment, receive the prescription, go to the pharmacy, and collect the medication.

The process involves various stages, including examination, testing, prescription, payment, and medication collection, with interactions between medical and pharmacy services.

2.4. Business Requirement Documents

Introduction: This section outlines the essential business requirements needed for the development of the Mobile Appointment Scheduling System for Thu Duc Regional General Hospital. These requirements aim to address the issues identified in section 1.4 regarding the current state and specific needs of the hospital. They will provide the foundation for the functionalities that the system must perform while supporting the goal of improving patient experience and optimizing hospital workflows.

Objectives

The main objectives of the system are:

- Provide patients with a convenient mobile application to schedule medical appointments.
- Offer a user-friendly interface for patients to select symptoms, preferred dates, times, and doctors.
- Enable patients to make secure online payments for confirmed appointments.
- Integrate the appointment scheduling system with the hospital's existing management system to ensure real-time updates on patient appointments and medical records.
- Streamline the check-in process by generating queue numbers for patients upon arrival at the hospital.

- Support prescription management with real-time updates to patient records during and after consultations.

Features

The system will include the following core features:

- User registration and authentication: Allow patients to create and manage profiles with secure hospital authentication.
- Appointment scheduling: Enable patients to select appointment dates, times, and doctors, with real-time availability displayed.
- Online payments: Provide integrated payment options for patients to confirm appointments online.
- Queue management: Automatically generate queue numbers upon patient check-in, reducing waiting times and improving patient flow.
- Prescription and record management: Allow doctors to update patient records and prescriptions, with real-time access for patients.
- Integration with hospital systems: Seamlessly connect with the hospital's existing management system to update appointments, medical records, and prescriptions.

Functional Requirements

- Enable user registration and patient profile management.
- Allow patients to schedule, view, and manage appointments, including cancellations.
- Implement secure online payment systems for booking confirmations.
- Provide access to both historical and upcoming appointments.
- Enable doctors to access and update patient medical records in real-time.
- Generate queue numbers upon check-in and notify patients of their queue status.
- Ensure real-time synchronization between the mobile application and the hospital management system.

Non-functional Requirements

- Security: Ensure robust data protection and encryption measures to safeguard patient information and payment details.

- Scalability: The system must be able to handle an increasing number of users and appointments without performance degradation.

- Reliability: The system must ensure minimal downtime, operating with high availability to guarantee consistent service.

Business Constraints

The following constraints should guide the system's development:

- The system must comply with healthcare regulations and data protection laws, such as the decree 13/2023/NĐ-CP (Government of Vietnam, 2023).

- The system must integrate with the hospital's existing management system to ensure seamless operation.

- The application must be compatible with popular mobile operating systems (iOS and Android).

In conclusion, the requirements outlined in this section are prerequisites for the successful development and implementation of the appointment scheduling system. They support the objectives established in section 1.5, which focus on building solutions to optimize the patient care process at Thu Duc Regional General Hospital. The connection between these business requirements and the current issues will be further expanded upon and analyzed in detail in section 2.5, where specific requirements will be described more clearly to aid in the design and development of the system.

2.5. Systems Requirements Documents

2.5.1. Functional Requirements

The functional requirements of the mobile appointment scheduling system for Thu Duc Regional General Hospital outline the essential functionalities that the system must provide to ensure it meets the operational needs of the hospital and the patients' demands for efficient and seamless services.

Table 2. 1. Functional Requirements (Source: Author)

ID	Name	Description		
		Input	**Process**	**Output**
F01	**User Registration and Profile Creation**	**Personal Information:** The system must collect detailed user information, including their full name, date of birth, gender, email address, phone number, hospital identification (ID), and any other required contact details. **Authentication Data:** Users should also create a secure password for future logins. If applicable, the system may also collect biometric information (e.g., fingerprints or face scans) for added	**Data Validation:** The system must validate the entered data, ensuring that fields like email, phone number, and hospital ID are correctly formatted and unique. For example, the email should follow standard formats, and the phone number should match regional or international formats. **Password Strength Check:** The system must enforce password complexity requirements, such as a mix of upper and lower case	**Registration Confirmation:** The system must provide a confirmation message to users upon successful registration, either as an in-app notification or an email/SMS. **Profile Update Confirmation:** When a user updates their profile, the system should send a confirmation message indicating that the changes have been saved.

		security. **Verification:** The system should include input fields for any verification codes sent via email or SMS to validate the user's identity during registration.	letters, numbers, and special characters. **Data Storage:** Upon successful validation, the system securely stores user information in a database. Passwords should be hashed using secure algorithms (e.g., bcrypt). **Profile Updates:** Users must have the ability to update their profile information, and the system must validate and securely update these records in real-time.	
F02	**Appointment Scheduling**	**Preferred Appointment Details:** Patients must select a	**Doctor Availability Check:** The system must query the	**Appointment Confirmation:** Upon successful booking, the

		preferred date, time, department, and doctor. Additionally, they may need to specify the type of appointment (e.g., consultation, follow-up, or specific tests). **Patient's Health Condition:** The system could allow patients to provide brief descriptions of their symptoms or medical conditions to help doctors prepare for the appointment. **Insurance Information:** If applicable, the system should capture insurance details to ensure payment processing or	hospital's doctor schedule to determine available time slots. It should take into account any public holidays, doctor vacations, or emergency unavailability. **Conflict Prevention:** The system must check for overlapping appointments to ensure that no two patients are scheduled for the same time slot with the same doctor. **Booking Confirmation:** Once a valid time slot is selected, the system must confirm the booking and reserve that slot in	system provides a confirmation message with the details of the scheduled appointment, including time, date, doctor's name, and department. This could be sent via in-app notifications, email, or SMS. **Available Time Slots:** If the initially selected time slot is unavailable, the system should display alternative available slots.

		claims are made accordingly.	the hospital's scheduling system.	
F03	**Appointment Management**	**User Actions:** Patients must be able to view, reschedule, or cancel appointments through the application. This includes selecting the appointment they wish to change or cancel from a list of their upcoming bookings. **Reschedule Requests:** When rescheduling, the patient must provide a new preferred date, time, doctor, or department.	**Real-Time Appointment Status Updates:** The system must process these actions in real-time, updating the appointment status in the hospital's database to reflect the changes. **Automatic Notifications:** Once an appointment is rescheduled or canceled, the system must automatically notify both the patient and the doctor of the change.	**Updated Appointment Status:** The system must immediately reflect any changes in the appointment status within the user's profile and provide confirmation via notifications. **Reminders:** The system should send reminders to patients before their appointment date, typically 24 hours in advance, via email or push notifications.
F04	**Payment System Integration**	**Payment Details:** The system must securely collect	**Secure Payment Processing:** The system must	**Payment Confirmation:** Once the

		payment information, such as credit card number, expiration date, CVV, and billing address. **Consultation Fees:** The system should display the total cost for the consultation, including any additional fees such as taxes or service charges. **Insurance Claims:** For insured patients, the system should gather insurance policy numbers and process claims if applicable.	securely handle payment transactions using encrypted communication channels (e.g., SSL). The payment gateway must process the transaction in real-time. **Fraud Detection:** The system should include basic fraud detection features, such as monitoring payment attempts for suspicious activity or mismatched addresses.	transaction is successful, the system must send a confirmation message to the patient, updating the appointment status to "Paid." **Failed Payment Notification:** If the payment fails, the system should provide an error message and allow the patient to re-enter their payment details.
F05	**Medical Records Access**	**Request for Records:** The system should allow patients to request access to	**Data Retrieval:** Upon a successful request, the system must query the hospital's database	**Medical Records Display:** The patient's medical records must be displayed within

		their medical history, including previous consultations, diagnoses, prescriptions, and test results. **Patient Authentication:** The system should verify the patient's identity before granting access to sensitive medical records, possibly using two-factor authentication (2FA).	to retrieve the patient's medical records. This must happen in real-time to provide accurate and up-to-date information. **Data Formatting:** The retrieved records should be formatted in a user-friendly way, using tables, graphs, or summaries where appropriate.	the application, allowing them to view, download, or print the information. **Notifications:** The system must notify the patient once their requested records are ready for viewing.
F06	**Real-Time Notifications and Updates**	**Triggers for Notifications:** Notifications should be triggered by various system events such as appointment changes, new test results, prescription updates, or	**Monitoring System Updates:** The system must monitor appointment bookings, status changes, and other updates in real-time to trigger relevant notifications.	**Real-Time Notifications:** Notifications should be sent to patients, doctors, and relevant staff as soon as updates occur. These notifications must be clear and concise, providing

		payment confirmations. **Communication Preferences:** The system should allow patients to choose how they prefer to receive notifications (e.g., email, SMS, or in-app).	**Notification Scheduling:** The system should automatically schedule notifications based on predefined rules, such as reminders sent 24 hours before an appointment.	relevant details (e.g., "Your appointment with Dr. Smith has been confirmed").
F07	**Integration with Hospital Management System**	**Patient Data:** The system must receive and process patient information, including appointment details, medical records, and billing information, from the hospital's management system. **API or Middleware Requests:** Inputs will consist of API	**Data Synchronization:** The system must ensure that all data, such as appointments, prescriptions, and medical records, are synchronized in real-time between the hospital's management system and the mobile application. **Conflict Resolution:** The system must	**Consistent Data:** The system must ensure that data in the mobile application reflects the hospital's most current information, preventing discrepancies. **Confirmation of Synchronization:** Both the hospital's system and the mobile app must provide confirmation that

		calls or middleware communications designed to retrieve and send data between the hospital's system and the mobile application.	resolve data conflicts (e.g., simultaneous updates) using predefined rules or algorithms to ensure data integrity. **Data Security:** Encryption and other security measures must be applied to protect patient information during transmission between systems.	data synchronization has occurred successfully.
F08	**Prescription Management**	**Prescription Details:** After consultations, doctors must enter prescription details into the hospital's system, including medication names, dosages, instructions, and any additional notes for the	**Prescription Storage:** The system must securely store the prescription data and link it to the patient's medical records. **Real-Time Availability:** Once entered by the	**Prescription Access:** The system must display the prescription information, allowing the patient to view and download it. Notifications should also be sent when new

		patient. **Patient Information:** The system must identify which patient the prescription is for, and link the prescription to their profile.	doctor, prescriptions must be immediately accessible to the patient through the application.	prescriptions are added. **Future Reference:** The system must ensure that patients can easily access their prescription history for future reference or reordering of medications.

2.5.2. Non-Functional Requirements

Non-functional requirements are crucial for ensuring that a system operates effectively, efficiently, and securely. Based on Circular 54/2017/TT-BYT, which outlines the criteria for applying information technology in healthcare facilities, these requirements emphasize key areas such as performance, security, usability, and compliance (The Ministry of Health, 2017). By focusing on these aspects, the system can achieve reliable operation while providing a smooth and user-friendly experience. This structured approach ensures that the system meets both operational standards and user expectations.

Table 2. 2. Non-Functional Requirements

ID	Type	Name	Description
NF01	**Performance requirements**	Enviroment	The system must be designed primarily as a mobile application, optimized for the app interface to ensure smooth navigation, fast load times, and a user-friendly experience.

NF02		Load handling	The system should support at least 1,000 concurrent users without performance degradation. Stress testing should be performed regularly to evaluate system behavior under heavy loads.
NF03		Speed and Responsiveness	The system should respond to user requests within 2-3 seconds for most operations (e.g., scheduling appointments, viewing medical records). Queries to the database for appointments or patient data should execute within 1-2 seconds.
NF04		System Availability	Ensure the system operates with 99.9% availability, allowing for minimal downtime only during maintenance windows scheduled in off-peak hours (e.g., weekends, late at night).
NF05		Data Throughput	The application must process at least 500 transactions per second (appointment bookings, updates, etc.) without bottlenecks.
NF06		Updates	The system should be updated quarterly with performance improvements and security patches. Critical security updates

			must be deployed within 24 hours of identification.
NF07	**Platform Constraints**	Supported Platforms	The application must be compatible with iOS and Android.
NF08		Backend Tools	Cloud-Based Infrastructure: Opt for cloud-based solutions (e.g., AWS, Google Cloud) to ensure flexibility in scaling resources based on demand.
NF09		Offline Access	Ensure offline functionality for some features, such as viewing past appointment details, in areas with poor internet connectivity. Sync data once the device is back online.
NF10		API Support	Ensure that the app integrates smoothly with third-party APIs for services like payment gateways, and health insurance verification.
NF11		Interoperability	The application must be capable of connecting and sharing data with other healthcare systems and software.

NF12	**Accuracy and Precision**	User Differentiation	Each user will be uniquely identified by unique hospital-provided ID.
NF13		Data Validation	Ensure all input fields, such as patient names, ID numbers, and contact information, follow strict validation rules to avoid errors (e.g., phone number format, correct date format, no special characters).
NF14		Session Handling	Maintain session accuracy, ensuring that the correct user data is accessed during interactions and preventing unauthorized session hijacking.
NF15		Error Tolerance	The system should handle input errors gracefully and provide detailed, user-friendly error messages (e.g., "Invalid input, please enter a valid phone number").
NF16		Duplicate Data Prevention	The system must check for duplicate entries, especially when handling medical records, ensuring no accidental duplication of patient information.

NF17		Precision in Scheduling	Implement precise time slot management for appointment scheduling, including handling overlapping appointments and cancellations automatically.
NF18		Alert System for Errors	The system should have a built-in *alert system* to notify administrators of unusual data patterns (e.g., duplicate medical records, appointment conflicts).
NF19		Logging	The system must log all user impacts on a centralized server for error analysis or system impact tracking as necessary.
NF20		Error Documentation	There must be a process for logging errors and their resolutions, particularly for safety and security-related issues during testing.
NF21	**Adaptability**	Modular Architecture	The system should have a modular design, allowing the addition or modification of features (e.g., new appointment types, notification methods) without requiring major rewrites of the codebase.
NF22		Device Adaptability	The app must automatically adjust its layout for different screen sizes and

			resolutions, ensuring a consistent experience across smartphones and tablets.
NF23		Scalable Infrastructure	As demand grows, the system must be easily scalable by adding new servers or cloud instances without requiring downtime.
NF24	**Security**	Data Copy Control Mechanism	Prevent physical data copy storage devices and install data copying prevention software.
NF25		Intrusion Prevention Firewall	Implement a specialized firewall between Internet, application servers, and internal network users to prevent unauthorized access.
NF26		Session Timeout	The system should automatically log users out after 15 minutes of inactivity to prevent unauthorized access.
NF27		Two-Factor Authentication (2FA)	2FA should be required for all administrative logins and for critical actions like viewing medical records or confirming appointments.

NF28		Audit Logging	A detailed audit log must be maintained, recording all actions performed on sensitive data, such as who accessed a record, when, and what changes were made.
NF29		Regular Security Audits	The system must undergo monthly security audits to identify vulnerabilities and ensure compliance with industry standards.
NF30		Data Encryption	Encrypt sensitive data to prevent data theft, managing decryption keys effectively.
NF31		Password Encryption	User passwords must be encrypted using salt and hash techniques to prevent password theft.
NF32	**Compliance**		The system must comply with healthcare regulations to ensure the confidentiality and privacy of patient data.

Part 3: System Analysis And Design

3.1. Overview and purposes

This part provides a comprehensive description of the system components and functionalities, including the Login/Signup process, User Account Management, Appointment Booking, Payment processing, Viewing Medical Records, and Tracking Hospital Visits. Each section details how these elements are integrated to ensure a seamless experience for users. The goal of part 3 is to establish a clear understanding of the core functions of the Mobile Appointment Scheduling System, highlight its capabilities, and demonstrate its role in improving healthcare access and efficiency.

The purpose of this part is to outline the system's architecture and workflow, illustrating how different features and processes interact to deliver convenient healthcare services. It serves to communicate the structure and behavior of the system to developers, stakeholders, and users, providing insight into how the system helps users schedule and manage their medical care efficiently. Each component is depicted using diagrams like BPMN, DFD, and use case diagrams, facilitating better understanding of the flow of data and user interactions within the system.

3.2. Deliverables (List and describe)

After part 3, the deliverables will include important documents that serve as the foundation for the development and deployment of the new system:

BPMN

BPMN (Business Process Model and Notation) is a tool that models business processes using standard symbols. It provides a clear visual representation of the new process by illustrating the actors involved, the main activities, the sequence of steps, and how data and information flow between these steps. Additionally, BPMN highlights the interaction points between users and the system, such as the start and end points of the process. This visual representation helps stakeholders easily understand the process, enabling them to make informed decisions on improvements and optimize the system more effectively.

DFD

Data Flow Diagrams (DFDs) are a tool for describing information systems by representing data flows and how data is processed. There are three levels of DFDs: DFD context level, DFD level 0, and level 1.

DFD context level: This is the highest level of DFD, representing the system as a single entity. It shows how the system interacts with external entities (like users or other systems) and clearly indicates the main data flows to and from these entities. DFD at this level provides an overview of the system and how it interacts with external actors.

DFD level 0: At this level, the system is broken down into its main functions. Each function handles a specific part of the system and is connected by data flows representing the information moving between them. DFD level 0 shows the main functions of the system, how data moves through the system, and how it's processed in the main processes.

DFD level 1: This is a more detailed version of the DFD level 0. Each process in the DFD level 0 is broken down into sub-processes to represent the system's activities in more detail. DFD level 1 describes more clearly how a specific process works within the system, providing the necessary details to implement the system or identify detailed system requirements.

Data dictionary

A Data Dictionary is a centralized repository that contains detailed information about the data in a system. It is used to collect, organize, and store detailed information related to all the data elements in the system. The Data Dictionary helps document elements such as:

Data Elements: The smallest meaningful units of data within the system.

Data Flows: Descriptions of how data moves between processes or data stores.

Data Stores: Locations where data is stored within the system.

Processes: Processes that handle and manipulate the data.

Entities: External or internal entities that interact with the system’s data.

A data dictionary helps developers and stakeholders easily access, understand, and manage the data used in the system, ensuring consistency during system implementation and maintenance. Functions of the Data Dictionary:

Documenting Data Elements: It provides a way to document all the details about each data element, helping users understand the nature and purpose of the data.

Documenting Data Flows: The Data Dictionary records how data flows through different processes in the system, ensuring that all data movement is logical and consistent.

Documenting Data Stores: Defines and describes the data stores where information is kept, making it easier to manage data storage.

Documenting Processes: It helps document the processes that manipulate and process data, ensuring that all procedures are clearly understood and correctly implemented.

Documenting Entities: Finally, the Data Dictionary captures and describes the entities interacting with the system, assisting in system management and improvement.

Use Case Diagrams

Use Case Diagrams are essential tools for illustrating the interactions between users (actors) and the system. These diagrams represent the system's functionality from a user's perspective, focusing on what the system does for each type of user.

Actors in the system include individuals or systems that interact with the new system, such as patients, doctors, or administrators. Use Cases describe specific functionalities or services offered by the system, such as scheduling appointments, viewing patient records, or generating reports. Relationships between actors and use cases demonstrate which functionalities are accessible by different users.

Use Case Diagrams assist stakeholders in visualizing how the system will operate, ensuring all users' needs are met while maintaining clarity during system development.

User Stories

User Stories provide a simplified description of the system's features from an end-user perspective, focusing on the needs and goals of each user type. They typically follow the format: "As a [user type], I want to [perform an action] so that [achieve a goal]."

User Types include different categories of individuals that interact with the system, such as patients, doctors, and administrative staff. Tasks describe the various actions users will perform, such as booking an appointment, viewing medical records, or processing payments. The Value Proposition highlights the benefit that each task provides to users, helping to prioritize the system’s functionalities during development.

User Stories are particularly beneficial for ensuring that developers and stakeholders align on the system's value to end users, facilitating a user-centric approach to development.

Activity Diagrams

Activity Diagrams provide a visual representation of workflows, showing the sequence of activities and decision points within a process. They illustrate the flow of tasks from start to finish and how different actors are involved.

These diagrams show each activity and decision point, along with the flow of control from one task to the next. Swimlanes are used to divide activities between different actors or system components, helping to visualize the distribution of responsibilities.

Activity Diagrams are valuable for understanding workflows, identifying bottlenecks, and optimizing processes to ensure efficiency and clarity in the system’s operations.

Sequence Diagrams

Sequence Diagrams model the dynamic behavior of the system by showing interactions between objects in a time-sequenced manner. They help visualize the order of operations that occur within the system during specific interactions.

Objects represent different components or entities in the system, such as classes, subsystems, or actors. Messages indicate the communication between these objects,

specifying which operations are invoked and in what order. Lifelines depict the existence of these objects over time, while messages illustrate the flow of control.

Sequence Diagrams help ensure that developers understand how different parts of the system interact, which is critical for the development, integration, and debugging of the system.

Class Diagrams

Class Diagrams represent the static structure of the system, detailing classes, attributes, methods, and relationships. They are key to visualizing the architecture of the system and how its components relate to one another.

Classes are the main components or entities, such as Patient, Doctor, and Appointment. Attributes and Methods define the properties and operations associated with each class. Relationships between classes include associations, inheritance, and dependencies, illustrating how different components are linked.

Class Diagrams help developers understand the relationships within the system and organize the system's structure effectively during coding, ensuring a well-organized and consistent implementation.

User Interface (UI) Mockups

UI Mockups provide a visual prototype of what the user will see when interacting with the system. They represent the visual layout and design of the system's screens and elements.

Screens and Layouts show how different pages (such as login screens, booking forms, or user dashboards) will appear. Interactive Elements include buttons, text fields, menus, and other components that users will use to navigate and interact with the system. The User Flow demonstrates the paths users take through the system, showing how different screens connect and the actions users can take.

UI Mockups are crucial for ensuring that the system provides an intuitive, user-friendly experience. They serve as a communication tool between stakeholders, designers, and developers, aligning everyone on the final look and feel of the system.

3.3. System overall

3.3.1 Brief description of the system

Login/Signup: Allows users to create a new account or log into an existing one using their phone number. An OTP code is sent for verification, simplifying access and enhancing security.

Manage User Account: Enables users to update personal information, such as contact details and payment methods, ensuring profiles are current and manageable.

Booking: Facilitates appointment scheduling by letting users select a preferred doctor, department, and time slot, improving access to healthcare services.

Payment: Integrates secure payment processing for appointments, supporting both direct payments and insurance claims for a seamless transaction experience.

View Medical Record: Allows patients to access their medical history, diagnoses, prescriptions, and test results, providing transparency and easy follow-up management.

Tracking: Sends real-time notifications and updates on appointment statuses, test results, and other important reminders, keeping patients informed throughout their healthcare journey

3.3.2. New system BPMN

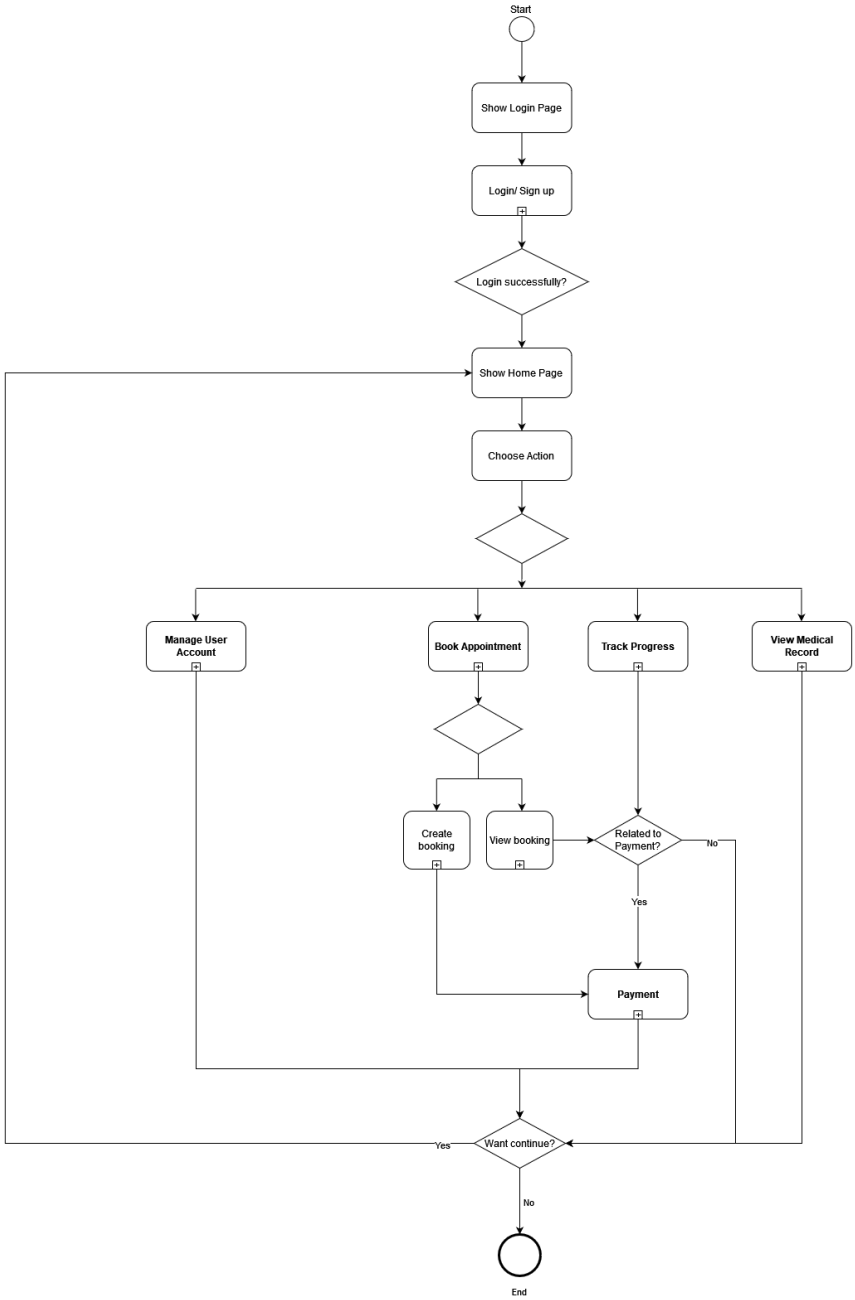


Figure 3. 1. Overall view of the BPMN (Source: Author)

The BPMN diagram provides an overall flow of user interactions with the system, covering major functions from login to specific actions related to appointments and records. Here's a more detailed breakdown:

Start and Login Process:

- The process begins with the user accessing the login page.

- The user can choose to either log in or sign up if they don't already have an account.
- If the login is successful, the user is directed to the home page. If not, they are prompted to try again.

Home Page and Action Selection:

- On the home page, the user is presented with various options. These include managing their account, booking an appointment, tracking progress, and viewing medical records.
- The system allows the user to choose one of these actions based on their needs.

Manage User Account:

- When selecting the "Manage User Account" option, the user is taken to a section where they can update their personal information, such as contact details or profile settings.
- After completing the updates, the user is directed back to the main action selection to continue or end the session.

Book Appointment:

- If the user chooses to book an appointment, they have two choices:
  - **Create Booking**: Initiates a new booking where the user can select a date, time, and department for their appointment.
  - **View Booking**: Allows the user to review or modify an existing booking.
- If the booking involves a payment (e.g., booking a paid service), the system redirects to the payment module to complete the transaction.

Track Progress:

- This option allows the user to monitor the status of their check-up or procedure.
- If any part of the tracking involves payments (for example, for ongoing services or due balances), the system navigates the user to the payment process to settle outstanding charges.

View Medical Record:

- This option provides the user access to their medical records, including detailed reports, lab results, or past visit summaries.
- After viewing or interacting with their medical records, the user can choose to return to the main menu for further actions.

Payment Process:

- For any action that involves a payment (either during booking or tracking), the system initiates the payment process.
- The user can complete the transaction using available payment methods. Upon successful payment, they receive confirmation, and the system updates the relevant status.

Session Continuation Check:

- After completing any action, the system checks if the user wants to continue.
- If the user opts to continue, they are redirected back to the home page to choose another action.
- If the user decides not to continue, the session ends, and the user is logged out.

3.3.3. DFD

a) Context diagram

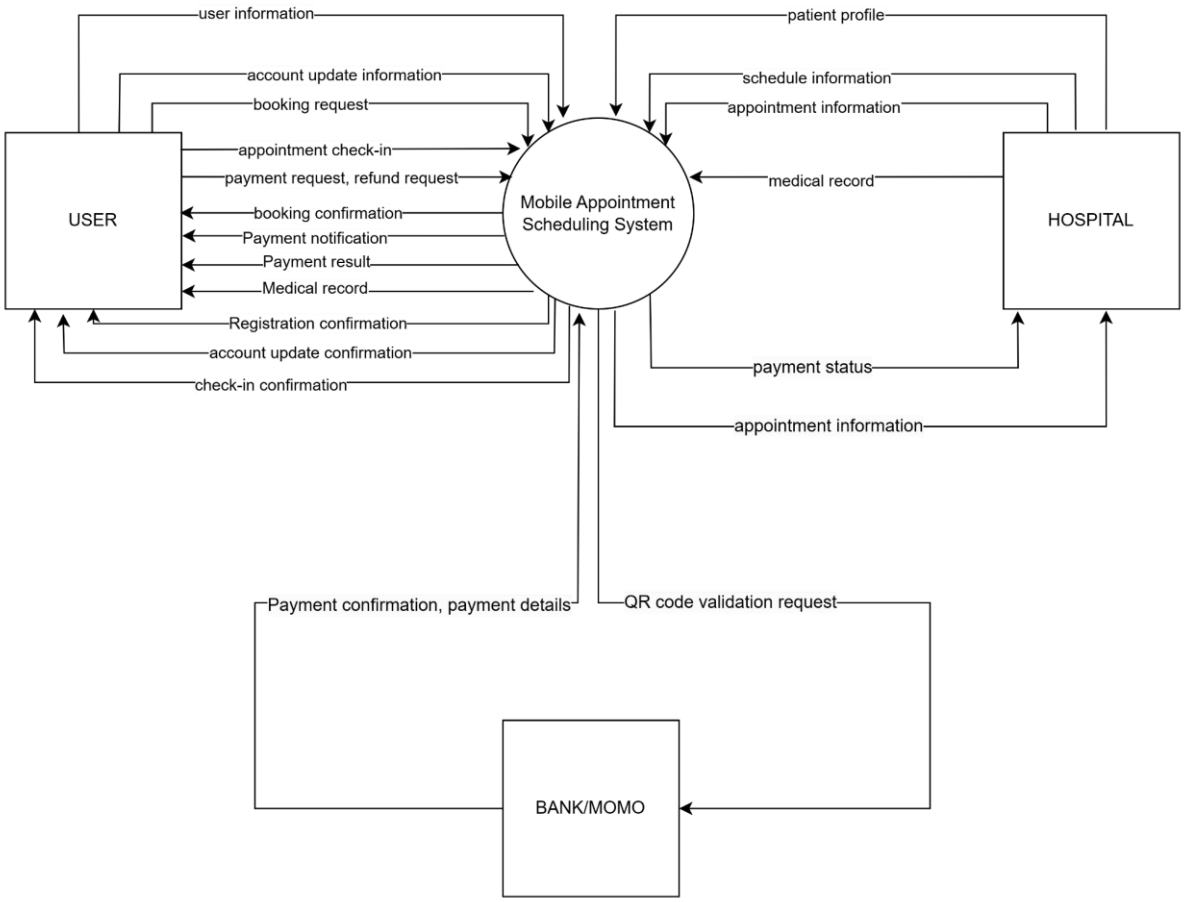


Figure 3. 2. Context diagram. (Source: Author)

The **Context Diagram of the Overall Process** provides an overview of the interactions within the Mobile Appointment Scheduling System.

1. **User**: Interacts with the system for various tasks such as booking appointments, checking in, updating account information, making payments, and requesting refunds. The system responds with confirmations, notifications, and medical record details.
2. **Hospital**: Supplies patient profile, schedule, appointment details, and medical records. It also updates the system with payment statuses and appointment confirmations.
3. **Bank/MoMo**: Processes payment transactions by validating QR codes and confirming payment details with the system.

b) Level 0 of DFD

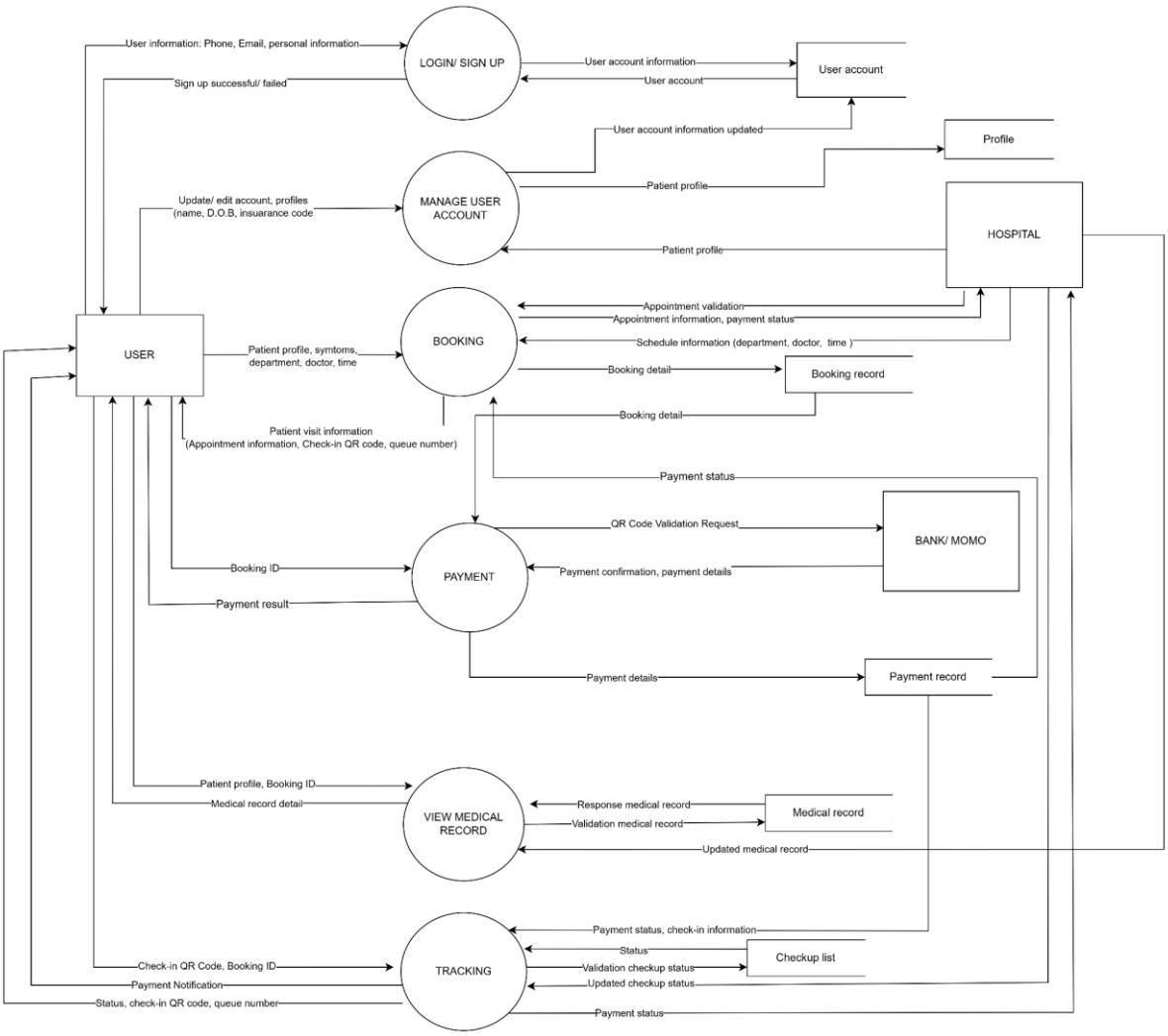


Figure 3. 3. Figure: DFD level 0 (Source: Author)

The **Level 0 DFD** breaks down the system into key processes:

- ***Login/Signup:*** Users create an account or log in. The system verifies credentials and returns a success or failure message.

- ***Manage User Account:*** Users can update their personal details, add or remove payment cards, and manage personal profiles.

- ***Booking:*** Users select a doctor, department, and time for their appointment, then confirm the booking.

- ***Payment:*** The system generates an invoice, processes payments through banks or MOMO, and confirms the payment status.

- ***View Medical Record***: Users can access their medical records, including examination results and prescriptions.

3.3.4. Overall Use Case

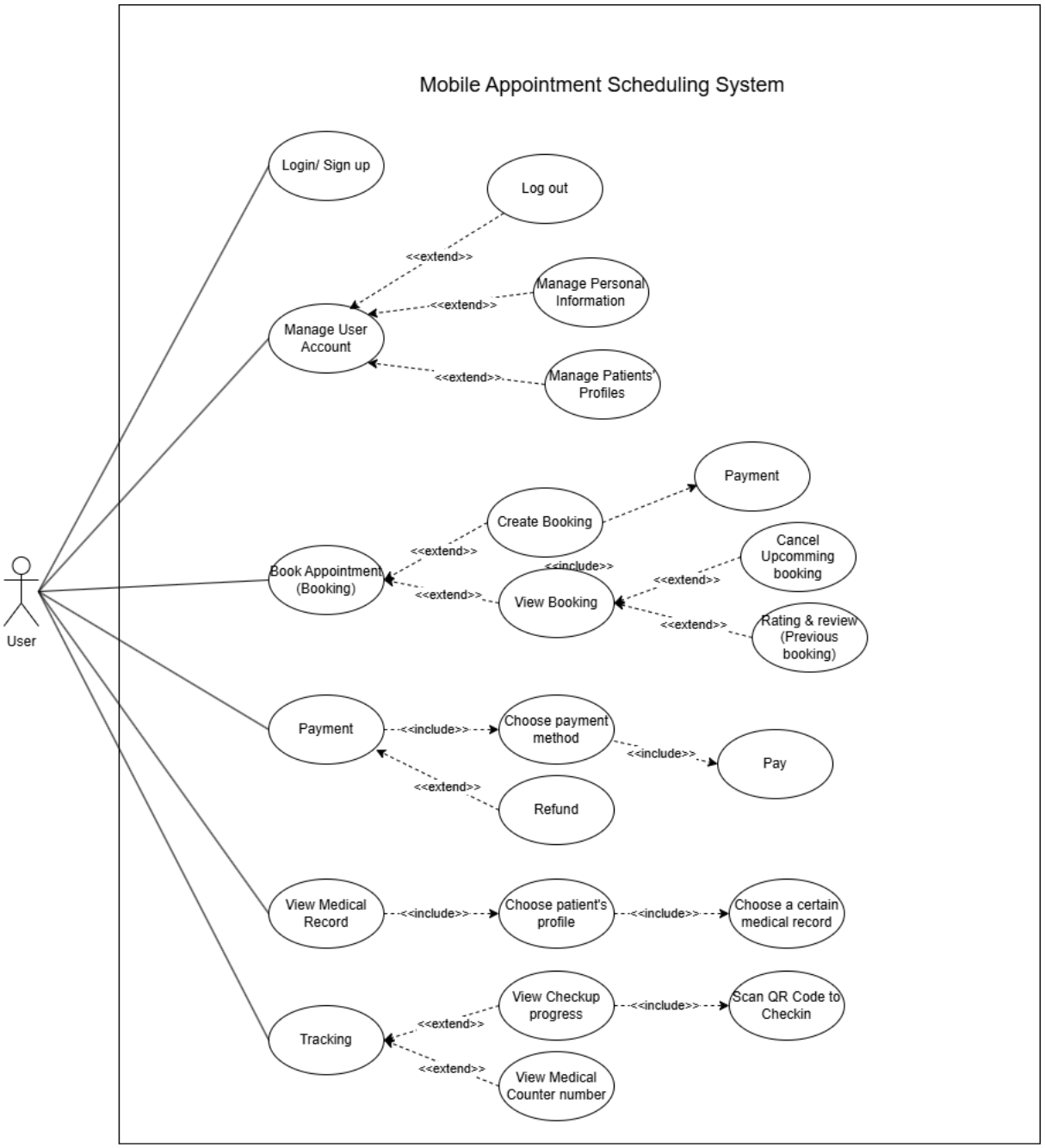


Figure 3. 4. Use case diagram of Mobile Appointment Scheduling System (Source: Author)

The use case diagram outlines the main functionalities of a **Mobile Appointment Scheduling System** for a user. The user can:

Login/Sign up and **Log out**.

Manage User Account, which includes managing personal information and patient profiles.

Book Appointment with options to create, view, cancel, and review past bookings.

Make Payments, with support for choosing payment methods, processing payments, and handling refunds.

View Medical Records by selecting a patient profile and specific records.

Track Appointments, including viewing checkup progress, medical counter numbers, and checking in via QR code.

3.3.5. Class diagram

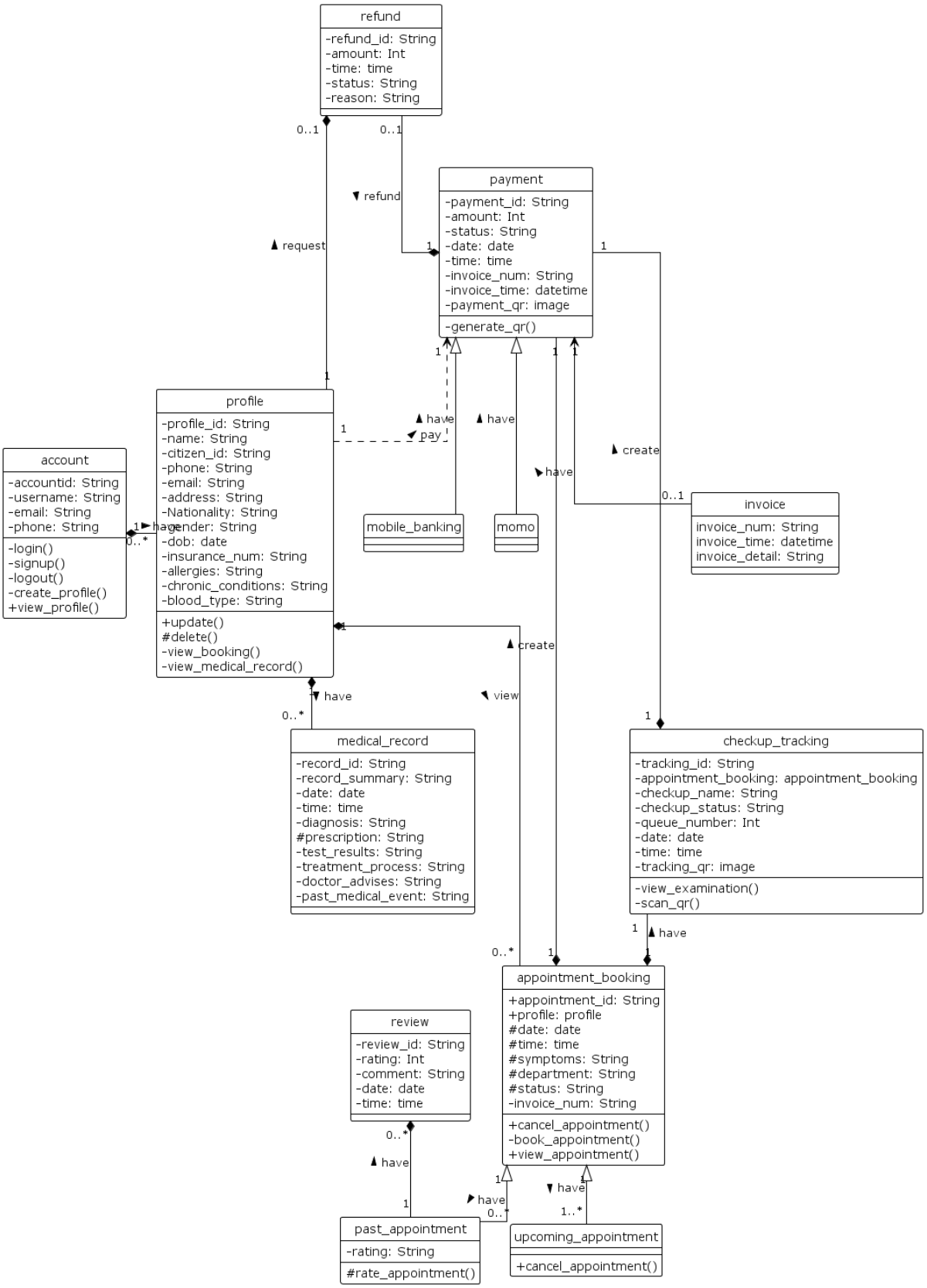


Figure 3. 5. Class diagram overview (Source: Author)

Class diagrams show objects as classes and relationships between them, this help development team have a greater understanding of the object in the actual system that they will make. The class diagram provides a structural overview of the **Mobile Appointment Scheduling System**, detailing classes, attributes, methods, and the relationships between them:

1. **Account**: Stores user details such as account ID, username, email, and phone, with methods for login, signup, logout, and profile management.
2. **Profile**: Contains personal information like profile ID, name, citizen ID, insurance number, and includes methods for updating, deleting, viewing bookings, and accessing medical records.
3. **Medical Record**: Holds health information including record summary, allergies, chronic conditions, prescriptions, and doctor advice, associated with a profile.
4. **Appointment Booking**: Manages appointment details such as appointment ID, profile, date, time, symptoms, and status, with methods to create, view, and cancel appointments.
5. **Payment**: Includes payment details like payment ID, amount, status, date, and time, with a QR generation method for mobile banking or Momo transactions.
6. **Checkup Tracking**: Tracks the progress of medical checkups with information like tracking ID, checkup name, status, queue number, and tracking QR, supporting viewing and scanning functionality.
7. **Review**: Allows users to rate and review past appointments, with attributes for review ID, rating, comments, date, and time.

3.3.6. Overall Activity Diagram

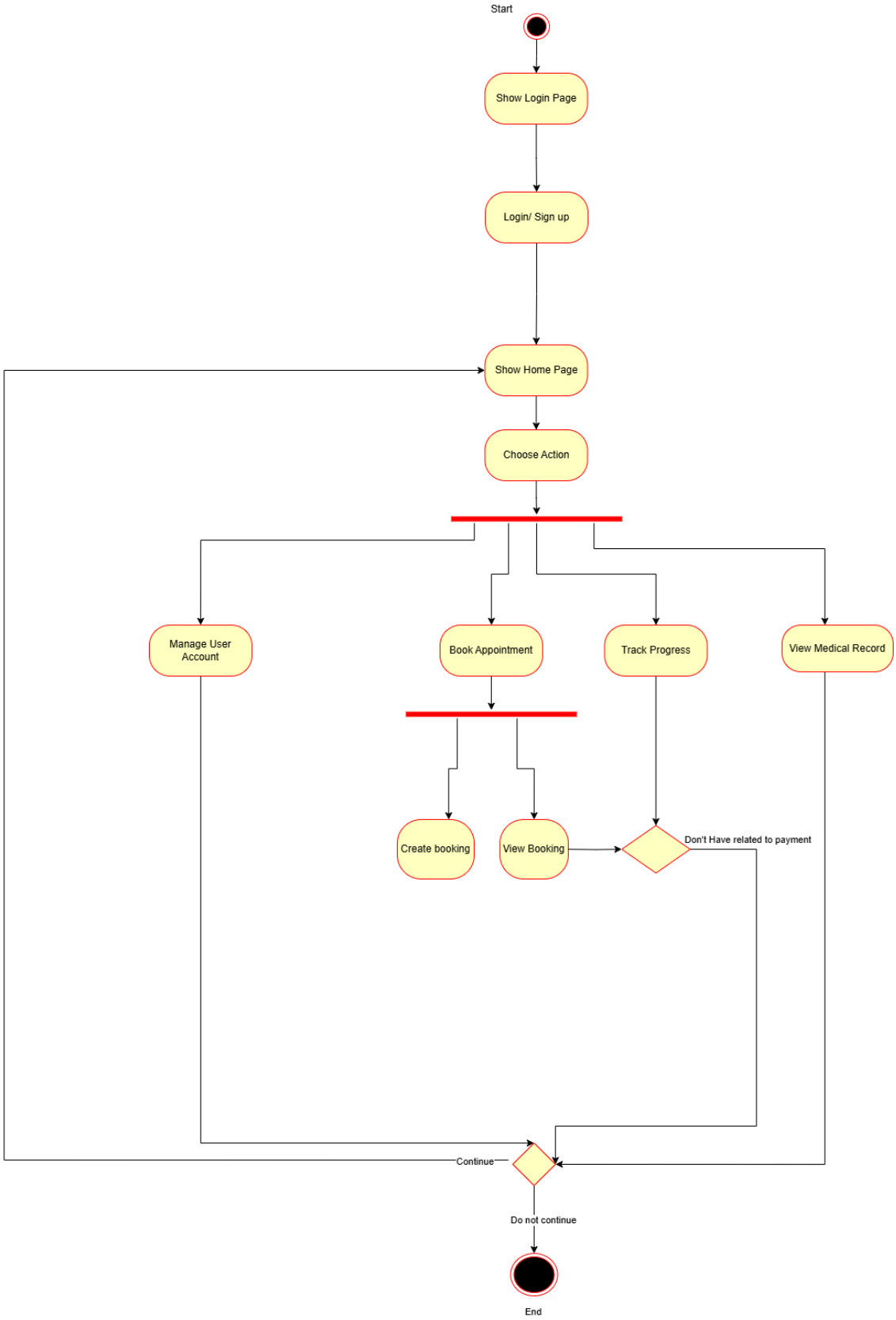


Figure 3. 6. Activity diagram of Mobile Appointment Scheduling System (Source: Author)

The activity diagram illustrates the workflow of a **Mobile Appointment Scheduling System**:

The process starts with displaying the login page, where the user can log in or sign up.

Upon successful login, the system shows the home page, allowing the user to choose an action.

The user has four main options:

- **Manage User Account**: Manage account details.
- **Book Appointment**: Create a new booking or view existing bookings.
- **Track Progress**: Track the status of appointments, with payment-related checks.
- **View Medical Record**: Access medical history records.

After performing an action, the user can decide to continue using the system or exit.

3.4. Detailed functions

3.4.1. Sign up & Login

a) Function introduce

Function meaning

Login is used to authenticate users. Users simply enter their phone number, and an OTP code will be sent for them to input in the next step, offering convenience and speed. This method eliminates the need to remember passwords or go through the hassle of resetting them. For first-time users, after entering the correct OTP, the application will request their name and citizen ID through a scanning process. Once logged in, the system will automatically log users in on future visits.

Business rules

The business rule for this function is the user's device must be already connected to the internet when logging in and and user must provide their phone number when signing up.

b) BPMN

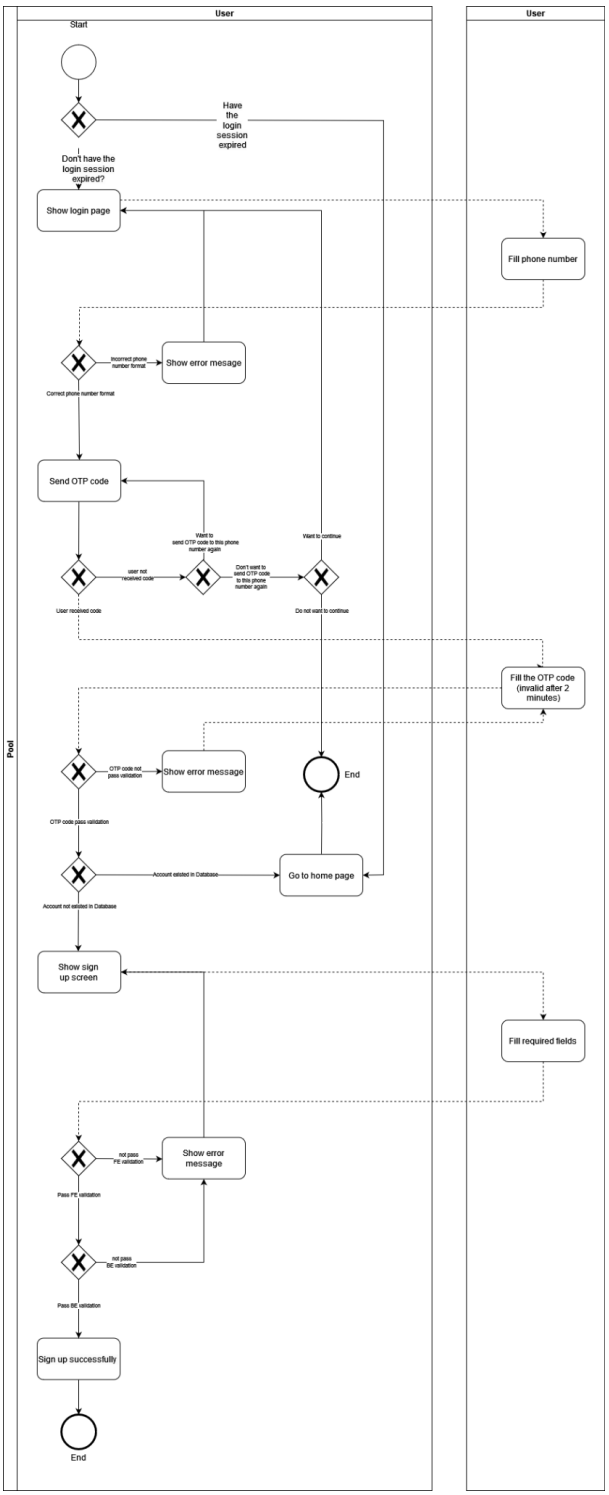


Figure 3. 7. BPMN of the Log In/Sign Up process (Source: Author)

The Log In/Sign Up process in a mobile application involving several steps:

Login: The user is shown the login page if the session is expired. They enter their phone number, which is validated for correct format. If valid, an OTP code is sent.

OTP Verification: The user inputs the OTP code within a specified time. If the OTP is correct and the account exists, the user is directed to the home page. Otherwise, an error message is displayed.

Sign Up: If no account exists, the user is shown the sign-up screen to fill in required details. After passing all validations, the user is successfully signed up.

c) DFD

- Users input personal information (name, phone, OTP).

- The system verifies the information and responds with success or failure.

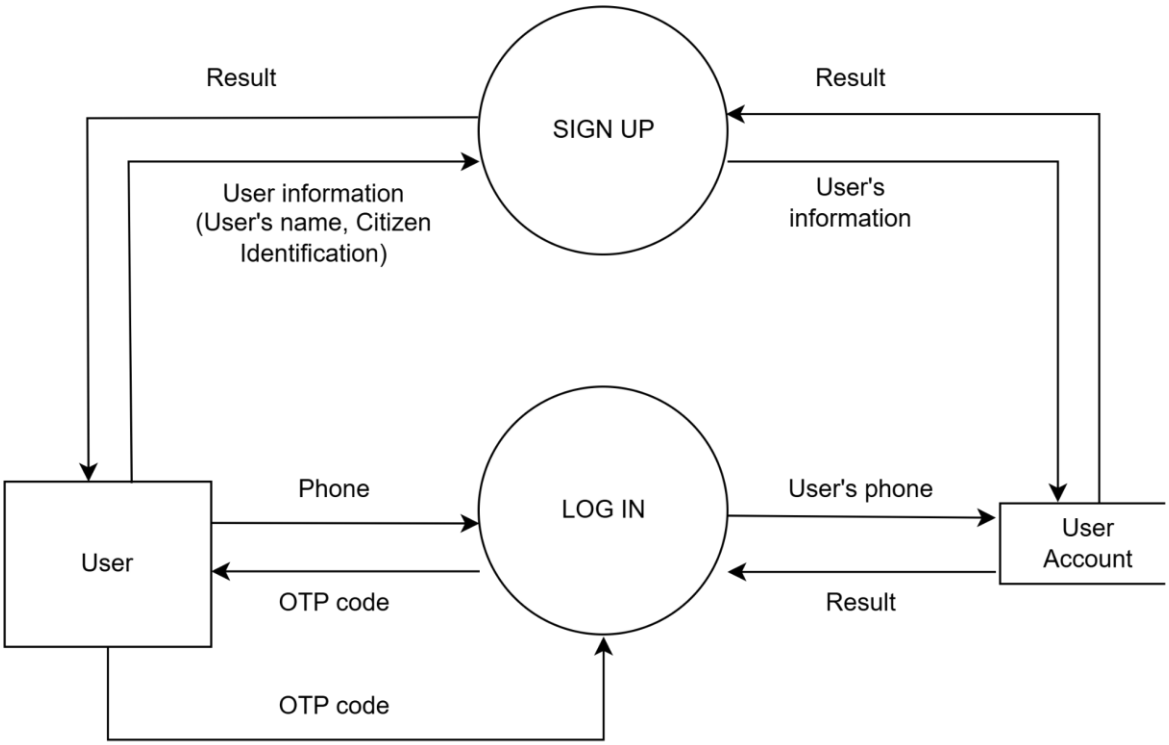


Figure 3. 8. Log In/Sign Up process DFD level 1 (Source: Author)

The Level 1 Data Flow Diagram (DFD) shows the main components of the **User Authentication Process**:

User: The entity that interacts with the system. The user can perform two primary actions based on their authentication needs.

Sign Up: This process allows a new user to create an account. It involves the collection of necessary registration details to establish a new user profile in the system.

Log In: This process is for returning users who want to access their existing account. It typically requires verification through credentials like a username and password.

d) Use case

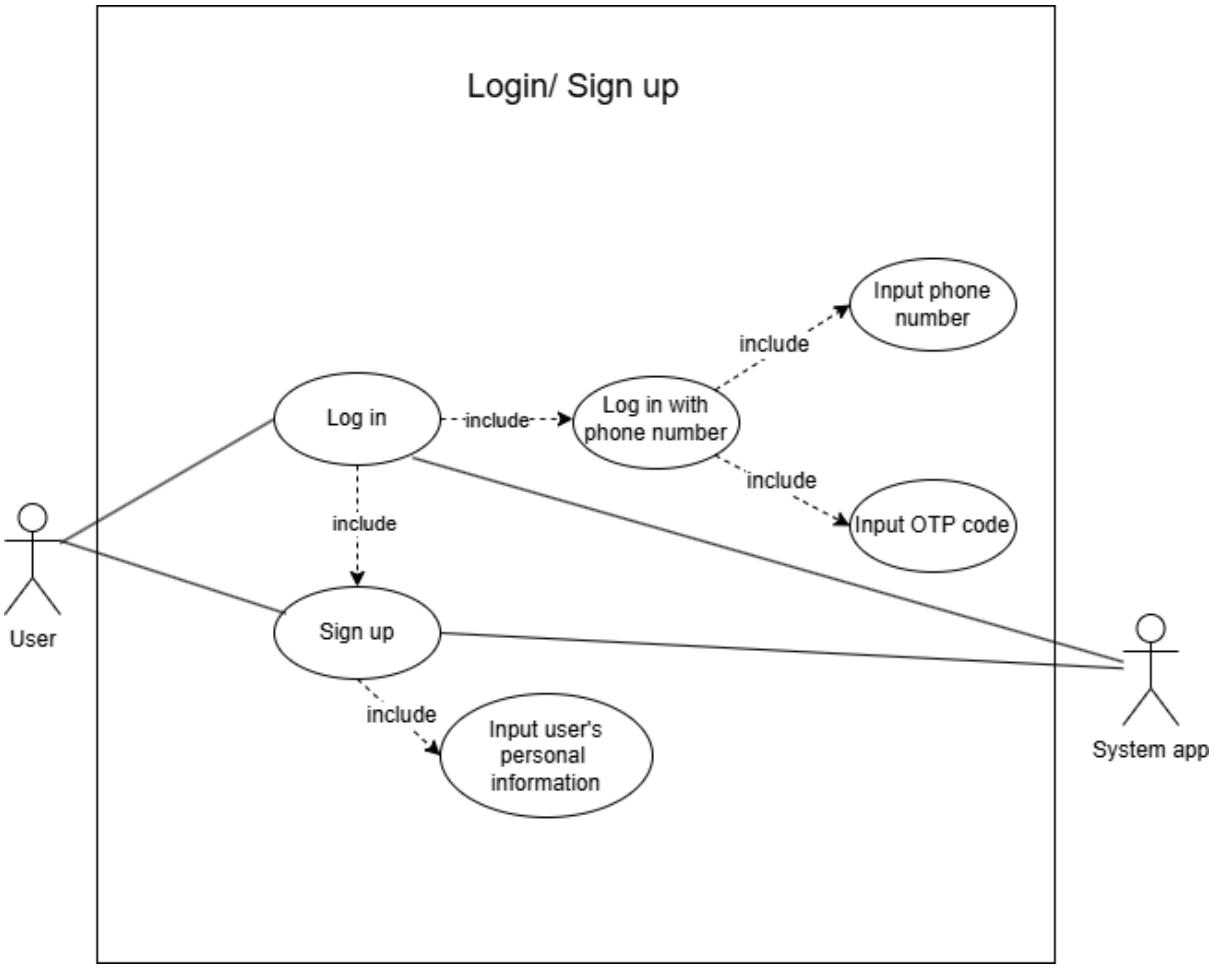


Figure 3. 9. Use case diagram of Log in/ Sign up process (Source: Author)

The use case diagram of **Log In/Sign Up** process with two main actors: the **User** and the **System App**.

Log In: For existing users, the system enables secure access through a phone-based authentication process. This includes:

Input Phone Number

Input OTP Code

Sign Up: For new users, the system offers a straightforward registration process where they provide essential personal information to create a new account.

Table 3. 1. Use case description of Log in process (Source: Author)

Use Case Name	Log in process	
Use Case No.	01	
Primary Actor	User	
Secondary Actor	Hosiptal system	
Use Case Story	This use case enables the user to login and access the accounts in mobile application	
Trigger	User access to the application	
Pre-Condition	User accounts already created and must be active User account has been authorized The user's device is already connected to the internet when logging in	
Post-Condition	User successfully logged into the application The system records successful logging in Activity Log	
Primary Flow (PF)	**Title:** User is able to login without any errors	
	Actor Action	**System Response**
	1. User access to the application	
	2. User fills in their phone number	3. System validates phone number. If valid and don’t exist in database,

		system sends OTP Code to user's phone number. If valid and don't exist in database, go to Sign up process. If invalid, go to Alternate Flow 1 (AF1).
	4. User fills in OTP Code	5. System validates OTP Code. If valid, go to next step If invalid, go to AF2.
		6. The system authenticates the login information successfully and allows the user to access the application
		7. The system records successful logging in Activity Log
Alternate Flow 1 (AF1)	**Title:** Invalid phone number	
	Actor Action	**System Response**
		1. System displays the following inline error message: Số điện thoại không hợp lệ. Vui lòng kiểm tra và nhập lại số điện thoại khác.
	2. Go to Step 1 of PF	
Alternate Flow 2 (AF2)	**Title:** Invalid OTP Code	
	Actor Action	**System Response**
		1. System displays the following inline error message: Mã OTP không hợp lệ.

		Vui lòng kiểm tra và nhập lại số mã OTP hoặc yêu cầu mã OTP khác.
	2. User refills in OTP Code	3. System validates OTP Code. If valid, go to next step. If invalid, go to step 6 PF.
	4. User request a new OTP Code	5. System sends another OTP Code to user’s phone number.
	6. Go to step 2 of AF2.	7. Go to step 3 of AF2

Table 3. 2. Use case description of Sign up process (Source: Author)

Use Case Name	Sign up
Use Case No.	02
Primary Actor	Guest
Secondary Actor	Hosiptal system
Use Case Story	This use case enables the user to sign up to the application and have an user account.
Trigger	Guest access to the application
Pre-Condition	The guest's device is already connected to the internet when signing up
Post-Condition	Guest successfully created an account in the application. The system records successful lsigning up in Activity Log.
Primary Flow	**Title:** Guest is able to sign up without any errors

(PF)	**Actor Action**	**System Response**
	1. Guest access to the application	
	2. Guest fills in their phone number	3. System validates phone number. If valid system sends OTP Code to the guest's phone number. If invalid, go to Alternate Flow 1 (AF1).
	4. Guest fills in OTP Code	5. System validates OTP Code. If valid, go to next step. If invalid, go to AF2.
		6. System authenticates the information successfully and requires sign up information.
	7. Guests fills in their sign up information and selects sign up command.	8. System authenticates the sign up information. If successful, creates an account for guest. If unsuccessful, go to Alternate Flow 3 (AF3).
	9. User can log in to use the application.	10. Go to Log in process.
Alternate Flow 1 (AF1)	**Title:** Invalid phone number	
	Actor Action	**System Response**
		1. System displays the following

		inline error message: Số điện thoại không hợp lệ. Vui lòng kiểm tra và nhập lại số điện thoại khác.
	2. Go to Step 1 of PF	
Alternate Flow 2 (AF2)	**Title:** Invalid OTP Code	
	Actor Action	**System Response**
		1. System displays the following inline error message: Mã OTP không hợp lệ. Vui lòng kiểm tra và nhập lại số mã OTP hoặc yêu cầu mã OTP khác.
	2. Guest refills in OTP Code	3. System validates OTP Code. If valid, go to next step. If invalid, go to step 6 PF.
	4. Guest request a new OTP Code	5. System sends another OTP Code to user's phone number.
	6. Go to step 2 of AF2.	7. Go to step 3 of AF2
Alternate Flow 3 (AF3)	**Title:** Invalid Personal information	
	Actor Action	**System Response**
		1. System displays the following inline error message: Thông tin không hợp lệ. Vui lòng kiểm tra và nhập lại.
	2. Go to step 7 of PF	

e) User story

Table 3. 3. User Story of Log in/ Sign up

Log in/ Sign up	
As a	New user or old user of the Mobile Appointment Scheduling System
I want	to be able to log in or sign up to the application just using my phone number
So that	I can create a new account or access my existing one. By simply entering my phone number, I'm automatically navigated to either the login or sign-up process, depending on whether I already have an account.
Acceptance Criteria	1. The system must display an input field for user to enter phone number on the login screen. 2. When user clicks the "Send OTP Code" button after entering a phone number, the system must validate that the input matches a predefined phone number format (10 digits). Old user flow: 3. If the phone number does not match the required format, the system must display an inline error message: "Số điện thoại không hợp lệ. Vui lòng kiểm tra và nhập lại số điện thoại khác." 4. If the phone number is valid and exist in database, the system must initiate the OTP (One-Time Password) generation and delivery process.

	5. Upon successful validation of the phone number, the system must generate a unique OTP code and send it to the user’s phone number. 6. The OTP code should be delivered to the user within 10 seconds of the request. 7. The system must present an input field for the user to enter the received OTP code on the OTP verification screen. 8. The OTP input must accept only numeric characters and verify that it matches the code sent to the user. 9. If the user enters an incorrect OTP, the system must display an inline message: "Mã OTP không hợp lệ. Vui lòng kiểm tra và nhập lại số mã OTP hoặc yêu cầu mã OTP khác." 10. If the correct OTP is entered, the system must allow the user to proceed to the main dashboard. New user flow: 11. If the entered phone number is valid and don’t exist in database, the system must display an message: “Số điện thoại chưa đăng ký tài khoản, vui lòng đăng ký tài khoản mới.” then, the user must be directed to a sign-up form. 12. The sign-up form must contain fields for required personal information. 13. The system must validate that all required fields in the sign-up form are completed. 14. Upon submission of valid sign-up information, the system must create a new user profile and store it in the database. Then must display message : "Đăng ký tài khoản thành công.”

	15. Then user be navigated to Log in process.

f) Sequence Diagram

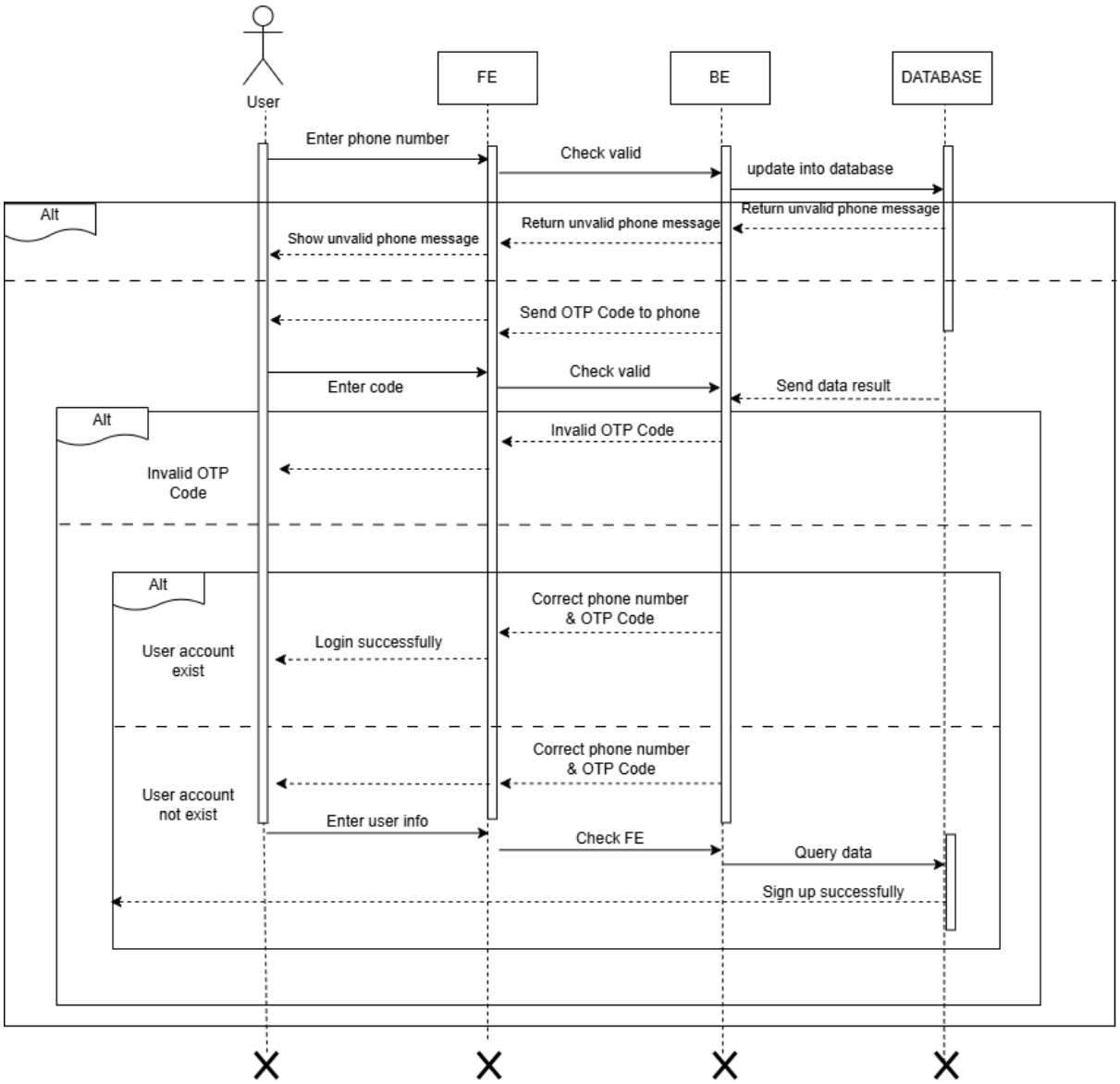


Figure 3. 10. Sequence diagram of Sign up/ Login (Source: Author)

The sequence diagram of **the Sign Up/Login** process, highlighting interactions between the **User**, **Front End (FE)**, **Back End (BE)**, and **Database**.

1. **Enter Phone Number**: The user inputs their phone number, which the FE checks for validity via the BE.

2. **OTP Code Verification**: If the phone number is valid, an OTP code is sent to the user's phone. The user enters the OTP, which the BE verifies.
    - If the OTP is invalid, an error message is returned.
3. **Account Check**: Upon successful OTP verification, the system checks if the user account exists.
    - If the account exists, the user is logged in.
    - If the account does not exist, the user provides additional information, and a new account is created and saved in the database.

g) Activity Diagram

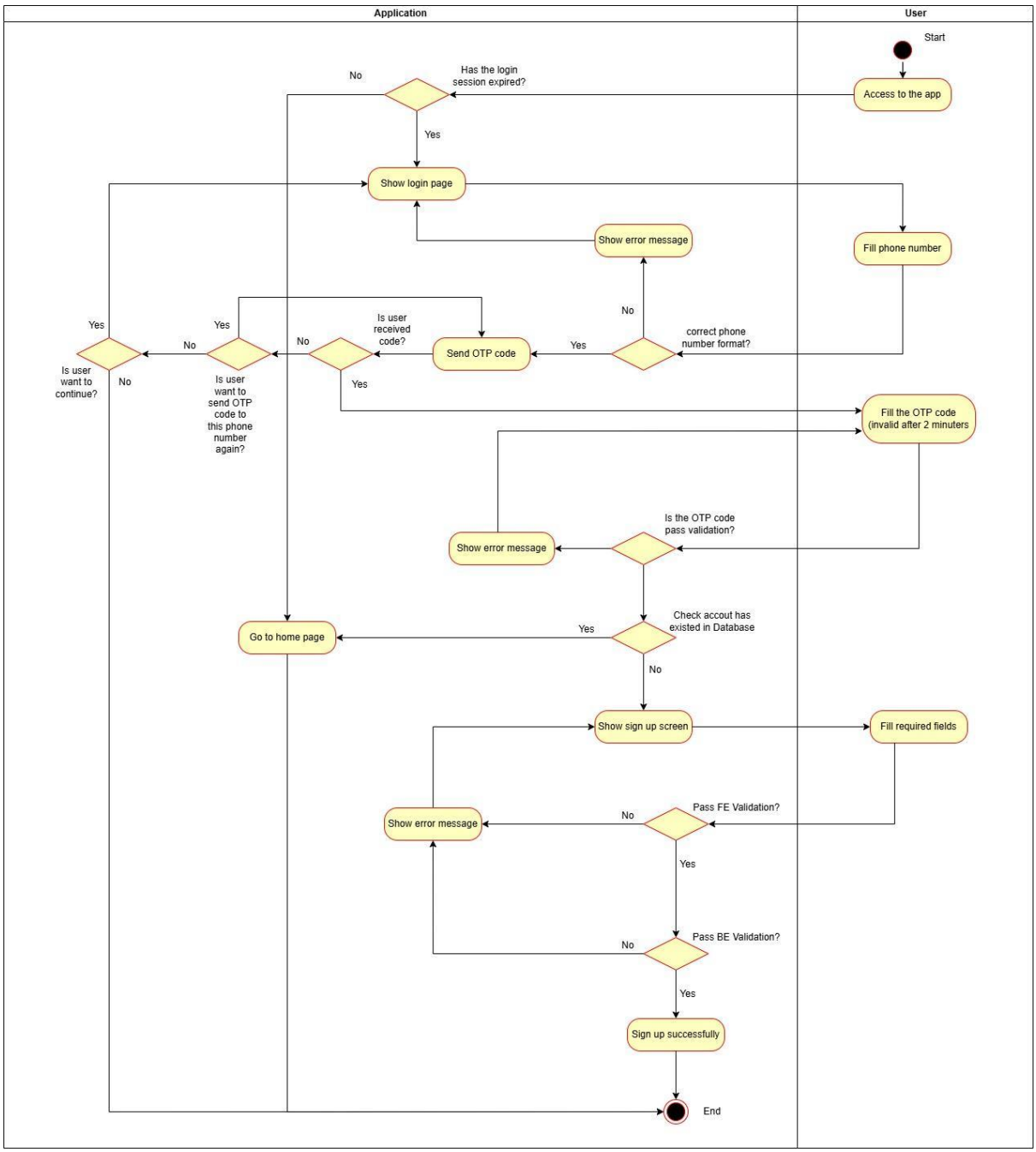


Figure 3. 11. Activity Diagram of Sign up/Log in (Source: Author)

The activity diagram of the **Sign Up/Login** workflow in an application:

1. **Session Validation**: The system first checks if the user's session is still active. If it has expired, the login page is displayed.
2. **Entering Phone Number**: The user inputs their phone number. If the format is incorrect, an error message is shown, prompting the user to correct it.

3. **Sending OTP**: Once a valid phone number is provided, the system sends an OTP code. The user must enter the OTP within 2 minutes. An invalid OTP triggers an error message.
4. **Account Existence Check**: The system checks if the account exists:
    - If it exists, the user is taken to the home page.
    - If it does not exist, the user proceeds to the sign-up screen to enter additional required information.
5. **Front-End and Back-End Validation**: The sign-up information goes through front-end and back-end validation checks. Successful validation at both levels completes the sign-up process.

h) Data Dictionary

Table 3. 4. Data Element Dictionary of function Sign up & Login (Source: Author)

Label	Alias	Type & Length	Default Value	Acceptable values	Source	Security	Responsible User(s)	Description/Comments
User name	Full Name	String (50 characters)	None	Valid names, no special characters	Input by user	Only user and Hospital Database Aministrator can view/edit	User	User's full name for personal information management
User Email	Email Address	Alphabetic (50 characters)	None	Valid email format	Input by user	Only user and Hospital Database Aministrator can view/edit	User	User's email address for communication

User Phone Number	Contact Number	Numeric (10 digits)	None	Valid 10-digit mobile phone numbers	Input by user	Only user and Hospital Database Aministrator can view/edit	User	Phone number for authentication and OTP code receipt
OTP Code	One-Time Password	Numeric (6 digits)	None	Valid 6-digit code	Generated by system	Only user can view, system-generated	Hospital Information System Hospital Database Aministrator istrators	One-time authentication code sent to user's phone
User ID	Account ID	Numeric (10 digits)	None	Valid value generated by the system	System	Only Hospital Database Aministrator can view/edit	System and user	Uniquely represents the user's account in the system
Login Result	Authentication Result	Boolean	FALSE	True if authentication is successful, False if not	System processed	Secure, only relevant to system and user	System automatic and user	Authentication result after user enters OTP code
Signup Result	Registration Status	Boolean	FALSE	True if registration is successful, False if not	System processed	Secure, only relevant to system and user	System automatic	The result of the user's account registration process

Table 3. 5. Data flows dictionary of function Sign up & Login (Source: Author)

Label	Description	Alternate Name(s)	Origin	Destination	Record	Volume and Frequency
User's Information Flow	User's information (name, citizen ID) sent for registration.	User Data Flow	User	Sign Up Process	User's Name, Citizen ID	Based on user registration attempts per day.
Phone Flow	User's phone number used for login verification.	Phone Number Flow	User	Log In Process	Phone	Each time a user attempts to log in.
OTP Code Flow	OTP code sent to user for verification purposes.	Verification Code Flow	Log In Process	User	OTP Code	Each time the user initiates a login request.
Result Flow	Represents the outcome of Sign Up or Log In.	Status Flow	Sign Up/Log In Process	User	Result	Each time the system processes a request.

Table 3. 6. Data stores dictionary of function Sign up & Login (Source: Author)

Label	Description	Alternate Name(s)	Attributes	Volume and Frequency
User Account	Stores all personal information of the user, including name, date of birth, address, contact information.	User Profile	User ID, Name, Date of Birth, Address, Contact Info	50,000 - 500,000 records, updated whenever the user updates personal information or payment information.

Table 3. 7. Data process dictionary of function Sign up & Login (Source: Author)

Process Name	Description	Process Number	Input Data Flows	Output Data Flows	Process Description
Sign Up	Registers new users by creating an account based on provided user information.	1.1	User Information	Result	Receives user's name and citizen ID, validates input, and stores information in the User Account data store.
Log In	Authenticates user credentials using phone number and OTP code.	1.2	Phone, OTP Code	Result	Receives phone and OTP code, verifies OTP, checks user existence, and provides access if valid.

Table 3. 8. Data entity dictionary of function Sign up & Login (Source: Author)

Entity Name	Description	Input Data Flows	Output Data Flows
User	Represents the patient or user interacting with the system. They can sign up for an account, log in, schedule appointments, and make payments.	- Personal information for Sign Up (name, phone number, email, citizen ID). - Phone number and OTP for Login. - Appointment information (patient profile, symptoms, time, doctor). - Payment method.	- Sign Up result (successful or failed). - Login result (successful or failed). - Invoice. - Appointment result (QR code, appointment information). - Medical record information. - Payment status.

i) User Interface

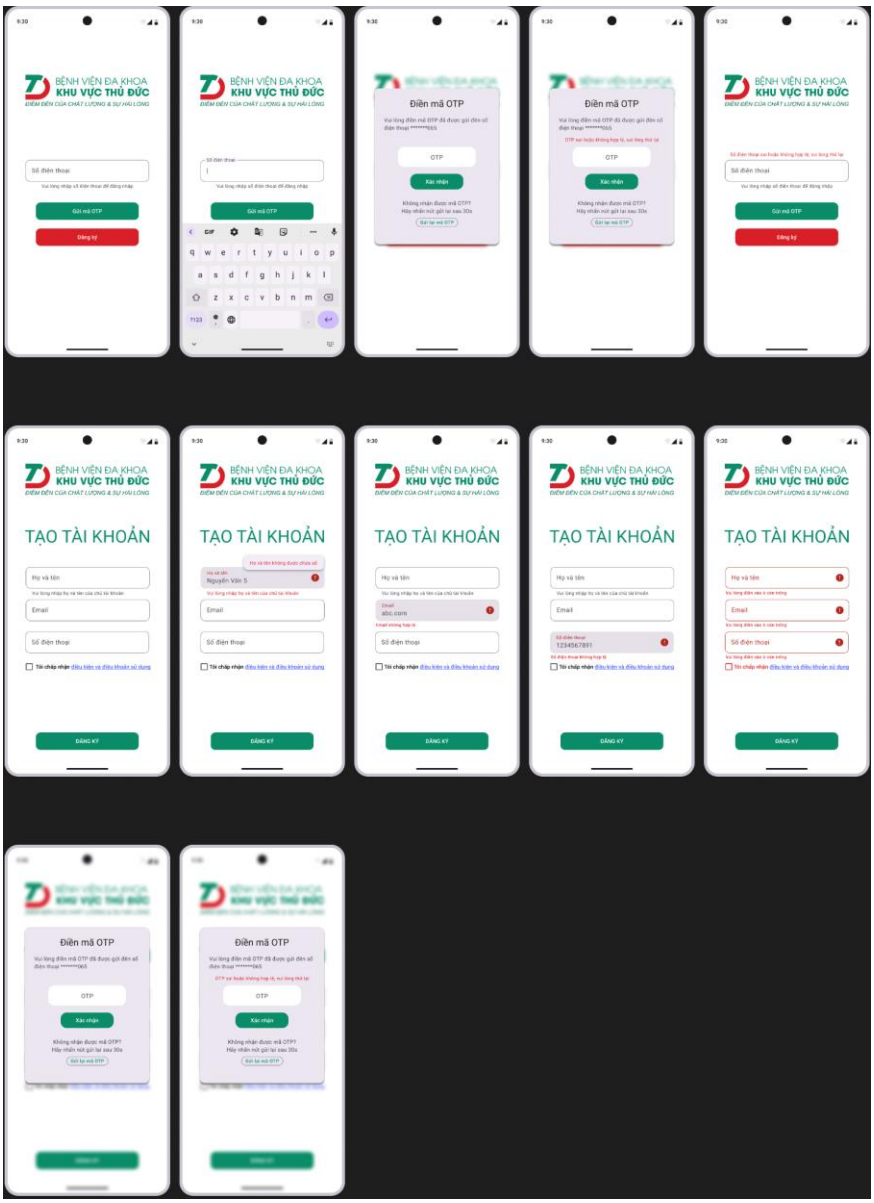


Figure 3. 12. User interface of the login and sign up (Source: Author)

This figure shows a sign-in and sign-up flow for a healthcare-related application. Here's a breakdown of the key screens and functions:

Sign-In Screen:

- The user is prompted to enter their phone number.
- There's an option to receive an OTP (One-Time Password) to proceed with the sign-in.

- An error message is displayed if the phone number is invalid or missing.

OTP Verification Screen:

- After entering the phone number, the user is directed to an OTP entry screen.
- The OTP screen includes an option to resend the OTP after a timer expires.
- Users can validate the OTP to complete the login process.
- Error messages are shown for incorrect or expired OTP entries.

Sign-Up Screen:

- Users can create a new account by entering their full name, email, and phone number.
- There are options to agree to the terms of service and privacy policy.
- The screen displays various validation messages to guide the user:
  - Ensuring the name and email fields are correctly filled.
  - Ensuring the phone number follows a valid format.
- Errors are highlighted in red for fields with incorrect or missing information.

Each screen is designed with a consistent, user-friendly layout, focusing on validation and error handling to ensure users input their details correctly. The interface uses clear text fields and validation feedback to guide users through the process.

j) Validation

Table 3. 9. Data validation of Login/Sign up function

Failed Validation	Error Message Displayed

All required data is entered	<Data Field Name> is a required field.
Username is in a valid format	Name is in a invalid format. Please enter a valid username.
Email address is in a valid format	Please enter a valid email address.
Email address is unique	An account with this email already exists.
Terms and conditions are accepted	You must accept the terms and conditions to continue.
User phone number is in a valid format	Phone number is in a invalid format. Please enter a valid phone number.
OTP code is valid	OTP code is wrong or invalid. Please try again.

3.4.2. Manage User Account

a) Function introduce

Function meaning

This feature allows users to manage personal information, patient profiles, and log out. Users can update their details and add payment cards as needed. To enhance versatility, the application supports a one-to-many relationship between a single account and multiple patient profiles. This means users can add multiple patient profiles to track medical records and book appointments on their behalf. Additionally, the application enables users to update patient profiles whenever changes occur.

Business rules

The business rule for this function is one account can have no more than one patient profile.

b) BPMN

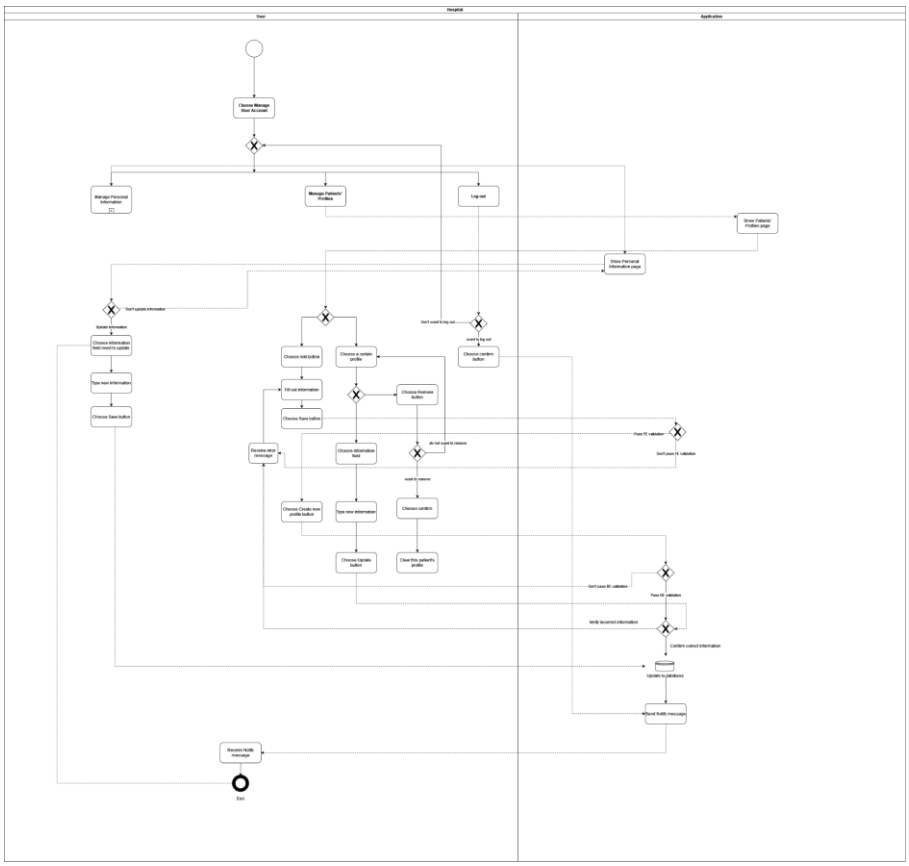


Figure 3. 13. BPMN of Manage User Account process (Source: Author)

The BPMN diagram of the **Manage User Account** process involving interactions between the **User**, **System**, and **Database**.

1. **User Account Access**: The user begins by choosing to manage their account, which may involve logging in if they are not already authenticated.
2. **Account Options**: After accessing the account, the user can select from various management options, such as updating personal information, changing account settings, or managing security details.
3. **Decision Points**: Each action leads to specific decision points or validations (e.g., verifying identity for sensitive changes).
4. **Update and Confirmation**: Upon completing a selected action, the system updates the database as needed and confirms the changes to the user.

c) DFD

- Users can view, update, or remove personal information and profiles.

- The system verifies the updated information, stores it in the database, and confirms the changes.

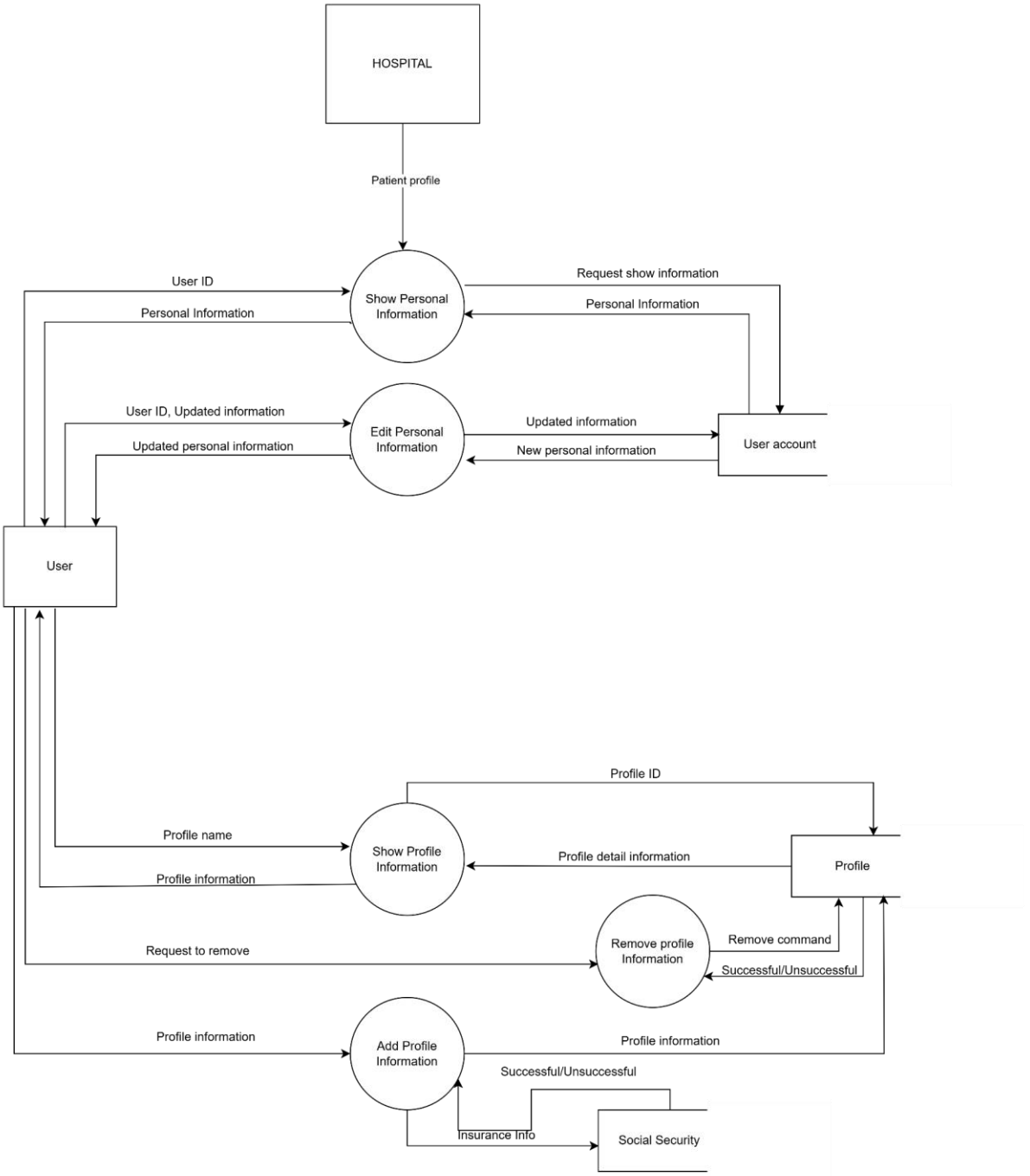


Figure 3. 14. User Account process DFD level 1 (Source: Author)

The Level 1 Data Flow Diagram (DFD) for the "Manage User Account" process shows the key interactions involved in managing a user's profile within a hospital system:

1. **User**: The user initiates actions to view, add, edit, or remove personal information within their account.
2. **Show Profile Information**: This process displays the user's profile data, retrieved from the User Account data store and shared with the user.
3. **Add Profile Information**: This process allows the user to add new profile information, which is then stored in the Profile data store and synchronized with Social Security records.
4. **Edit Personal Information**: This process enables the user to update existing personal information, with updates stored in the Profile data store and shared with the hospital to ensure up-to-date records.
5. **Remove Profile Information**: This process allows the user to delete specific profile data, which is removed from the Profile data store and reflected across associated records.
6. **Hospital**: The hospital has access to patient profiles for viewing or updating patient-related records, ensuring consistency in patient information.

d) Use case

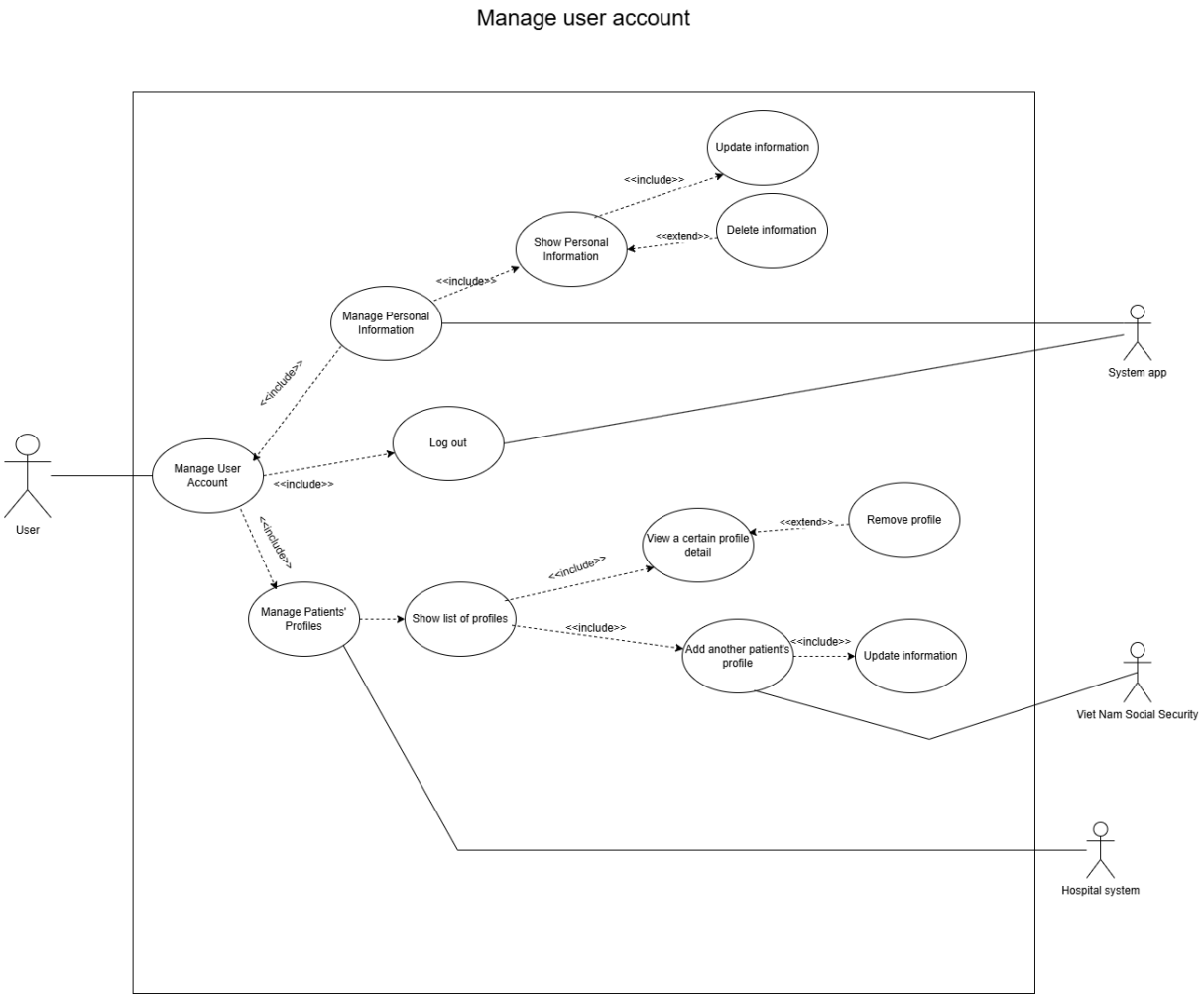


Figure 3. 15. Use case diagram of Manage user account (Source: Author)

The use case diagram for **Manage User Account** shows the main actions a **User** can perform within their account:

1. **Manage Personal Information**: The user can view, update, or delete personal information.
2. **Manage Patients' Profiles**: The user can view a list of profiles, add new profiles, view specific profile details, and remove profiles.
3. **Log Out**: The user can log out from their account.
4. **External Interactions**:
    - **System App**: Supports the user in managing account-related actions.
    - **Viet Nam Social Security** and **Hospital System**: These external systems provide integration for profile updates and validation.

Table 3. 10. Use case description of Manage user account process (Source: Author)

Use Case Name	Manage User Account	
Use Case No.	03	
Primary Actor	User	
Secondary Actor	Hosiptal system, Vietnam Social Security	
Use Case Story	This use case enables the user to manage their user account	
Trigger	User access User Account management	
Pre-Condition	User Account exists and must be active. User has logged in the account. User's device is already connected to the internet when logging in.	
Post-Condition	Information is updated User logs out of account successfully (if user logs out). The system records successful logout in Activity Log (if user logs out). User can create new patient profiles.	
Primary Flow (PF)	**Title:** User is able to manage user account without any errors	
	Actor Action	**System Response**
	1. User access User Account management	2. System shows list of functions for patient to choose.
	3. User chooses Manage	4. Show Personal

	personal information. If, go to step 6. If user choose Log out, go to step 16	information and allow user to update or delete information.
	5. User completes updating or deleting information.	6. System display the following message: “Chỉnh sửa thành công”/“Xóa tài khoản thành công.” Then system updates new information or delete old information in database.
		7. System update these activity into Activity Log
Alternate Flow 1 (AF1)	**Title:** User choose Manage Patient’s Profile	
	Actor Action	**System Response**
	1. User chooses Manage Patient’s Profile	2. System shows list of profiles for patient to choose.
	3. User chooses a function. If user want to remove the profile or edit information in it, go to step 4. If user want to add a patient’s profile, go to step 8.	
	4. User chooses a profile.	5. System shows Patient’s information.

	6. User removes the profile or edits information in it.	7. System updates new information or delete the profile in database.
	8. User choose to add a patient's profile	9. System requires user to fill in patient's information form.
	10. User fills in the form.	11. System validates user's information and confirm Insurance number with Vietnam Social Security. If the patient's information is valid, display the following message: Tạo hồ sơ mới thành công and creates new profile in database. If the patient's information is invalid, go to AF3.
Alternate Flow 2 (AF2)	**Title:** User chooses Log out	
	Actor Action	**System Response**
	1. User chooses Log out	2. System asks user to confirm log out.
	3. If user confirms Yes, then go to step 4. If user confirms No, then go to step 1 of PF	4. System logs out the current user account and return to homepage.

Alternate Flow 3 (AF3)	**Title:** Type invalid information	
	Actor Action	**System Response**
		1. System displays the following inline error message: Thông tin không hợp lệ. Vui lòng kiểm tra và nhập lại.
	2. Go to step 10 of AF1.	

e) User story

Table 3. 11. User Story of Manage User Account (Source: Author)

Manage User Account	
As a	user of the Mobile Appointment Scheduling System
I want	to be able to manage my account details, including my personal information and patient profiles.
So that	I can keep my information accurate and up to date within the application.
Acceptance Criteria	1. The system must provide an "Account Settings" option on the main dashboard for users who are logged in. 2. When the user selects "Account Settings," the system must display the Account settings screen, showing following options: a. Manage Personal Information b. Manage Patients' Profiles c. Log out Manage Personal Information flow: 3. The system must allow users to edit their personal information.

	4. When the user makes changes to their profile information and clicks "Save," the system must validate that all required fields are filled and that inputs are in the correct format. 5. If validation is successful, the system must update the user's profile information in the database and display a confirmation message: "Thông tin đã được cập nhật thành công." If validation fails, the system must display an error message: " Thông tin không hợp lệ. Vui lòng kiểm tra và nhập lại." indicating the field(s) that need correction. Manage patient profiles flow: 6. When user selects function "Manage Patient Profile", system must show a list of patient profile. 7. When a user chooses a profile, the system will enable user to new remove a profile or update information in it, system must validate that all required fields are filled and that inputs are in the correct format. 8. If validation is successful, the system must update the user's profile information in the database and display a confirmation message: "Thông tin đã được cập nhật thành công"/ "Tài khoản đã xóa thành công". If validation fails, the system must display an error message: " Thông tin không hợp lệ. Vui lòng kiểm tra và nhập lại." indicating the field(s) that need correction. Log out flow: 9. When the user selects "Log Out", then system logs out the user and returns to the homepage. And logs the log-out event in the Activity Log.

f) Sequence Diagram

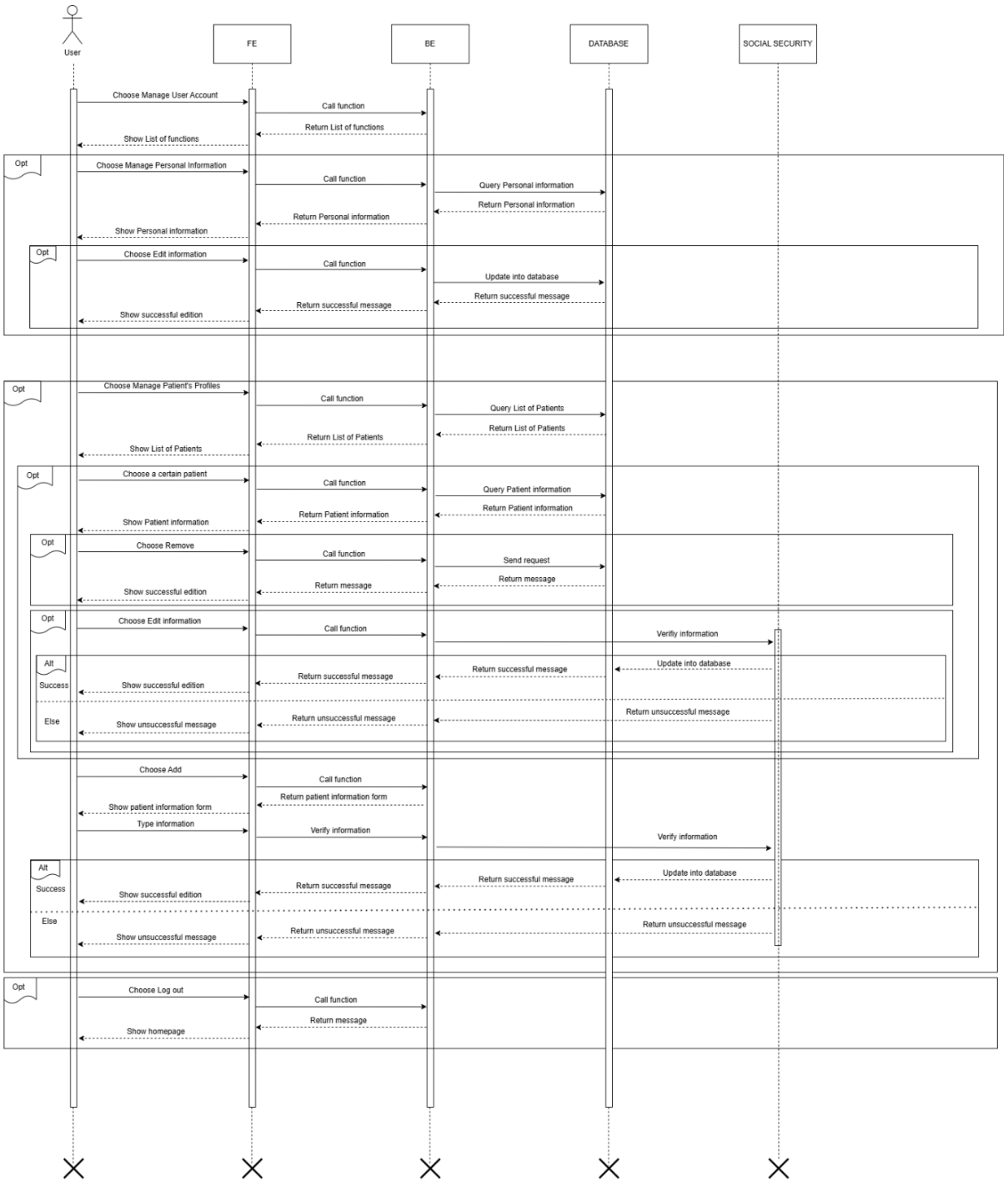


Figure 3. 16. Sequence Diagram of Manage User Account (Source: Author)

The sequence diagram for **Manage User Account** illustrates the interactions between the **User**, **Front End (FE)**, **Back End (BE)**, **Database**, and **Social Security** system in managing account and profile details:

1. **Manage Personal Information**: The user can view and edit their personal information. Edits are saved to the database, with confirmation messages returned upon success.
2. **Manage Patient Profiles**: The user can view a list of patient profiles, select specific profiles, and perform actions like viewing, editing, or removing profiles. Each action triggers data queries or updates in the database, with verification from Social Security as needed.
3. **Add Profile**: When adding a new profile, information is verified before being saved to the database. Success or error messages are returned based on the outcome.
4. **Log Out**: The user can log out, returning to the homepage.

g) Activity Diagram

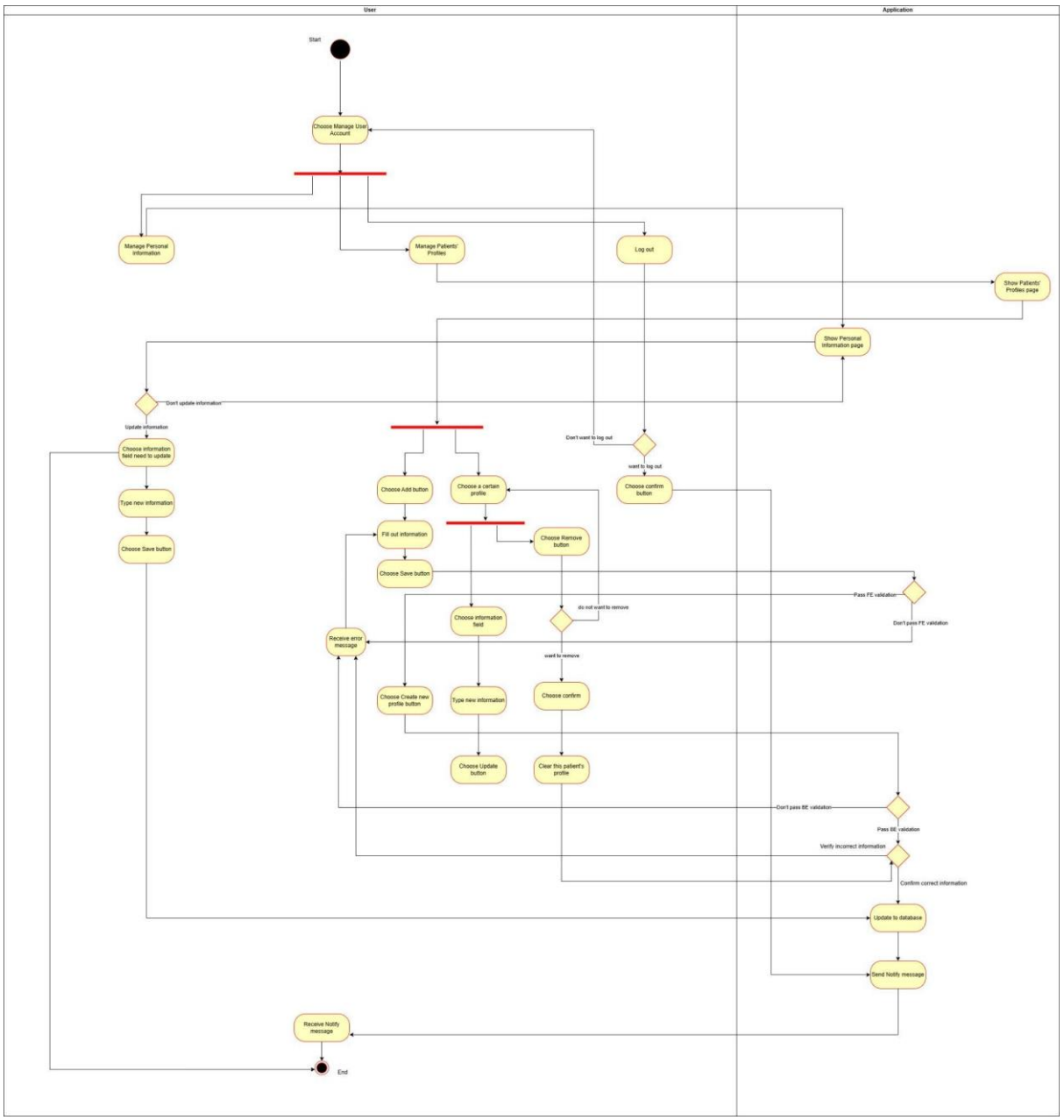


Figure 3. 17. Activity Diagram of Manage User Account (Source: Author)

The activity diagram for **Manage User Account** details the actions a **User** can take to manage their account and profile information within the **Application**:

Choose Manage User Account: The user initiates account management and selects one of the following options:

- **Manage Personal Information**: Allows the user to view, edit, and save personal information.

- **Manage Patient Profiles**: The user can add, view, edit, or remove patient profiles.
- **Log Out**: The user can choose to log out of the system.

Manage Personal Information: In this flow, the user can update personal details, which are saved after front-end and back-end validation.

Manage Patient Profiles: The user can add or edit patient details, remove profiles, and confirm updates. Each action involves data validation and updates to the database.

Process Completion: Once updates are successfully saved, the system confirms the changes with a success message.

h) Data Dictionary

Table 3. 12. Data elements dictionary of function Manage User Account (Source: Author)

Label	**Alias**	**Type & Length**	**Default Value**	**Acceptable values**	**Source**	**Security**	**Responsible User(s)**	**Description /Comments**
Patient ID		Numeric (10 digits)	None	Valid value generated by the system	System	Only Hospital Database Aministrator can view/edit	System and user	Represents the patient profile in the system
Patient Name	Patient Name	String (50 characters)	None	Valid names, no special characters	Input by user	Only user and Hospital Database Aministrator can view/edit	User	Patient's full name for patient information management
Patient Citizen Identification	ID Number	Numeric (12 digits)	None	Numeric string of 9-12 characters	Input by user	Only user and Hospital Database Aministrator can view/edit	User	Citizen ID patient identity verification

Patient Insurance Number		Numeric (20 digits)	None	Valid insurance number	Input by user	Only user and Hospital Database Aministrator can view	Insurance Department	Unique identifier for patient's insurance policy; used for billing and claims verification.
Patient Phone Number	Contact Number	Numeric (10 digits)	None	Valid 10-digit mobile phone numbers	Input by user	Only user and Hospital Database Aministrator can view/edit	User	Phone number for comunicatio n of patient
Patient Email	Email Address	String (50 characters)	None	Valid email format	Input by user	Only user and Hospital Database Aministrator can view/edit	User	Patient's email address for communicat ion and vertify information
Patient Gender	Sex	String (6 characters)	None	Nam, Nữ, Khác	Input by user	Only User and Hospital Database Aministrator can edit	Hospital Information System Hospital Database Aministrato r istrators	Patient's gender
Patient Age		Numeric (3 digits)	None	Numeric string of 1-3 characters	Input by user	Only User and Hospital Database Aministrator can edit	Hospital Information System Hospital Database Aministrato r istrators	Patient's age

Patient Date of Birth	Birth Date	Date (YYYY-MM-DD)	None	Valid date	Input by user/Hospital Database Aministrator	Only user and Hospital Database Aministrator can edit	Hospital Information System Hospital Database Aministrator istrators	Patient's date of birth
Patient Address	Residential Address	Alphanumeric (255 characters)	None	Full address	Input by user	Only user and Hospital Database Aministrator can	User, Hospital Database Aministrator	Patient's residential address
Patient Nationality	String (50 characters)	None	List of Countries	User Input	Only user and Hospital Administrator	User, Hospital Database Administrator		Patient's country of origin or citizenship.
Patient Allergy	Allergy Information	String (255)	None	Known allergens (e.g., peanuts, shellfish, pollen)	User Input, Medical History	Sensitive Data (encrypted)	Doctor, Medical Staff	Information about any allergies the patient may have.
Patient Blood Type	Blood Group	String (3)	None	A, B, AB, O (with Rh factor),None	Medical Record	Sensitive Data (encrypted)	Doctor, Medical Staff	Patient's blood type, used for medical procedures and emergencies.

Table 3. 13. Data flows dictionary of function Manage User Account

Label	**Description**	**Alternate Name(s)**	**Origin**	**Destination**	**Record**	**Volume and Frequency**
Patient Profile	Patient's profile information	N/A	Hospital	Show Personal Information	Patient Profile Record	Generated each time the user requests

						personal information
Personal Information	User’s personal information	N/A	Show Personal Information	User	Personal Information Record	Retrieved each time the user views personal information
Request Show Information	Request to display user information	N/A	User Account	Show Personal Information	Information Request Record	Sent whenever there is a request to view personal information
Updated Information	Updated personal information	N/A	Edit Personal Information	User Account	Updated Personal Information Record	Sent when the user updates their information
New Personal Information	New personal information of the user	N/A	User Account	Edit Personal Information	New Personal Information Record	Updated when the user changes their information
Profile ID	Unique identifier for the user's profile	N/A	User Account	Profile	Profile Identification Record	Sent when accessing user profile information
Profile Detail Information	Detailed information of the user's profile	N/A	Profile	Show Profile Information	Profile Detail Record	Retrieved when displaying profile information
Profile Name	User’s profile name	N/A	User	Show Profile Information	Profile Name Record	Requested when viewing profile information
Profile Information	Information about the profile	N/A	Show Profile Information	User	Profile Information Record	Retrieved when displaying profile information
Request to Remove	Request to delete profile information	N/A	User	Remove Profile Information	Removal Request Record	Sent each time a request to delete a profile is made
Remove Command	Command to delete profile information	N/A	Profile	Remove Profile Information	Deletion Command Record	Executed when the delete command is issued

Success/Unsuccessful	Result of the profile deletion process	N/A	Remove Profile Information	User	Deletion Status Record	Returned after the profile deletion process is completed
Add Profile Information	Information to add a user profile	N/A	User	Add Profile Information	New Profile Record	Sent when the user adds profile information
Insurance Info	User's insurance information	N/A	Add Profile Information	Social Security	Insurance Information Record	Accessed when the user adds insurance to their profile
Social Security	Social security information	N/A	Social Security	Add Profile Information	Social Security Record	Sent when adding social security information
Successful/Unsuccessful	Result of the profile addition process	N/A	Add Profile Information	User	Addition Status Record	Returned after the profile addition process is comple

Table 3. 14. Data store dictionary of function Manage User Account (Source: Author)

Label	**Description**	**Alternate Name(s)**	**Attributes**	**Volume and Frequency**
Profile Information	Stores detailed user profile information, including insurance and social security details.	Insurance and Social Security Data	Profile ID, Profile Name, Insurance Info, Social Security	10,000 - 100,000 records, updated when the user adds or edits a profile.
Social Security	Stores user's social security information for insurance and related services.	SSN Data Store	Social Security Number (SSN), Insurance Coverage Info	10,000 - 50,000 records, updated infrequently, depending on changes in user social security information.

Table 3. 15. Data Process dictionary of function Manage User Account (Source: Author)

Process Name	Description	Process Number	Input Data Flows	Output Data Flows	Process Description
Show Personal Information	Displays the user's personal information.	2.1	User ID	Personal Information	Receives the user's ID, retrieves, and displays personal information from the User Account data store.
Edit Personal Information	Allows the user to edit their personal information.	2.2	User ID, Updated Information	Updated Information, New Personal Info	Receives updated information, validates it, and stores the new information in the data store.
Show Profile Information	Displays the user's profile information.	2.3	Profile Name	Profile Detail Information	Receives the profile name and displays detailed profile information from the Profile data store.
Remove Profile Information	Allows the user to remove profile information.	2.4	Remove Command	Success/Fail, Profile Information	Deletes the user’s profile information from the data store after verifying the request.
Add Profile Information	Allows the user to add new profile information.	2.5	Profile Information	Success/Fail, Profile Info, Insurance Info, Social Security	Receives new profile information, validates it, and stores it in the Profile data store (including

					insurance and social security info).

Table 3. 16. Data entity dictionary of function Manage User Account (Source: Author)

Entity Name	Description	Input Data Flows	Output Data Flows
Hospital	Manages and stores patient information, medical records, and appointment details	Patient profile, Request show information, Booking ID	Updated personal information, Profile details, Appointment confirmation status
User	Represents the patient or individual interacting with the hospital system for personal and medical purposes	User ID, Profile name, Insurance Info, Booking ID	Personal Information, Queue number, Updated profile information, Appointment request

i) User Interface

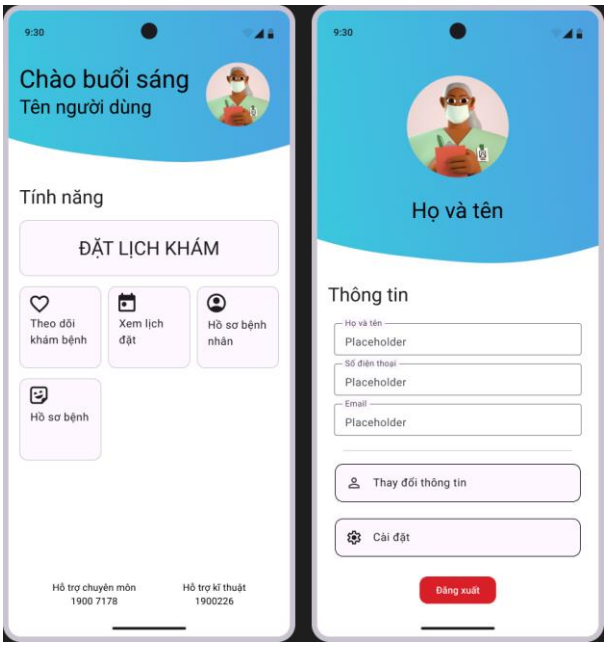


Figure 3. 18. User interface of the main screen and profile (Source: Author)

The figure shows two main screens of a healthcare-related application include:

Home Screen:

- Displays a personalized greeting ("Chào buổi sáng") with the user's name, creating a welcoming experience.
- Main features for user interaction:
  - **Book Appointment**: A primary button for scheduling medical appointments.
  - **Track Consultations**: Allows users to monitor their medical visits.
  - **View Appointments**: Provides an overview of booked appointments.
  - **Patient Profile**: Likely stores the user's health records and personal details.
  - **Medical Records**: Grants access to detailed medical history.
- At the bottom, there are contact options for **Specialist Support** and **Technical Support** with respective hotline numbers.

Profile Screen:

- Shows a user profile section with a circular avatar placeholder and the user's name.
- Fields to display and update personal information:
  - **Full Name**
  - **Phone Number**
  - **Email**
  - **Citizen Identification Number (CCCD)**
  - **Health Insurance Number (Số BHYT)**
  - **Gender** (options include Male, Female, Other)
  - **Date of Birth** (selectable via a calendar)
  - **Address**

        - **Nationality**
        - **Blood type**
        - **Allergy**
    - Includes two action buttons:
        - **Edit Information**: Likely opens an editable form to update profile details.
        - **Settings**: Presumably for app or account settings.
    - A **Log Out** button is located at the bottom for signing out.

This interface is designed with a clear, user-friendly layout, emphasizing ease of access to essential features and personalization.

The interface profile management system within a healthcare application, specifically focused on patient profiles. Here's a breakdown of each screen and its function:

Manage Profiles Screen:



Figure 3. 19. User interface of manage profile (Source: Author)

- Shows a list of existing patient profiles with name and gender displayed.
- A floating “+” button allows users to add a new profile.

Create Profile Screen:

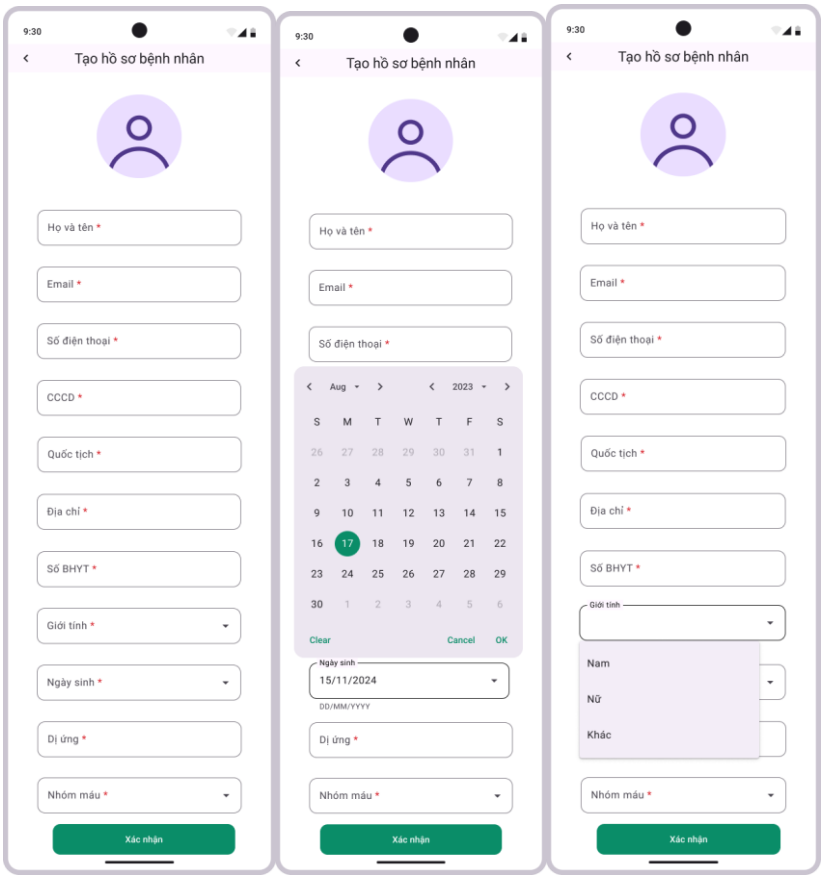


Figure 3. 20. User interface of creating profile information (Source: Author)

- Users can create a new patient profile by filling out the following fields:
  - **Full Name**
  - **Email**
  - **Phone Number**
  - **Citizen Identification Number (CCCD)**
  - **Health Insurance Number (Số BHYT)**
  - **Gender** (options include Male, Female, Other)
  - **Date of Birth** (selectable via a calendar)
  - **Address**
  - **Nationality**
  - **Blood type**
  - **Allergy**
- A “Submit” button is provided to save the profile.

Update Profile Screen:

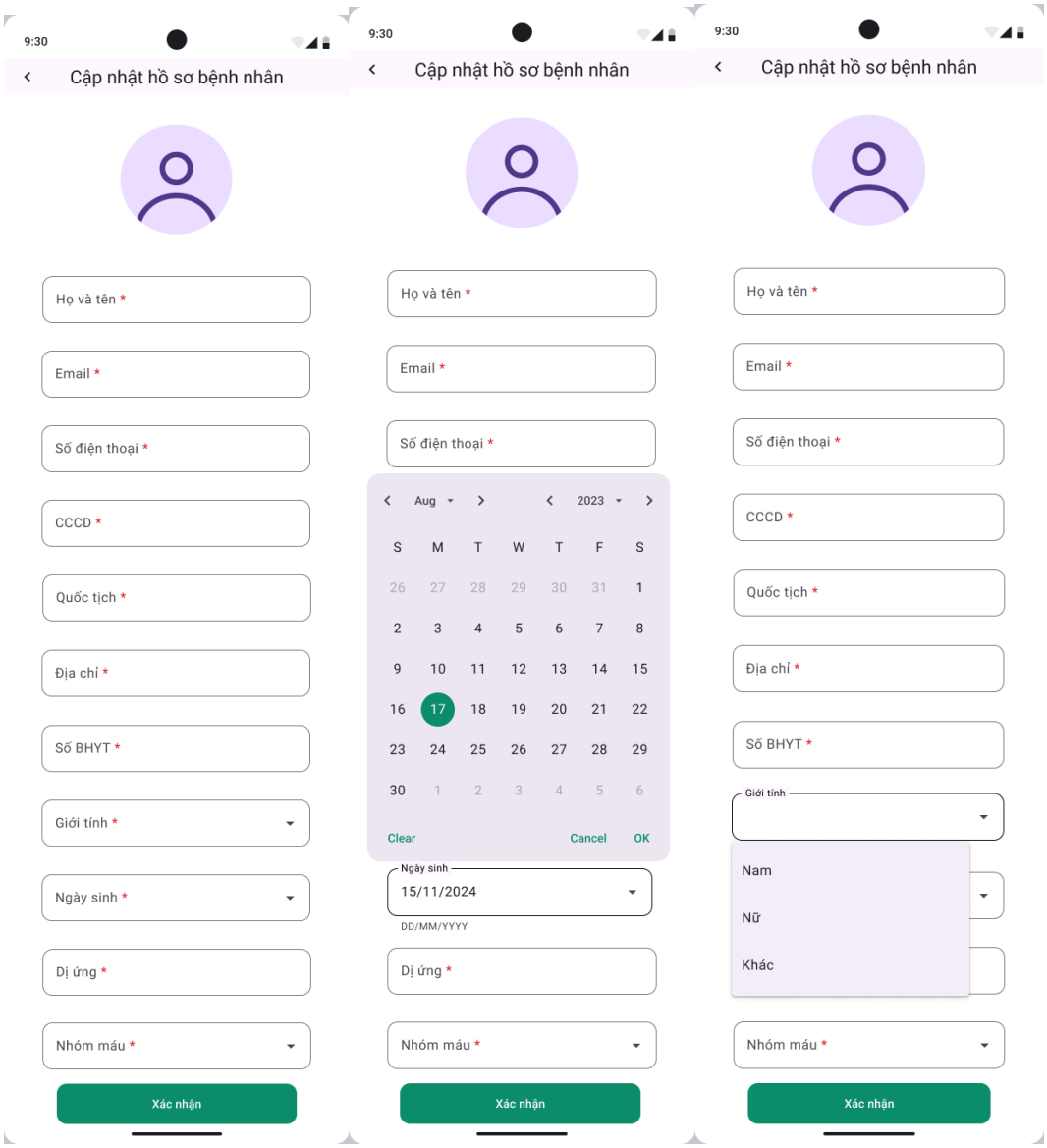


Figure 3. 21. User interface of Updating profile information (Source: Author)

- Similar to the Create Profile screen, but designed for editing an existing patient profile.
- Users can modify any of the fields and submit the changes.

View Profile Screen:

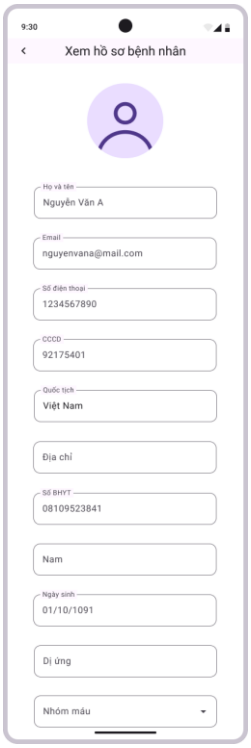


Figure 3. 22. User interface of viewing profile (Source: Author)

- Displays detailed information of a specific patient profile, including name, email, phone number, ID, health insurance number, gender, and date of birth, address, nationaltiy.
- The user can view the profile but cannot edit it from this screen.

Profile Management Actions:

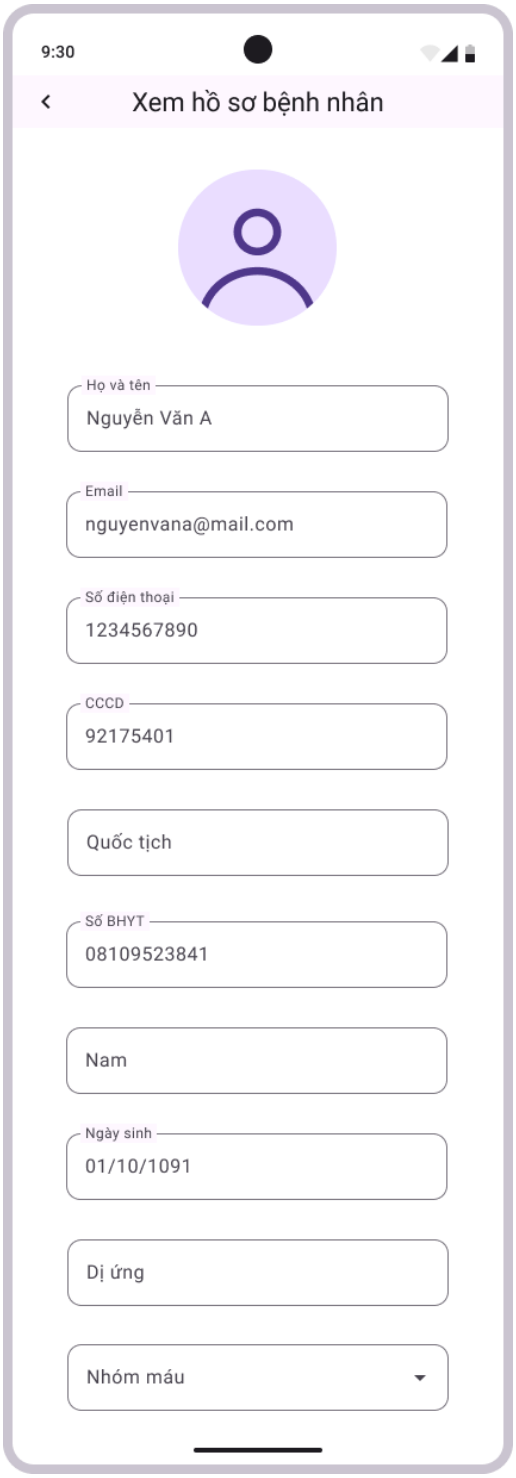


Figure 3. 23. User interface of options for profile (Source: Author)

- Options to **Edit Profile** and **Delete Profile** are available within the profile management list.
- Editing takes the user to the Update Profile screen, while deleting prompts a confirmation dialog.

Delete Screen:

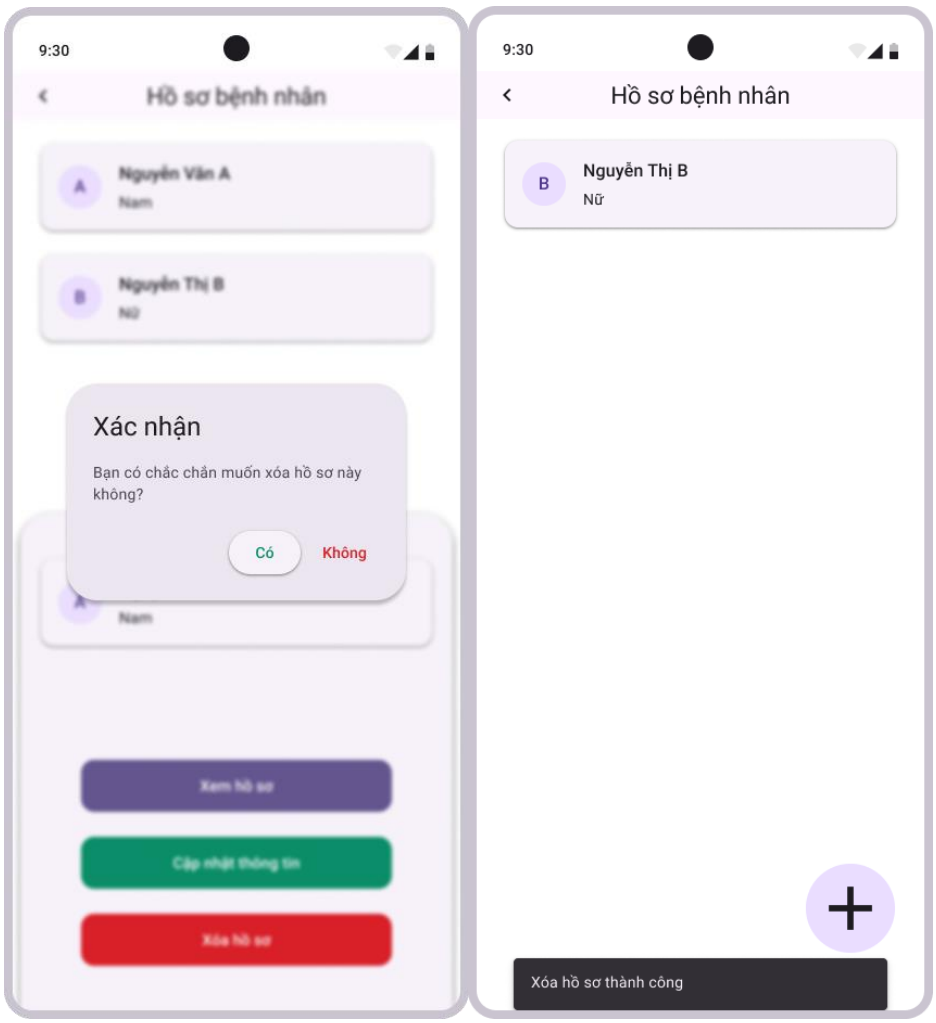


Figure 3. 24. User interface of confirming profile deletion (Source: Author)

- Asks the user to confirm whether they want to delete the selected profile.
- Options are provided to confirm (Yes) or cancel (No) the deletion.
- After successful deletion, the screen returns to the Manage Profiles list with the selected profile removed.

j) Validation

Table 3. 17. Data validation of Manage user account function (Source: Author)

Failed Validation	**Error Message Displayed**
All required data is entered	<Data Field Name> is a required field.

First and last names are in a valid format	Please enter valid names without special characters.
Phone number is in a valid format	Phone number is in a invalid format. Please enter a valid phone number.
Address fields are complete	Address is incomplete. Please provide all fields.
Citizen Identification Number (CCCD) is in a valid format	Citizen Identification Number is in a invalid format. Please enter a valid Citizen Identification Number
Health Insurance Code is in a valid format and match the Citizen Identification Number	Health Insurance Code is in a valid format or not found under the Citizen Identification Number. Please enter a valid Health Insurance Code

3.4.3 Appointment Booking

a) Function introduce

Function meaning

The booking function enables patients to schedule medical appointments with ease. Patients can search for available slots by doctor, specialty, or preferred time, allowing them to select the most convenient date and time. This feature helps patients save time and reduces hassle, while also enhancing hospital efficiency and improving patient satisfaction.

Business rules

The business rule for this function is the user cannot edit appointments that they have made, they must cancel the appointment and make a new appointment. If the user does not select a doctor or a time period, the application will choose a random doctor or time for the user.

b) BPMN

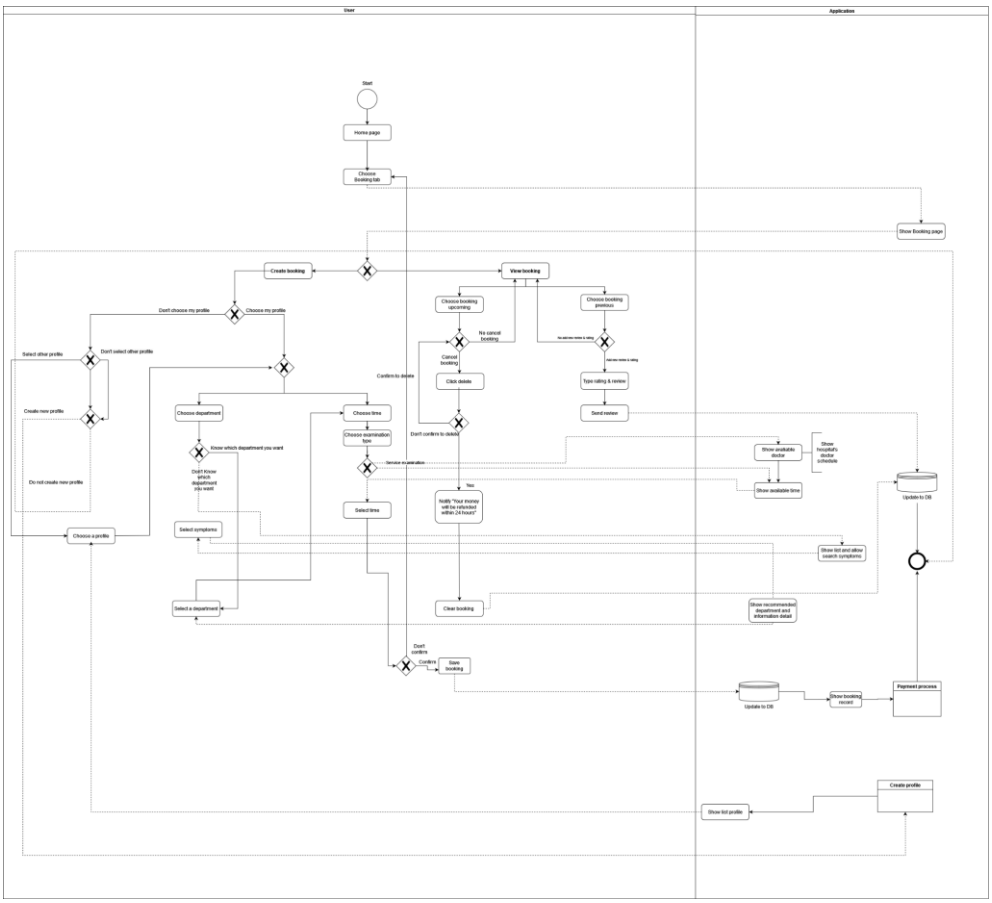


Figure 3. 25. BPMN of Appointment Booking process (Source: Author)

The BPMN diagram for the **Appointment Booking** process details the steps a **User** follows to create and manage bookings within the **Application**:

Create Booking: The user selects or creates a profile, chooses a department, and picks a date and time for the appointment. The system verifies the availability and confirms the booking.

View Booking: The user can view existing bookings, modify details, or cancel if necessary. Each action includes options to update or send notifications.

Update and Cancel: If changes are made, the system updates the database and sends any required notifications.

Payment Processing: If the booking requires payment, the user proceeds through a payment process, after which the system confirms the booking.

c) DFD

- Users choose symptoms, department, doctor, and appointment time.

- The system checks the doctor's schedule, confirms the booking, and updates the appointment status.

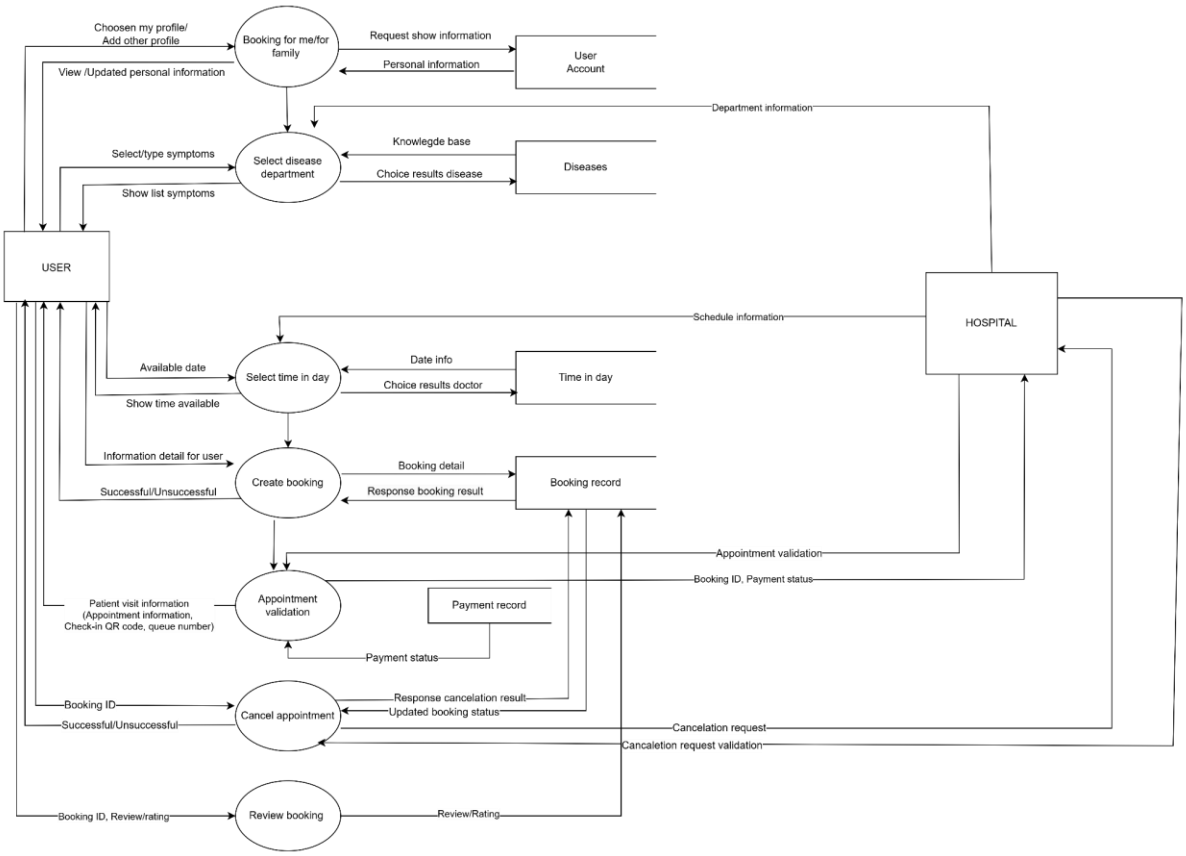


Figure 3. 26. Appointment Booking process DFD level 1 (Source: Author)

The Level 1 Data Flow Diagram (DFD) for the **Appointment Booking** process outlines the key actions a **User** takes and interactions with the system:

Booking for Self or Family: The user selects whether the appointment is for themselves or a family member.

Select Department and Time: The user chooses a medical department and selects a preferred time for the appointment.

Create Booking: The system processes the booking, retrieves schedule information, and provides the booking result.

Appointment Validation: The booking is validated, and the user receives details like check-in QR code and queue number.

Payment Record: If payment is required, the payment status is recorded and associated with the booking.

Cancel Appointment: The user can cancel the booking, and the system updates the status.

Review Booking: The user can rate and review the completed appointment.

d) Use case

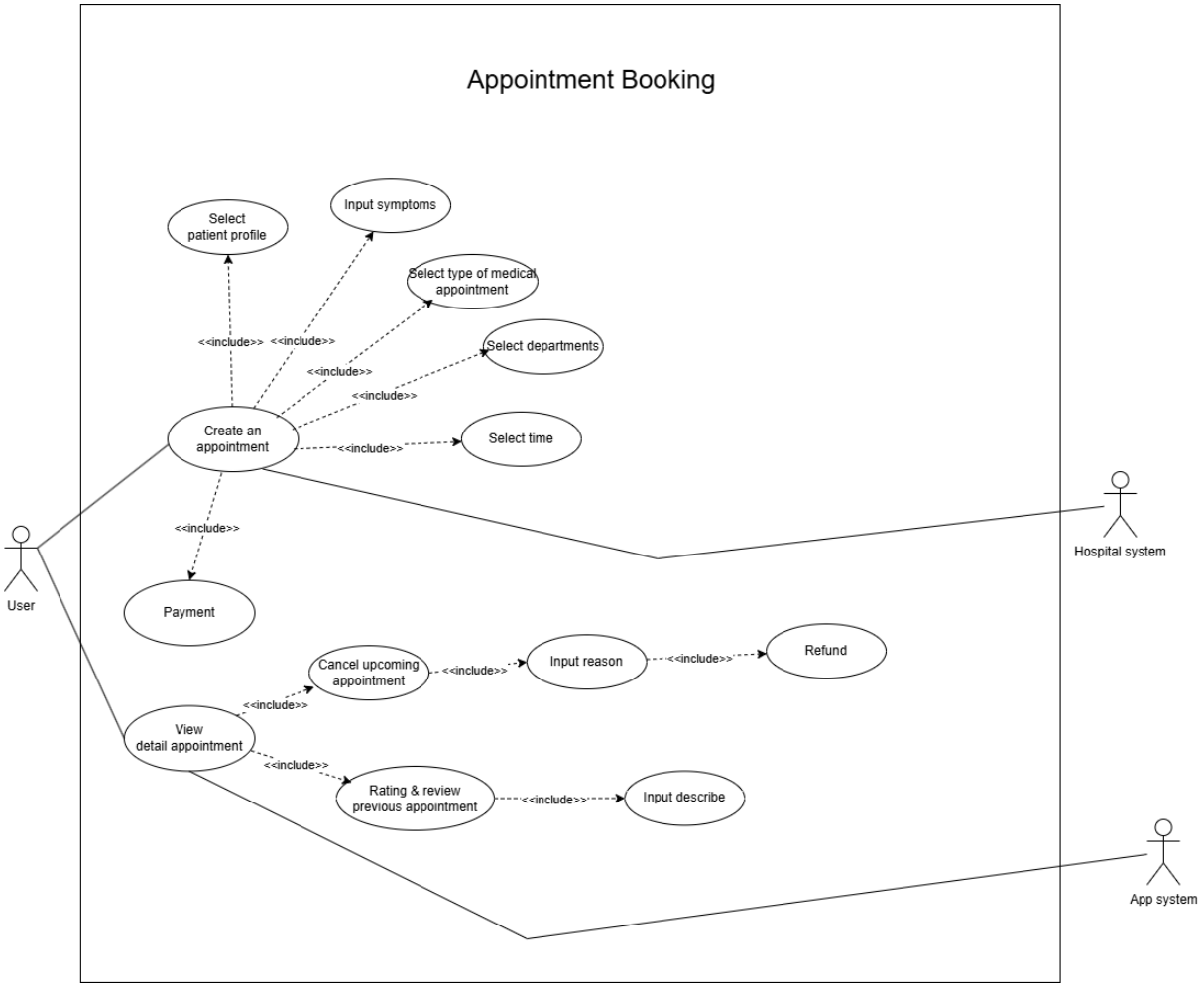


Figure 3. 27**.** Use case diagram of Appointment booking (Source: Author)

The use case diagram for **Appointment Booking** outlines the interactions between a **User**, the **Hospital System**, and the **App System**:

Create an Appointment: The user can select a patient profile, symptoms, type of medical appointment, department, and time to schedule an appointment.

Payment: The user proceeds with payment if required by the appointment process.

View Detail Appointment: The user can view details of a scheduled appointment.

Cancel Upcoming Appointment: The user can cancel an appointment, providing a reason, with the option for a refund if applicable.

Rating & Review Previous Appointment: After the appointment, the user can leave a rating and review, including descriptive feedback.

Table 3. 18. Use case description of Appointment booking process (Source: Author)

Use Case Name	Appointment booking	
Use Case No.	04	
Primary Actor	User	
Secondary Actor	Hospital system	
Use Case Story	This use case enables the user to make, view, cancel a booking.	
Trigger	User accesses the Appointment booking process.	
Pre-Condition	User Account exists and must be active. User has logged in the account. User's device is already connected to the internet when logging in.	
Post-Condition	All the information related to the booking has to be updated in application database and also send to hospital system.	
Primary Flow (PF)	**Title:** User is able to do all functions without any errors.	
	Actor Action	**System Response**
	1. User access Appointment booking	2. System shows a list of patient's profiles for user to choose.

	3. User choose a patient's profile or add a new patient's profile and choose a function. If user chooses Create an appointment, go to step 4 If user chooses View detail appointment, go to step 9. If user wants to create a new patient's profile go to function "Add another patient's profile"	4. System shows all the options related to the appointment for user to choose.
	5. User selects the preferred option he/she wants (select symptom, doctor, department, time) and selects Book appointment command.	6. Systems shows booking detail and requires user to pay for the booking.
	7. User pays for the booking	8. System confirms the booking and creates a booking QR code for user. System save the booking detail in database and send it to the hospital system.
		9. System shows list of upcoming and previous appointments.
	10. User chooses an appointment and a function. If user wants to cancel an upcoming appointment, go to	11. Systems requires user to fill in a cancellation reason form.

	step 11. If user wants to review and rate previous appointments, go to step 15.	
	12. User completes filling in the form and selects Send command.	13. System displays the following message: Hủy cuộc hẹn thành công. Phí khám bệnh sẽ được hoàn lại trong vòng 24 giờ. System deletes the appointments in database.
		14. System refunds the booking fee for user.
		15. Systems requires user to fill in a review form and rate the appointment.
	16. User completes filling in the form and selects Send Review command.	17. System displays the following message: Gửi đánh giá thành công. System updates the review in database and displays the review in review section.

e) User story

Table 3. 19. User Story of Appointment Booking (Source: Author)

Appointment Booking	
As a	user of the Mobile Appointment Scheduling System
I want	to be able to create, view, cancel and manage my medical appointments within the application.
So that	I can keep track of my appointments efficiently and make changes when needed.
Acceptance Criteria	1. When user selects the "Appointment Booking" function from the main dashboard, the system must display the appointment booking screen with a list of patient's profile and following 2 main options: a. Create appointment b. View appointment Create appointment flow: 2. When the user selects "Create appointment" function then chooses a patient's profile and selects "Create appointment" the system must display all the options (symptoms, department, doctor, time) for user to choose from. 3. System must allow the user to select symptoms he/she has from the list of specific symptoms. 4. Users must be able to select a department they want to visit and also the type of medical appointment: standard consultation or premium service. 5. System must allow the user to select an available time from the calendar displayed.

	6. After date, time, and doctor selection, user clicks "Confirm", system must display a summary of the appointment details and creates a QR code for user to pay for the booking. 7. After user pays for the booking, system confirms the appointment by displaying a message: "Cuộc hẹn đã được đặt thành công" and update the appointment information in database. View appointment flow: 8. When the user selects “View appointment” function, system must display 2 lists of upcoming and previous appointments. 9. If user selects an appointment from the upcoming appointments list, he/she can cancel it by completing the cancellation form provided by the system. 10. If the user selects an appointment from the previous appointments list, they can write a review and rate it by completing the review section displayed by the system."

f) Sequence Diagram

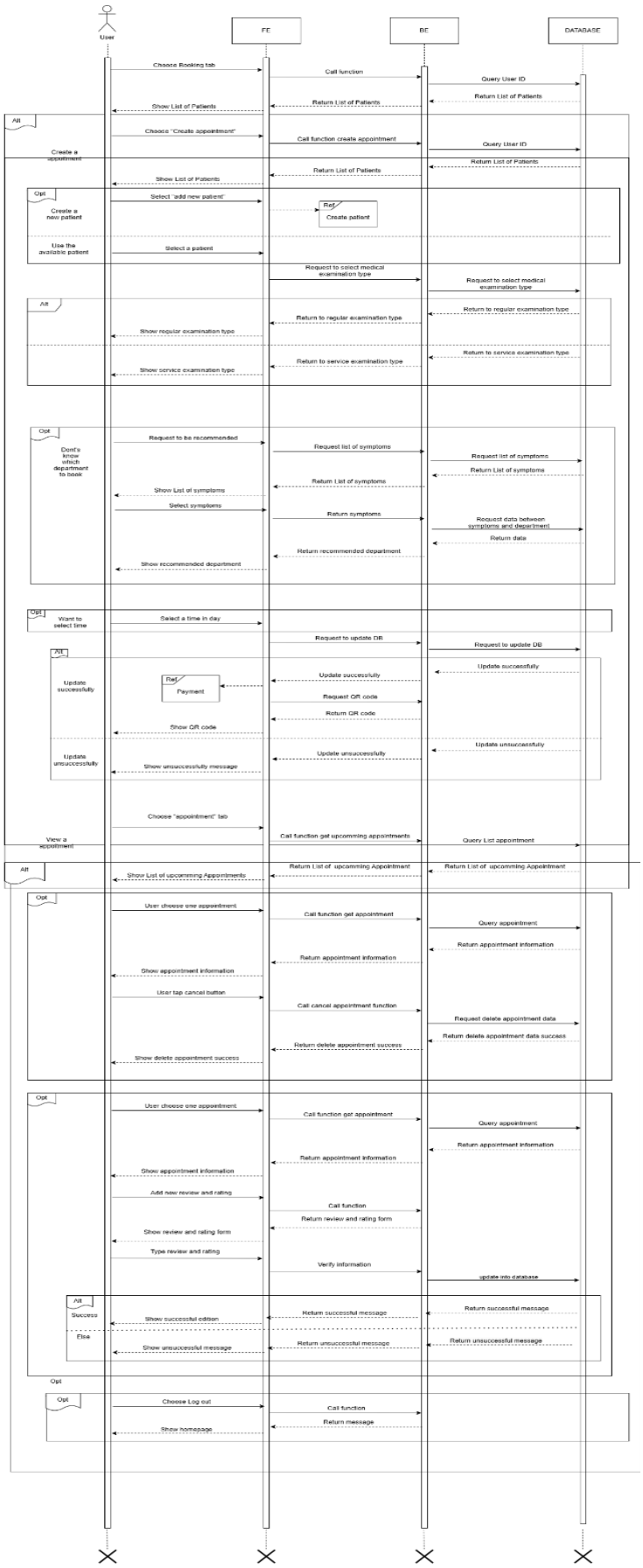


Figure 3. 28. Sequence Diagram of function Booking (Source: Author)

The sequence diagram for the **Booking Function** illustrates the process flow involving the **User**, **Front End (FE)**, **Back End (BE)**, and **Database**:

Create Appointment:

- The user selects the booking tab and chooses to create an appointment.
- The system retrieves and displays a list of patients, allowing the user to select or add a new patient.
- The user specifies the type of medical examination, symptoms, department, and time.
- The BE and Database handle updates and store appointment details, generating a QR code if payment is required.

View Appointment:

- The user can view a list of upcoming appointments.
- Selecting an appointment provides detailed information and options to cancel if necessary.

Add Review and Rating:

- After an appointment, the user can add a review and rating.
- The system verifies and stores the feedback.

Log Out:

- The user logs out, returning to the homepage.

g) Activity Diagram

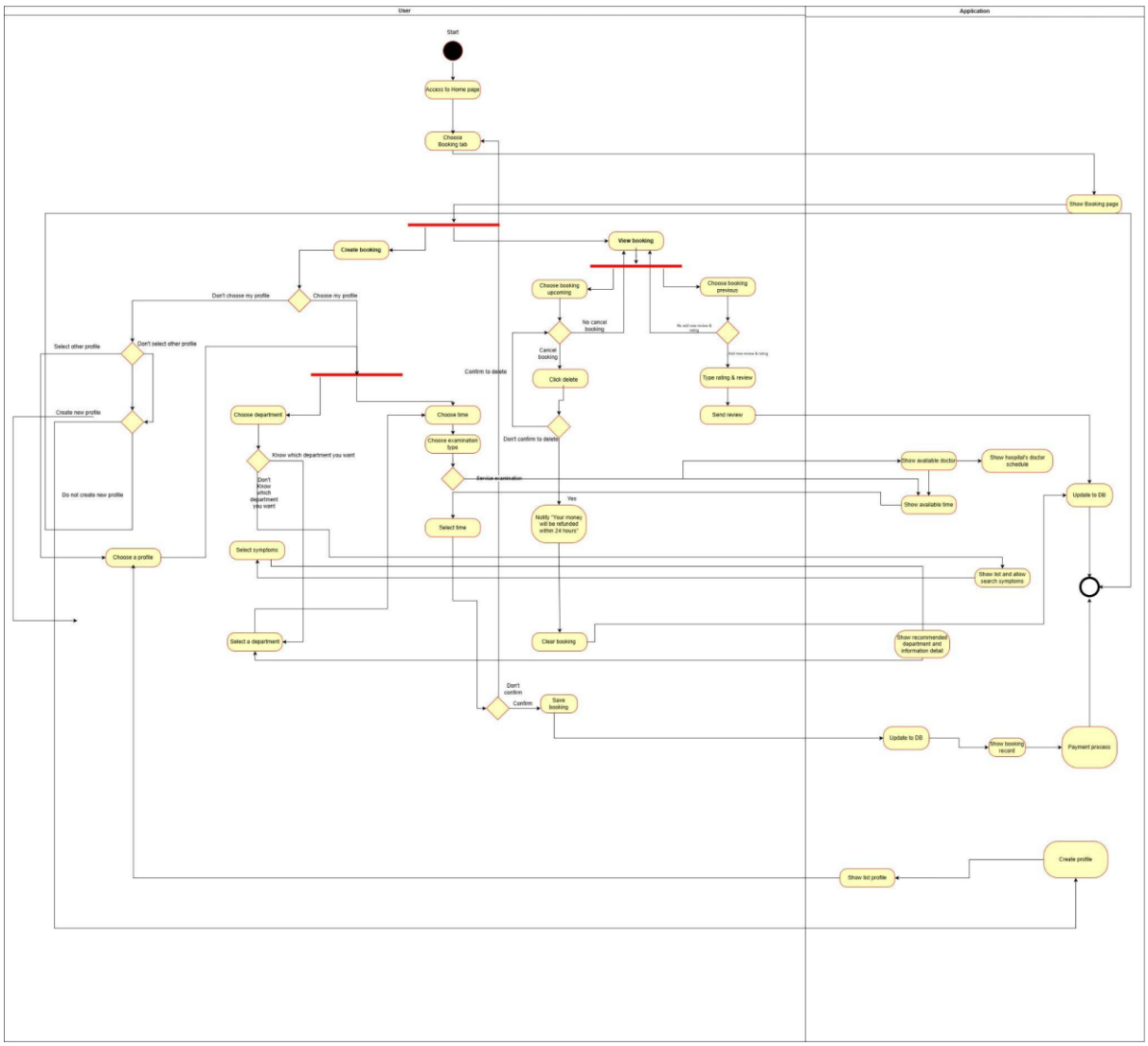


Figure 3. 29. Activity Diagram of Book Appointment (Source: Author)

The activity diagram for **Book Appointment** outlines the user journey and application processes for creating and managing appointments:

Initiate Booking: The user selects the booking option from the home page.

Profile Selection: The user can choose an existing profile or create a new one for the appointment.

Appointment Details: The user selects the department, specifies symptoms, and chooses a time slot.

Confirm Booking: The system checks availability and confirms the booking. If necessary, a payment process is initiated.

View and Manage Bookings: The user can view current bookings, rate past appointments, or cancel upcoming ones.

Update Records: The application updates the database with booking details, cancellations, or feedback.

h) Data Dictionary

Table 3. 20. Data elements dictionary of function Booking (Source: Author)

Label	Alias	Type & Length	Default Value	Acceptable values	Source	Security	Responsible User(s)	Description/ Comments
Appointment ID		String (20)	Auto-generated	Alphanumeric; unique identifier	Appointment Scheduling System	Hospial staff	Hospial staff	Unique identifier for each appointment booking, used to track and manage individual appointments.
Time Slots	Appointment Slots	String (10)	None	HH:MM-HH (e.g., "09:00-10:00")	Scheduling System	View access for scheduling and medical staff	Staff, Scheduling Team	Defined time slots available for patient appointments.
Department ID		Numeric (10 digits)	None	Valid department ID	Hospital System	Only user and Hospital Database Administrator can view	Medical Department	Identifier for the department where the user has an appointment.
DepartmentName	Department	Alphanumeric (50 characters)	None	Valid department name	Hospital System	Only user and Hospital Database Administrator can view	Medical Department	Name of the department associated with the user's appointment.

DepartmentIsWorking		Boolean	FALSE	True, False	Hospital System	Only user and Hospital Database Administrator can view	Medical Department	Indicates if the department is currently operational.
Invoice Number	Invoice No	Numeric (12 characters)	None	12-character string starting with INV	Payment process	Only accounting department can edit	Accounting department	Unique identifier for the invoice generated after the appointment.
Review ID	ReviewID	String (20)	System auto-generated	Alphanumeric; unique identifier	Feedback System	Customer Service	Customer Service	Unique identifier for each review submitted
Rating	Rating	Integer (1)	None	1-5	User Input	Public; View access for authorized users	Customer Service	Numeric rating provided by the user, typically ranging from 1 (lowest) to 5 (highest).
Comment	ReviewText	String (500)	None	Text comment, max 500 characters	User Input	Public	Customer Service	User-submitted feedback comment, offering details about their experience, limited to 500 characters.
Review Date	ReviewDate	Date	None	Standard date format (YYYY-MM-DD)	Application System	Public	Customer Service	Date when the review was submitted, used to track feedback over time.
Review Timestamp	ReviewTime	Time	None	Standard time format (HH:MM)	Application System	Public; View access for authorized users	Customer Service	Time when the review was submitted

Table 3. 21. Data flows dictionary of function Booking (Source: Author)

Label	Description	Alternate Name(s)	Origin	Destination	Record	Volume and Frequency
Request show information	Request sent by the user to display personal or family information.	Request sent by the user to display personal or family information.	User	Booking for me/for family (3.1)	User request information	5,000 requests/day – Generated every time a user views or updates information.
Select/type symptoms	Symptoms entered by the user for selecting the department.	Symptoms entered by the user for selecting the department.	User	Select disease department (3.2)	Symptoms data	2,000 records/day – For every department selection based on symptoms.
Show list symptoms	System retrieves and displays relevant symptoms based on user input.	System retrieves and displays relevant symptoms based on user input.	Knowledge base	User	Symptoms list	2,000 queries/day – Accessed as needed when users enter symptoms.
Choice results disease	Displays the selected disease or department based on symptoms.	Displays the selected disease or department based on symptoms.	Select disease department (3.2)	Diseases	Selected disease	2,000 results/day – Generated whenever users select a department based on symptoms.
Doctor's info	Doctor's details (name, working hours, etc.) displayed to the user.	Doctor's details (name, working hours, etc.) displayed to the user.	Doctor	Select doctor (3.3)	Doctor information	4,000 lookups/day – Accessed every time a doctor is selected by a

						user.
Choice results doctor	Selected doctor's information confirmed and sent for further booking.	Selected doctor's information confirmed and sent for further booking.	Select doctor (3.3)	User	Doctor choice	4,000 results/day – Generated for each doctor selection.
Show available time	Displays available time slots based on selected date and doctor.	Displays available time slots based on selected date and doctor.	Select time in day (3.4)	User	Available time	3,500 records/day – Displayed to the user for each selected doctor and date combination.
Choose time	Confirmation of selected time slot.	Confirmation of selected doctor and time slot.	Select time in day (3.4)	Create booking (3.5)	Doctor and time details	3,500 confirmations/ day – Passed to create a booking for each selected time slot.
Booking's information	Booking details (doctor, time, etc.) sent for booking confirmation.	Booking details (doctor, time, etc.) sent for booking confirmation.	Create booking (3.5)	Booking record	Booking details	3,000 bookings/day – Stored for each confirmed booking.
Updated booking status	Status update after the user cancels a booking.	Status update after the user cancels a booking.	Cancel booking (3.6)	Booking record	Updated status	500 updates/day – Updated every time a booking is canceled.

Review/Rating	User's review or rating for the completed booking.	User's review or rating for the completed booking.	User	Review booking (3.7)	Review and rating	500 reviews/day – Submitted after each completed booking or appointment.
Request show information	Request sent by the user to display personal or family information.	Request sent by the user to display personal or family information.	User	Booking for me/for family (3.1)	User request information	5,000 requests/day – Generated every time a user views or updates information.
Booking ID	Provide Booking ID to choose		User	Choose Unpaid Checkup from List Process	List of Booking ID	Accessed when a user requests to pay
Unpaid Checkup List	List of unpaid checkups selected for generating an invoice	Unpaid Checkup Data	Choose Unpaid Checkup List Process	Create Invoice Process	List of Unpaid Checkup	Accessed when a user requests to pay
Invoice	Generated invoice based on selected checkups	Invoice Document	Create Invoice Process	Account Receivable data store	Invoice Number, Invoice Date, Description of Service, Goods	Created after unpaid checkups are selected
Invoice detail	Provide detail information about Invoice		Account Receivable data store	Choose payment method process	Invoice Number, Invoice Date, Description of Service, Goods	

Table 3. 22. Data store dictionary of function Booking (Source: Author)

Label	Description	Alternate Name(s)	Attributes	Volume and Frequency
Disease	Stores information about diseases related to the checkup, used to validate booking and payment details.	Diagnosis Data Store	Disease ID, Disease Name, Symptoms	100,000 - 500,000 records, updated with new disease records as needed.
Doctor	Stores information about the doctors involved in the booking, including availability and details.	Physician Information Store	Doctor ID, Doctor Name, Department, Available Working Days	1,000 - 10,000 records, updated when doctor information changes or when checked by users.
Time in Day	Stores information about the available time slots during the day for bookings and checkups.	Appointment Slot Data Store	Time Slot ID, Available Time, Doctor ID	100,000 - 1,000,000 records, frequently updated based on doctor schedules and user bookings.
Booking Record	Stores the record of the booking, including booking details, payment status, and refund history.	Appointment Data Store	Booking ID, Booking Date, Time Slot, Payment Status	50,000 - 500,000 records, updated every time a booking is created, modified, or payment status changes.
Payment Record	Stores all payment-related details		Payment status, Invoice ID, Total amount	5,000 - 50,000 records, updated per completed transaction

Table 3. 23. Data process dictionary of function Booking (Source: Author)

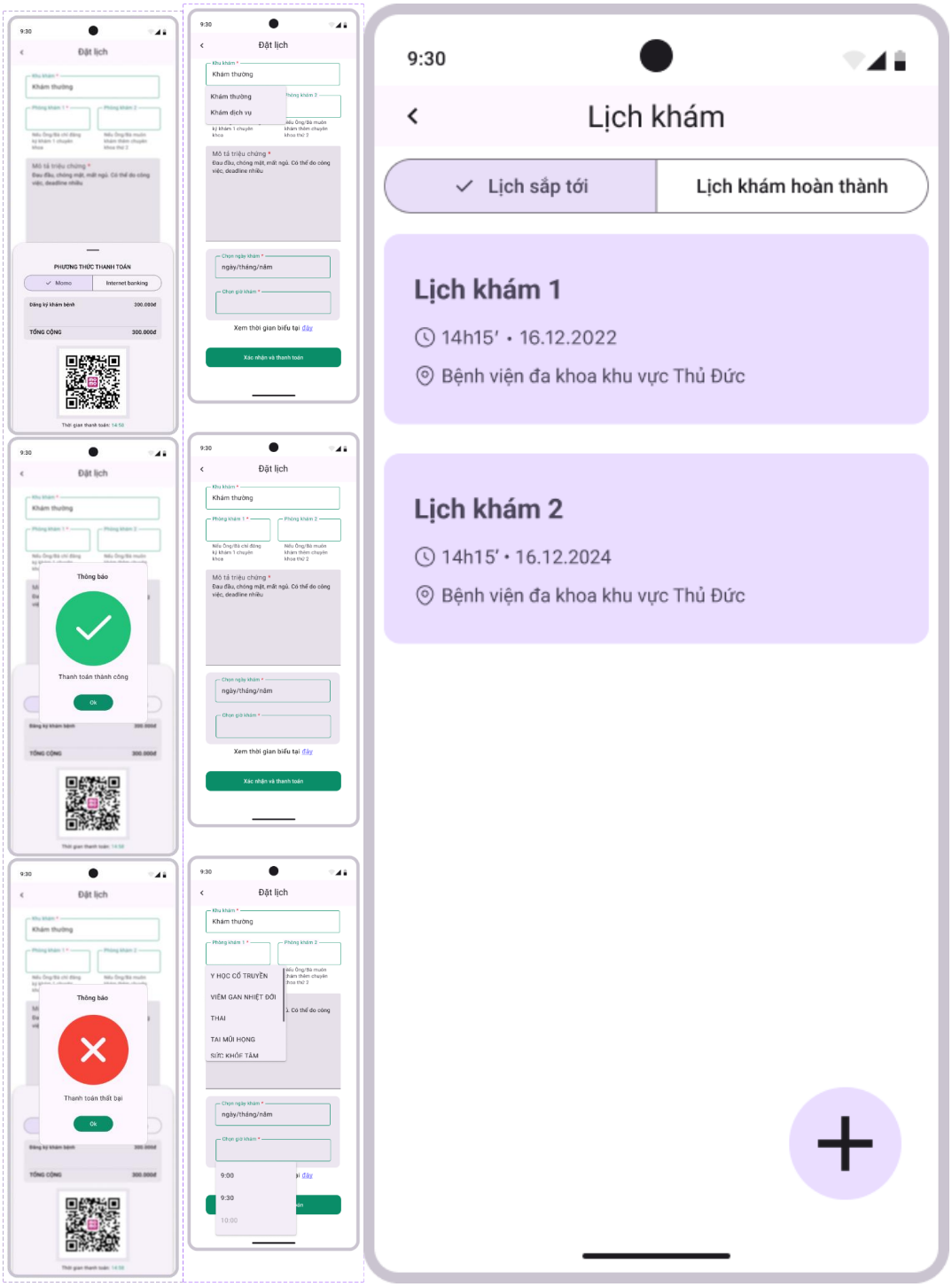
Process Name	Description	Process Number	Input Data Flows	Output Data Flows	Process Description
Choose Unpaid Checkup	User selects an unpaid checkup from the list	4.1	Booking record	Unpaid checkup list	Retrieves list of unpaid checkups associated with a

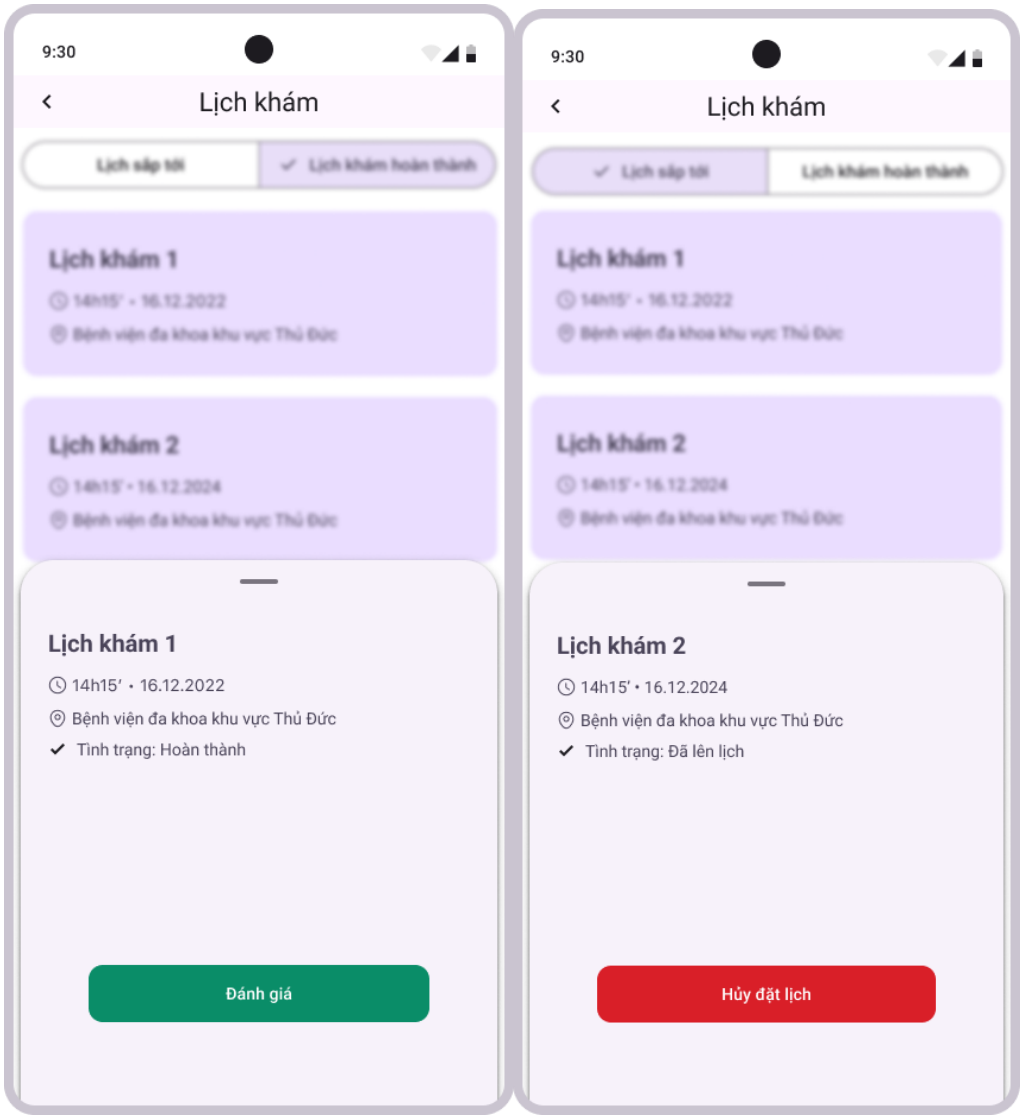
					booking record for user selection
Create Invoice	Generates an invoice based on selected checkup	4.2	Unpaid checkup list	Invoice, Account receivable	Creates an invoice for the selected unpaid checkup and adds it to account receivables
Choose Payment Method	Allows user to select a payment method	4.3	Invoice, Selected payment method	Selected payment method	Prompts user to choose between available payment methods (Bank or MoMo)
Generate Dynamic QR Code	Creates a QR code based on selected payment method	4.4	Selected payment method, Invoice	Payment QR code, QR Code Validation Request, QR Code Validation Result	Generates a QR code for the payment amount and invoice ID, and initiates QR code validation with Bank/MoMo
Apply Payment	Validates and applies payment based on QR code scan	4.5	QR Code Validation Result, Payment QR code, Payment Confirmation	Payment status, Payment record, Account receivable, Payment status update	Confirms payment by validating QR code with Bank/MoMo and updates payment status
Refund	Processes refund if the payment failed	4.6	Payment status, Refund Request	Refund Confirmation, Updated payment details	Initiates refund process by requesting the bank or MoMo to refund the transaction

Table 3. 24. Data entity dictionary of function Booking (Source: Author)

Entity Name	Description	Input Data Flows	Output Data Flows
Hospital	Manages patient information, medical records, and facilitates appointment scheduling and validation	Patient profile, Department information, Diseases, Doctor information, Schedule information, Booking ID, Payment status, Cancelation request validation	Updated personal information, Appointment validation, Response booking result, Updated booking status, ReviewRating
User	Represents the patient or individual interacting with the hospital system for personal, appointment, and medical purposes	User ID, Personal Information, Doctor's information, Available date, Date info, Booking detail, Payment status, Booking ID, ReviewRating	Choice results disease, Choice results doctor, Choice results time, Response booking result, Updated booking status, Queue number, Successful/Unsuccessful

i) User Interface


9:30
Lịch khám
Lịch sắp tới
Lịch khám hoàn thành
Lịch khám 1
14h15' • 16.12.2022
Bệnh viện đa khoa khu vực Thủ Đức
Lịch khám 2
14h15' • 16.12.2024
Bệnh viện đa khoa khu vực Thủ Đức
Đặt lịch
Thông báo
Thanh toán thành công
Thanh toán thất bại


9:30
Lịch khám
Lịch sắp tới
Lịch khám hoàn thành
Lịch khám 1
14h15' - 16.12.2022
Bệnh viện đa khoa khu vực Thủ Đức
Lịch khám 2
14h15' - 16.12.2024
Bệnh viện đa khoa khu vực Thủ Đức
Lịch khám 1
14h15' • 16.12.2022
Bệnh viện đa khoa khu vực Thủ Đức
Tình trạng: Hoàn thành
Đánh giá
9:30
Lịch khám
Lịch sắp tới
Lịch khám hoàn thành
Lịch khám 1
14h15' - 16.12.2022
Bệnh viện đa khoa khu vực Thủ Đức
Lịch khám 2
14h15' - 16.12.2024
Bệnh viện đa khoa khu vực Thủ Đức
Lịch khám 2
14h15' • 16.12.2024
Bệnh viện đa khoa khu vực Thủ Đức
Tình trạng: Đã lên lịch
Hủy đặt lịch

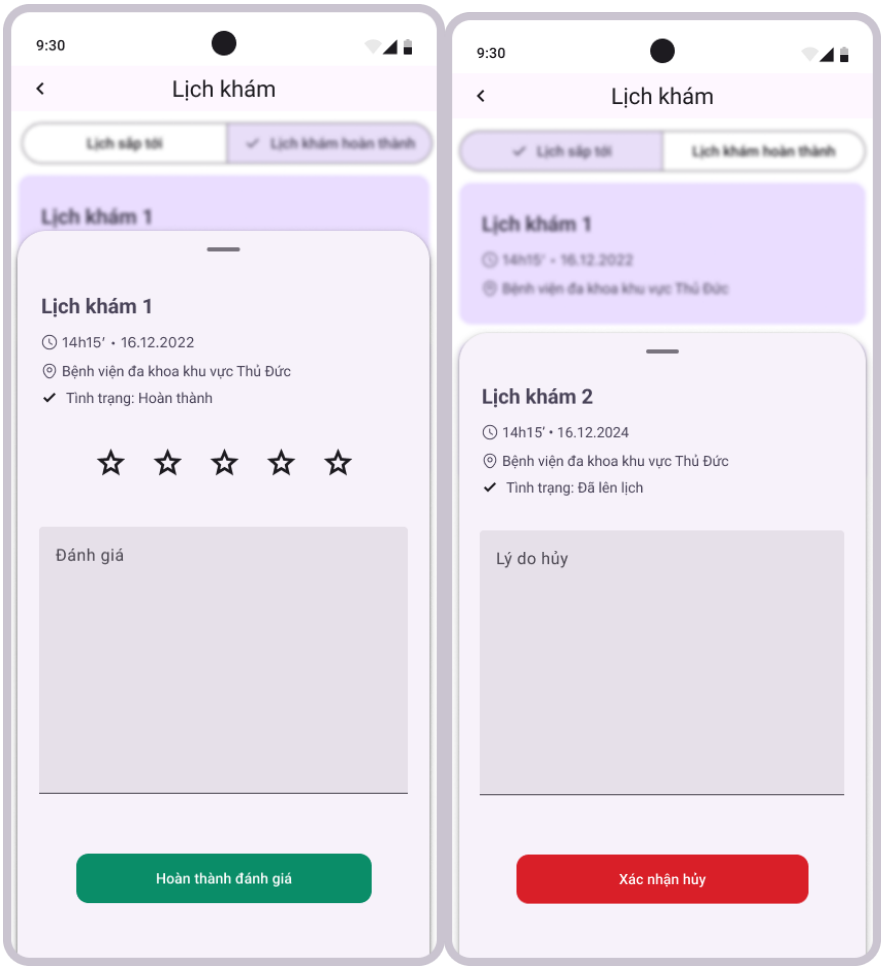


Figure 3. 30. User interface of booking appointment (Source: Author and link)

This figure demonstrates the process of booking a medical appointment and handling payment within a healthcare application. Here's a description of each screen:

Appointment Booking Screen:

- The user selects options for:
  - **Patient Profile**: The name of the patient booking the appointment.
  - **Type of Consultation**: The type of medical service (e.g., general consultation).
  - **Room Selection**: Users can choose from available rooms (e.g., Room 1 or Room 2).

- **Symptoms Description**: A text box allows users to describe their symptoms or specific concerns (e.g., headache, dizziness, or fatigue).
- **Date and Time**: Users select a date and time for the appointment from a dropdown or calendar.

○ The screen includes a link to view the available schedule and a "Confirm and Pay" button to proceed with the booking.

Payment Method Screen:

○ After confirming the appointment, the user is taken to the payment screen.
○ Payment options include methods such as MoMo and Internet banking.
○ The screen displays the total cost and provides a QR code for completing the payment.
○ The payment time is indicated at the bottom.

Successful Payment Notification:

○ A green confirmation screen appears, notifying the user of a successful payment.
○ The message includes a checkmark symbol and a confirmation message, "Payment Successful," with an "Ok" button to proceed.

Failed Payment Notification:

○ If the payment fails, a red error notification appears.
○ The screen shows an "X" icon with the message "Payment Failed" and an "Ok" button to dismiss the message.

j) Validation

Table 3. 25. Data validation of Appointment booking function (Source: Author)

Failed Validation	**Error Message Displayed**

All required data is entered	<Data Field Name> is a required field.
User profile is created before booking	Can’t be able to book an appointment because user have to choose profile first to view or create booking
The payment must be successful for the booking to be created	The booking will not be displayed until the payment is successful

3.4.4. Payment

a) Function introduce

Function meaning

The hospital's payment function can be automated to enhance efficiency and accuracy. Automated payment systems minimize the risk of fraud and errors, while also improving patient satisfaction and increasing revenue collection. The application enables users to pay for booking fees and post-examination expenses, such as medication, directly through the system.

Business rules

The business rule for this function is booking fees and expenses after medical examination can only be paid through mobile banking and MOMO e-wallet.

b) BPMN

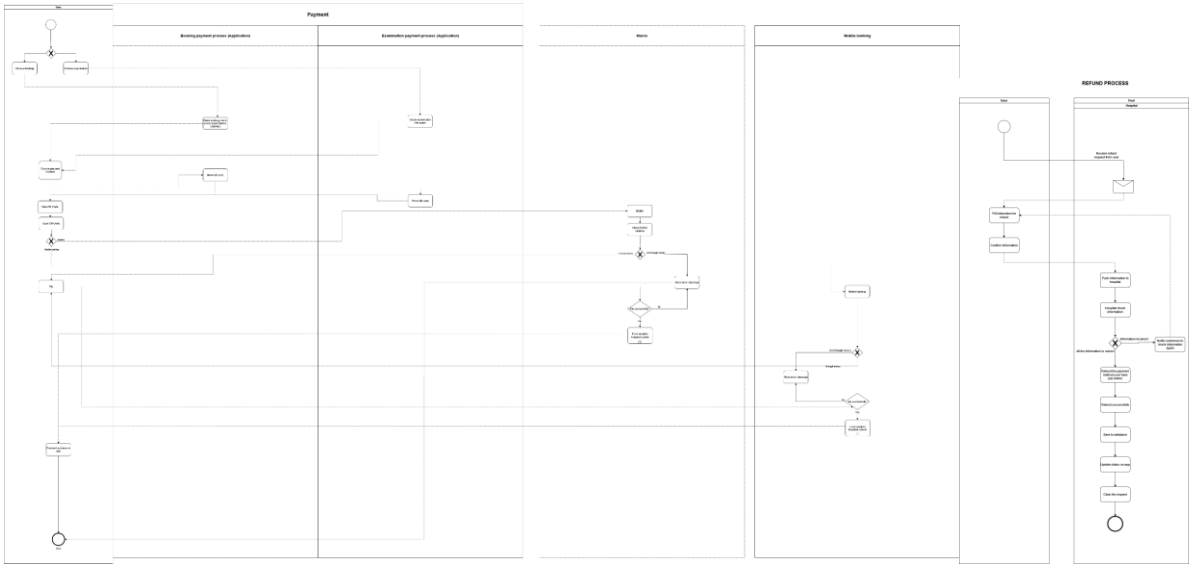


Figure 3. 31. BPMN of the Payment process (Source: Author)

The BPMN diagram for the **Payment Process** outlines the steps involved in managing payments for an appointment booking:

Initiate Payment: The process begins with the user initiating payment after booking an appointment.
Payment Verification: The system verifies the payment details, checking for correctness and sufficiency.
Confirm Payment: Once verified, the payment is confirmed, and a receipt is generated.
Handle Errors: If there is an issue with the payment (e.g., insufficient funds or incorrect details), the system notifies the user to resolve it.
Finalize Appointment: Upon successful payment, the appointment is confirmed, and payment status is updated in the booking records.

The BPMN diagram for the **Refund Process** mentions the steps taken when a user requests a refund:

Request Refund: The user initiates a refund request.
Submit Request: The system collects details of the refund request and submits it for processing.
Review and Approval: The refund request is reviewed, and if valid, it proceeds to approval.
Refund Processing: Upon approval, the refund is processed through the payment system.
Notify User: Once the refund is completed, the user is notified of the successful refund.

c) DFD

- Users select a payment method (bank, MOMO).

- The system validates the payment details, processes the payment, generates an invoice, and confirms the payment status.

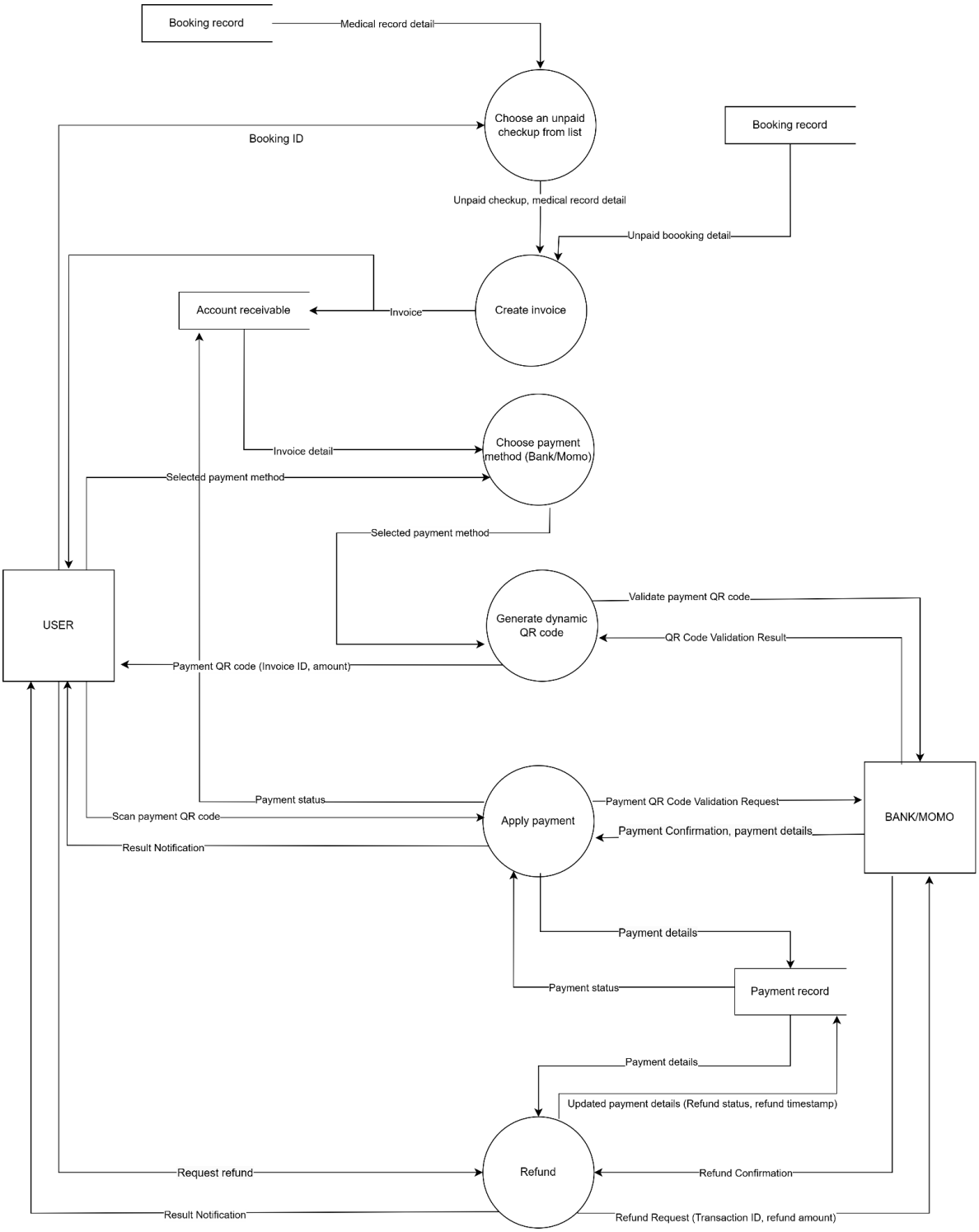


Figure 3. 32. Payment process DFD level 1 (Source: Author)

The Level 1 Data Flow Diagram (DFD) for the **Payment Process** illustrates the key steps involved when a **User** makes a payment through the **Bank/Momo** system:

Choose Unpaid Checkup: The user selects an unpaid checkup from a list.

Create Invoice: The system generates an invoice with booking and medical record details.

Choose Payment Method: The user selects a payment method (e.g., Bank or Momo).

Generate Dynamic QR Code: A QR code is generated for the selected payment method.

Apply Payment: The user scans the QR code to apply payment, and the payment is validated with the bank.

Payment Status: The system updates the payment status and notifies the user of the result.

Refund (if applicable): The user can request a refund, which is processed, and the user is notified of the refund status.

d) Use case

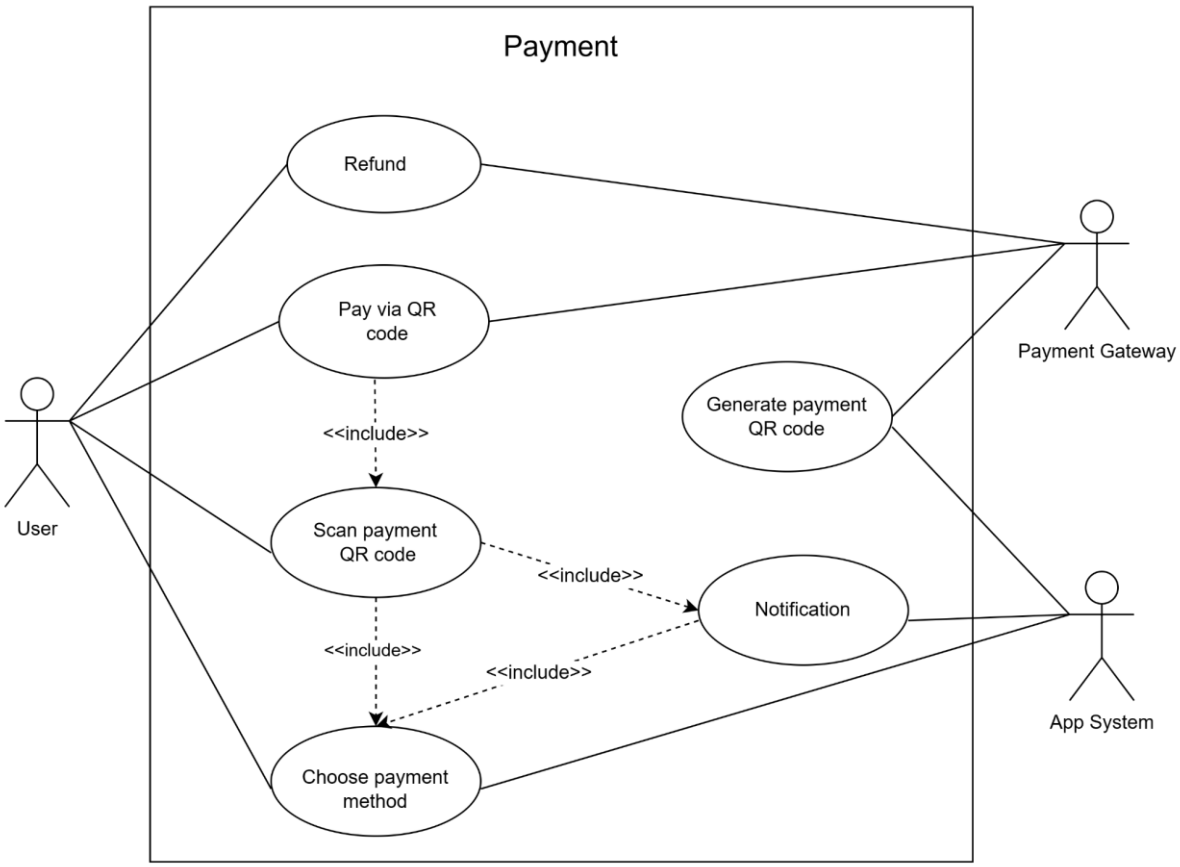


Figure 3. 33. Use case diagram of Payment (Source: Author)

The use case diagram for **Payment** outlines the main actions a **User** can perform and interactions with the **Payment Gateway** and **App System**:

Choose Payment Method: The user selects a preferred payment method.
Generate Payment QR Code: The system generates a QR code for the payment, facilitated by the Payment Gateway.
Pay via QR Code: The user initiates payment by scanning the QR code.
Scan Payment QR Code: This includes verifying the payment through the QR code.
Notification: The App System sends a notification confirming the payment status to the user.
Refund: If necessary, the user can request a refund, which involves the Payment Gateway.

Table 3. 26. Use case description of Payment process (Source: Author)

Use Case Name	Payment
Use Case No.	05
Primary Actor	User
Secondary Actor	Payment gateway, app system
Use Case Story	This use case enables the user to complete the payment process, including applying any discounts, calculating the total amount, and generating or validating a QR code for payment.
Trigger	User initiates payment for a service (from Check-up and Hospital interaction) or booking process.
Pre-Condition	User accounts exist and must be active. User has logged in the account. User has a valid booking or service to pay for. Internet connection is active.

	Users have Momo e-wallet or Internet bank service.	
Post-Condition	Payment is successfully processed. The system records the transaction in the application database (Activity Log) retrieved for the Booking Appointment process and Check-up and Hospital interaction process. Users got the booking or service.	
Primary Flow (PF)	**Title:** User completes payment without errors.	
	Actor Action	**System Response**
	1. User initiates the payment process by clicking the button “Thanh toán ngay”.	2. The app system includes "Chọn cổng thanh toán" and prompts the user to select a payment option “Momo” or “Internet Banking”.
	3. User selects a payment method.	4. The app system calculates the total amount.
		5. The app system generates a dynamic QR code containing the payment details and sends it to the payment gateway for validation. If the payment gateway is not available, go to AF2.
		6. The payment gateway validates the QR code and returns it to the system if validation is successful.

		7. System displays the validated QR code to the user with messages “Mã thanh toán” có hiệu lực trong 15 phút. If the payment QR code is not used in 15 minutes, go to the AF1.
	9. User opens Momo/Internet banking app và scans the payment QR code.	10. System validates the payment.
		11. System sends a payment notification to the user “Thanh toán thành công” and records the transaction in the Activity Log.
Alternate Flow 1 (AF1)	**Title:** QR Code Expired (15-Minute Timeout)	
	Actor Action	**System Response**
	1. User don’t scan QR code or scans the QR code using a compatible application after 15 minutes.	2. System checks the QR code validity and detects that it has expired (exceeded 15-minute limit).
		3. System displays an error message to the user: "Mã QR đã hết hạn. Vui lòng tạo mã QR

		mới để tiếp tục thanh toán."
	4. User requests a new QR code.	5. System regenerates and sends a new QR code to the payment gateway for validation.
		6. Upon successful validation, the system displays the new QR code to the user.
	7. Go back to step 9 of PF.	
Alternate Flow 2 (AF2)	**Title:** Payment Gateway Unavailable Due to Error or Maintenance	
	Actor Action	**System Response**
	1. The app system sends the QR code to the payment gateway for validation.	2. The payment gateway fails to validate the QR code due to an error or maintenance issue.
		3. System displays an error message to the user: "Thất bại khi liên kết với cổng thanh toán. Vui lòng thử lại sau hoặc chọn cổng thanh toán khác."
	4. User chooses to try again or selects a different payment gateway.	5. If the user retries, the system re-sends the QR code for validation.
		6. If the user selects a different payment gateway, the system redirects them to choose an

		alternate gateway and continues from step 2 of PF.
	7. Go back to step 9 of PF.	

e) User story

Table 3. 27. User Story of Payment Process

Payment	
As a	user of the Mobile Appointment Scheduling System
I want	to complete payments for my bookings or services quickly and securely within the app, using either Momo e-wallet or Internet Banking.
So that	I can confirm my appointments and hospital interactions without delays or complications.
Acceptance Criteria	1. The app system provides a clear and accessible "Thanh toán ngay" button from the booking or check-up list page. 2. On clicking on “Thanh toán ngày”, the app system shall be able to provide user to choose between Momo e-wallet and Internet Banking as payment methods. 3. The app system counts total payment amount, including any discounts or additional fees, should be displayed clearly before the user proceeds to payment. 4. The system should generate a dynamic QR code containing the payment information and display it to the user. 5. User user Momo/Internet banking app to scan the payment QR code. 6. Upon successful payment after the user scans the QR code, the system should display a confirmation message and send a notification to the user.

	7. Error handling: 7.1. If the QR code expires after 15 minutues, the system should display an error message “Mã thanh toán hết hạn” and allow the user to generate a new QR code. 7.2. If the payment gateway is unavailable, the system should display an error message and suggest alternative payment methods or retry the payment.

f) Sequence Diagram

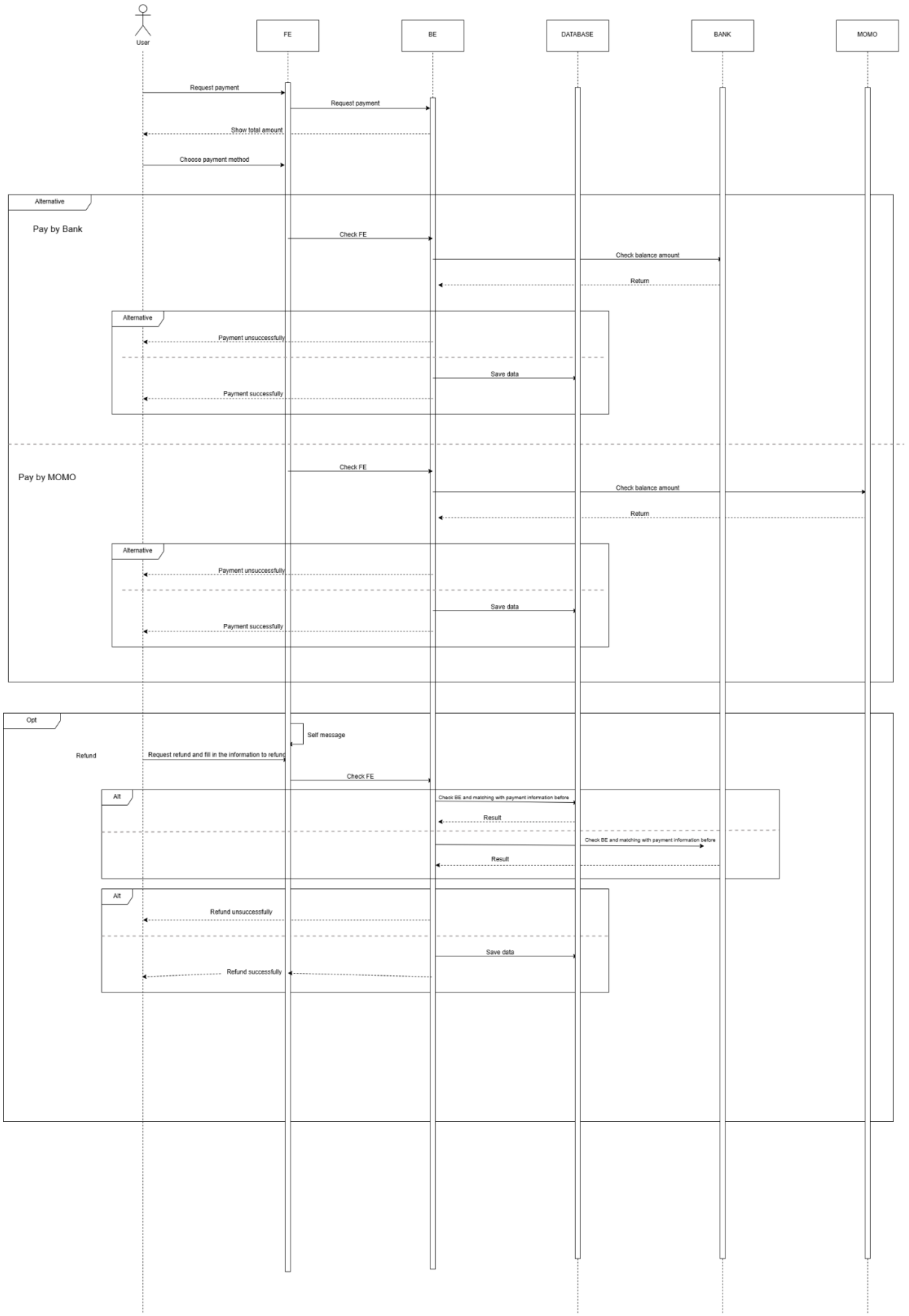


Figure 3. 34. Sequence Diagram of Payment (Source: Author)

The **Sequence Diagram of Payment** illustrates the process flow for handling payments, including different payment methods and refund requests:

Initiate Payment: The user starts the payment process, choosing their preferred method (bank or MOMO).

Payment by Bank or MOMO:

- The system checks the front-end (FE) for payment information.
- The backend (BE) communicates with the database to verify the balance.
- Upon successful verification, the system confirms payment, otherwise, it returns a failure message.

Refund Process:

- The user requests a refund, filling in the required details.
- The system verifies eligibility and processes the refund if approved.
- A success or failure message is returned to the user based on the outcome.

g) Activity Diagram

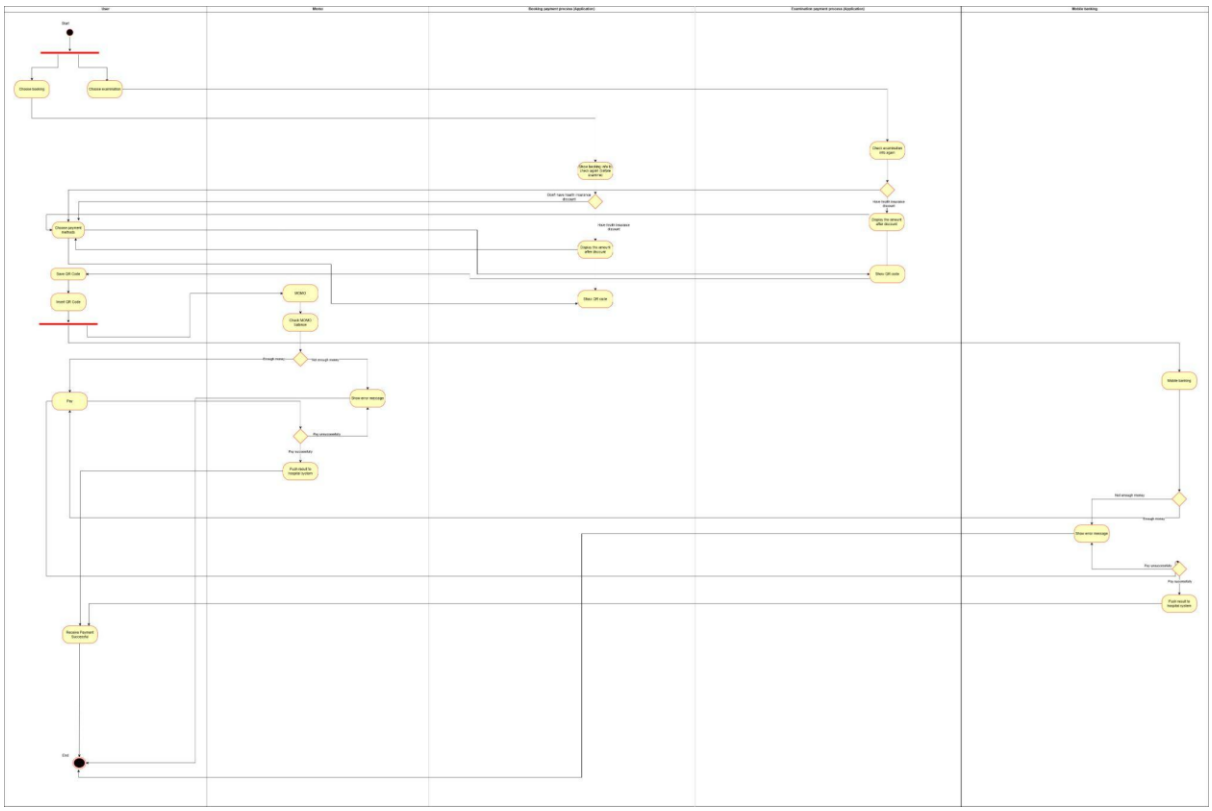


Figure 3. 35. Activity Diagram of Payment (Source: Author)

The **Activity Diagram of Payment** illustrates the steps in the payment process:

1. **Payment Selection**: The user chooses a payment method.
2. **Payment Process**: The system verifies payment details, processes the transaction, and checks for success or failure.
3. **Error Handling**: If the payment fails, an error message is shown, and the user may retry.
4. **Completion**: Upon successful payment, confirmation is sent, and records are updated.

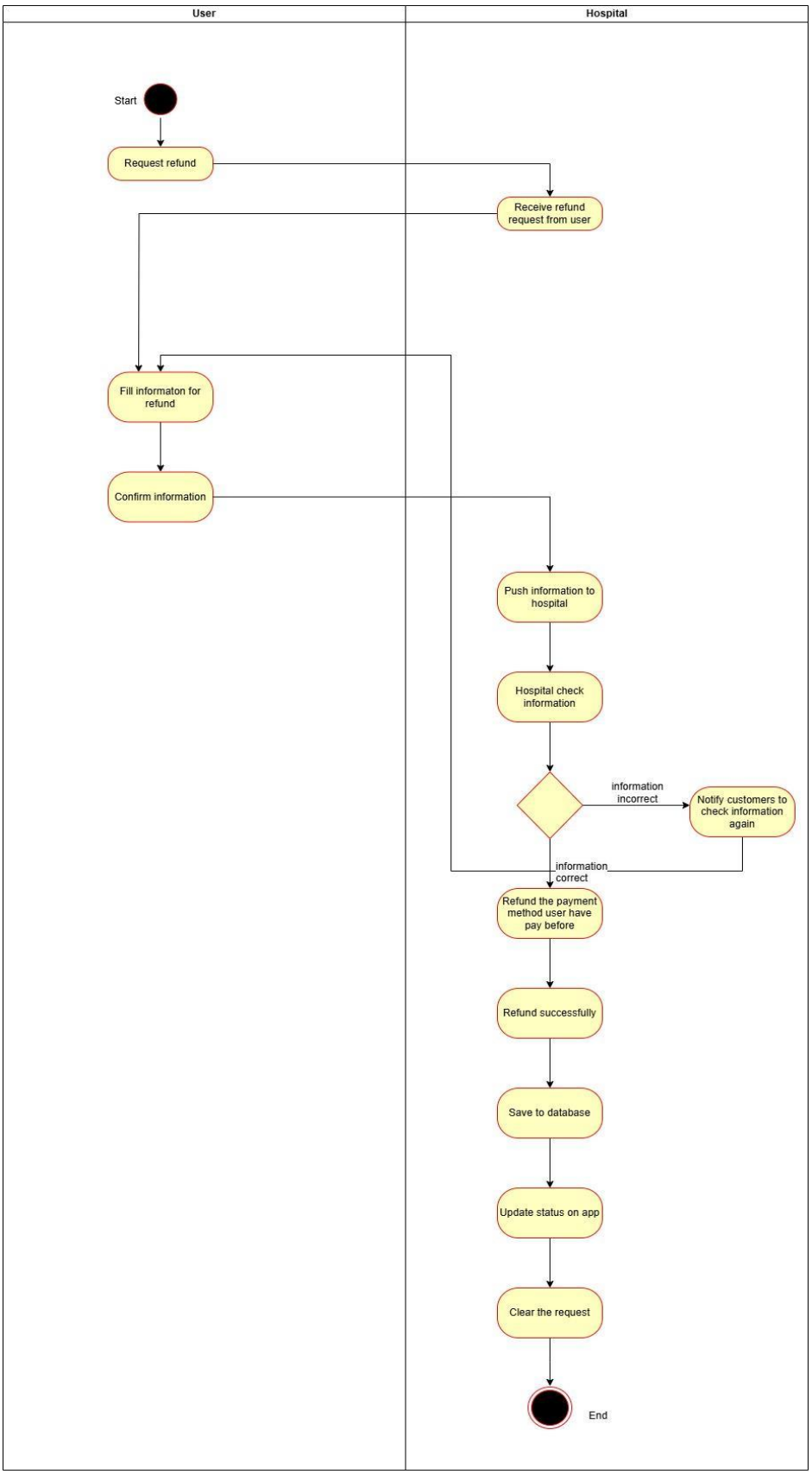


Figure 3. 36. Activity Diagram of Refund (Source: Author)

This Activity Diagram mentions the refund process:

Request Refund: The user initiates a refund request and fills in the required information.

Hospital Verification: The hospital receives and checks the provided information.

- If information is incorrect, the user is notified to recheck.

Process Refund: Once verified, the refund is issued through the original payment method.

Finalize: The refund is recorded in the database, the app status is updated, and the request is cleared.

h) Data Dictionary

Table 3. 28. Data elements dictionary of function (Source: Author)

Label	Alias	Type & Length	Default Value	Acceptable values	Source	Security	Responsible User(s)	Description/Comments
Payment ID		String (30)	Auto-generated by system	Alphanumeric; unique identifier	Payment Processing System	Accounting department	Accounting department	Created in the "Apply Payment" step, this ID is used to verify and process real-time payments with the bank or payment gateway.
Timestamp of Payment	TransactionTimestamp	Timestamp	None	Standard datetime format (YYYY-MM-DD HH:MM)	Payment Processing System	Accounting department	Accounting department	The exact time a payment transaction request was made, used for audit and

								verification.
Payment Method	How to pay	String (20 characters)	None	“Banking” or “Momo”	Payment process	Accounting department	Accounting department	The method used for payment
Payment Amount	Total payment	Float	None	Non-negtive number	Payment process	Accounting department	Accounting department	The total amount being paid
Payment status	Transaction Status	Boolean	Completed, Failed	System	Payment process	System	System	Current status of the payment transaction (Failed, completed).
Invoice Number	Invoice No	Numeric (12 characters)	None	12-character string starting with INV	Payment process	Only accounting department can view	Accounting department	Unique identifier for the invoice generated after the appointment.
Invoice Date	Invoice issued Date	DateTime	None	Acceptable values would include both date and time	Payment process	Accounting department	Accounting department	The date when the invoice was issued
Payment QR Code	Pay Code	QR code	None	Valid QR code format	Payment process	Accounting department	Accounting department	The dynamic QR code for patient to scan when

								paying an invoice
Refund ID	Refund Transaction ID	Numeric (10 digits)	None	Valid transaction ID format	System-generated	Accounting department	Accounting department	Unique identifier for tracking the refund transaction.
Refund Amount	Refund Amount		Float	None	Non-negative number	System	Accounting department	The total amount being refunded
Refund Timestamp	RefundTimestamp	Timestamp	None	Standard datetime format (YYYY-MM-DD HH:MM)	Payment Processing System	Accounting department	Accounting department	The time a refund request was processed.
Refund Status	RefundStatus	String (15)	"Pending"	Enum: Pending, Approved, Rejected, Processed	Payment Processing System	Confidential; View access for finance team	Billing Dept, Customer Service	Indicates the current status of a refund

Table 3. 29. Data flow dictionary of function Booking (Source: Author)

Label	**Description**	**Alternate Name(s)**	**Origin**	**Destination**	**Record**	**Volume and Frequency**
Choose My Profile / Add Other Profile	Selection of user profile for booking	N/A	User	Booking for me/for family	User Profile Record	Sent when selecting profile for booking
Request Show Information	Request to display personal information	N/A	Booking for me/for family	User Account	Personal Information Request	Sent upon user request for profile info

Personal Information	Personal data of the user or family member	N/A	User Account	Booking for me/for family	Personal Information Record	Retrieved when showing profile information
Select/Type Symptoms	Input or selection of symptoms by user	N/A	User	Select disease department	Symptom Input Record	Sent when symptoms are inputted
Show List Symptoms	Display of symptoms available for selection	N/A	Select disease department	User	Symptom List Record	Retrieved when user types/selects symptoms
Knowledge Base	Information source for symptoms and diseases	N/A	Diseases	Select disease department	Disease Knowledge Record	Accessed when selecting disease department
Choice Results Disease	Result of disease selection based on symptoms	N/A	Select disease department	User	Disease Selection Record	Returned after disease selection
Department Information	Information about hospital departments	N/A	Hospital	Select disease department	Department Information Record	Accessed when user selects department
Available Date	Dates available for booking	N/A	Hospital	Select time in day	Date Availability Record	Retrieved when selecting time
Show Time Available	Available times displayed based on date selection	N/A	Select time in day	User	Time Availability Record	Sent when user selects an available date
Date Info	Information about selected date	N/A	Select time in day	Hospital	Selected Date Record	Sent when user confirms booking date
Choice Results Time	Result of time selection	N/A	Select time in day	User	Time Selection Record	Returned after time is chosen
Information Detail for User	Detailed booking information for user	N/A	User	Create booking	Booking Information Record	Sent when creating a booking
Booking Detail	Comprehensive information for the booking	N/A	Create booking	Hospital	Complete Booking Record	Sent when creating a new booking

Response Booking Result	Result of booking process, including confirmation status	N/A	Hospital	Create booking	Booking Response Record	Returned after booking creation
Successful/Unsuccessful	Status of booking creation	N/A	Create booking	User	Booking Status Record	Returned to user after booking attempt
Appointment Validation	Validation of appointment details, including check-in info	N/A	Hospital	Appointment validation	Appointment Validation Record	Sent when validating an appointment
Patient Visit Information	Details for patient's visit, including QR code and queue info	N/A	Appointment validation	User	Patient Visit Record	Returned to user for appointment validation
Payment Record	Information about payment made for the appointment	N/A	Hospital	Appointment validation	Payment Transaction Record	Retrieved for appointment payment status
Payment Status	Status of payment (confirmed/unconfirmed)	N/A	Appointment validation	User	Payment Status Record	Returned to user after payment confirmation
Booking ID	Unique identifier for the booking	N/A	Cancel appointment	Hospital	Booking Identifier Record	Sent when user cancels a booking
Response Cancellation Result	Result of cancellation request	N/A	Hospital	Cancel appointment	Cancellation Response Record	Returned after cancellation request
Updated Booking Status	Updated status of the booking after cancellation	N/A	Cancel appointment	User	Updated Booking Status Record	Returned to user post-cancellation
Cancellation Request	Request to cancel an existing booking	N/A	User	Cancel appointment	Cancellation Request Record	Sent when user initiates cancellation
Cancellation Request Validation	Validation for booking cancellation	N/A	Cancel appointment	Hospital	Cancellation Validation Record	Sent for cancellation request validation

Booking ID, Payment Status	Information about booking ID and payment status	N/A	Hospital	Review booking	Booking and Payment Info Record	Sent when user wants to review booking
Review/Rating	User's review and rating for the booking	N/A	Review booking	Hospital	Review and Rating Record	Sent after the user provides feedback

Table 3. 30. Data store dictionary of function Booking (Source: Author)

Label	Description	Alternate Name(s)	Attributes	Volume and Frequency
Medical Record	Stores patient's medical history, treatments, notes		Patient Profile, Medical History	100,000 - 1,000,000 records (hospital-wide), updated per patient cycle

Table 3. 31. Data entity dictionary of function Booking (Source: Author)

Entity Name	Description	Input Data Flows	Output Data Flows
User	Represents the patient or user interacting with the system. They can make payments.	- Payment method.	- Payment status.
Bank/MoMo	Online payment system, processes transactions when the user pays via bank or MoMo e-wallet.	- Invoice ID. - Total amount. - Bank account information.	- Payment result (successful or failed).

i) User Interface

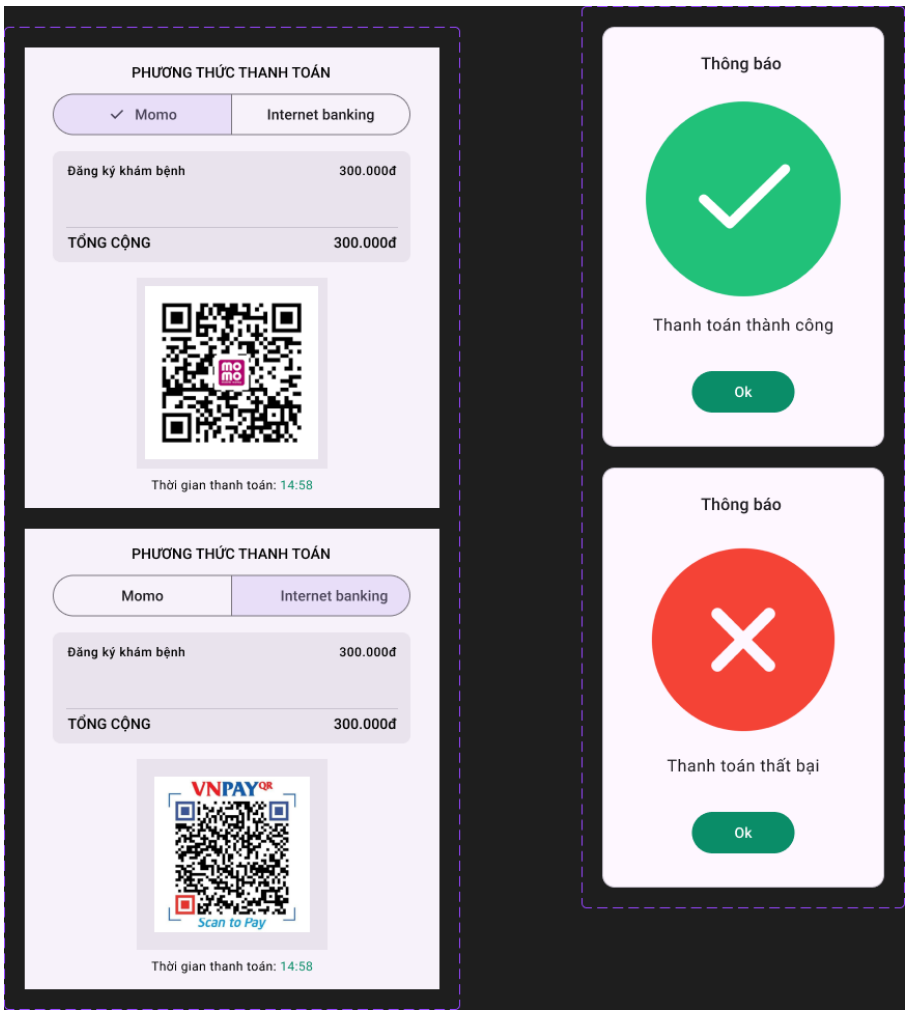


Figure 3. 37. User interface of Payment (Source: Author)

The figure shows the payment process within a healthcare application, with options for different payment methods and notifications for transaction outcomes. Here's a detailed description of each section:

1. **Payment Method Selection Screen**:
    - Users can choose between **MoMo** and **Internet Banking** as payment methods.
    - The screen displays the total cost of the appointment (300,000 VND).
    - A **QR code** is generated based on the selected payment method:

      MoMo QR Code for MoMo payments.

      VNPay QR Code for Internet Banking payments.
    - The payment timestamp is displayed at the bottom, indicating the time when the transaction was initiated.

2. **Payment Success Notification**:
    - A green notification screen confirms a successful payment.
    - The screen features a checkmark icon with the message “Payment Successful” and an **Ok** button to dismiss the notification.
3. **Payment Failure Notification**:
    - A red notification screen indicates a failed payment.
    - This screen displays an “X” icon with the message “Payment Failed” and an **Ok** button to close the notification.

j) Validation

Table 3. 32. Data validation of Payment function (Source: Author)

Failed Validation	Error Message Displayed
Payment considered successful when it is confirmed by hospital in 15 minutes	Payment is unsuccessful

3.4.5. View Medical Record

a) Function introduce

Function meaning

This function displays patient information recorded in the hospital's system after an examination. Details about the patient, the examination process, and their medical condition are recorded and updated almost immediately after the examination, making it easily accessible for the patient.

This function not only stores the most recent medical records but also retains all previous records, even those from before the patient started using the app, providing a comprehensive view of their health history. Users can manage and view the medical records of all patients they've added to their account in one place.

Business rules

The business rule for this function is the user must be logged in to view their medical records and only their medical records.

b) BPMN

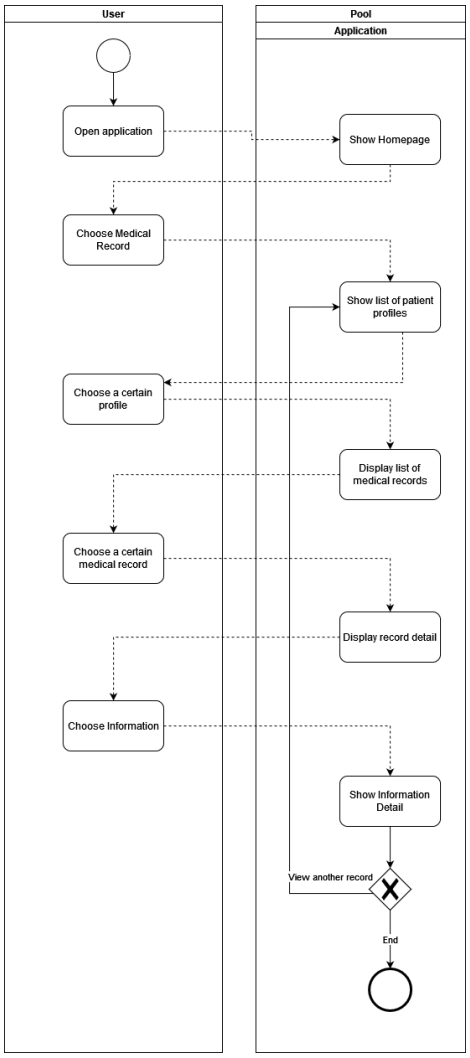


Figure 3. 38. BPMN of Medical Record Access process (Source: Author)

The **BPMN of Medical Record Access process** depicts the steps a user follows to access and view medical records:

Application Access: The user opens the application and reaches the homepage.

Navigate Records: The user selects the "Medical Record" option and chooses a patient profile.

View Records: After selecting a profile, a list of medical records is displayed. The user can select a specific record to view detailed information.

Detailed Information: The user can choose to view additional details within the record.

End Process: The process ends or allows the user to view another record.

c) DFD

- Users request access to their medical records.

- The system verifies the request, retrieves the records from the database, and displays the information.

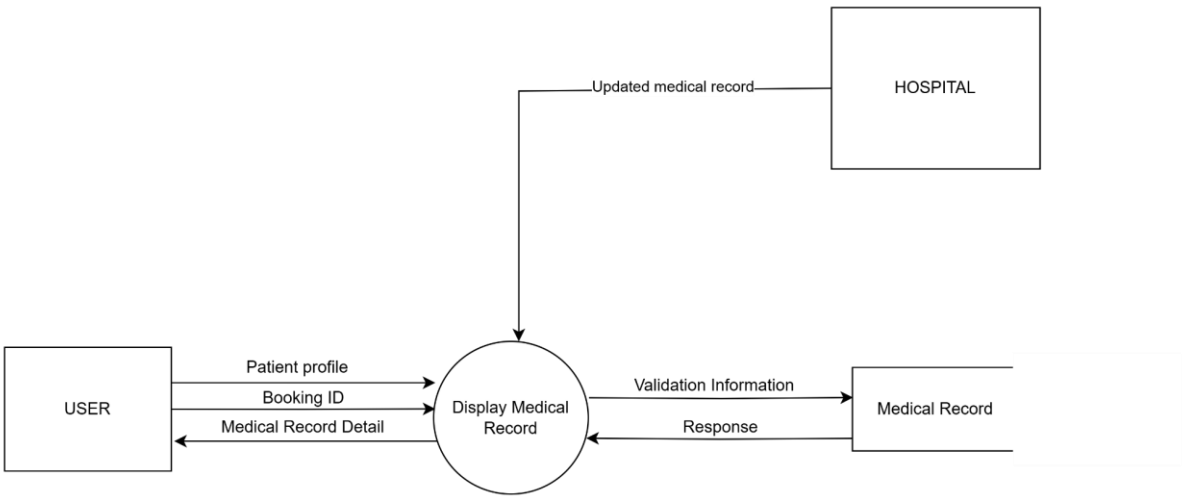


Figure 3. 39. Medical Record Access process DFD level 1 (Source: Author)

The **Medical Record Access process DFD level 1** illustrates how a user accesses medical record information:

1. **User Interaction**: The user provides a patient profile and booking ID to request medical record details.
2. **Display Medical Record**: The system retrieves and displays the medical record details based on the provided information.
3. **Hospital System Update**: The hospital system updates the medical record information, which is then reflected in the displayed data.
4. **Validation with Medical Record Database**: The system validates information with the medical record database, ensuring accuracy before providing a response back to the user.

d) Use case

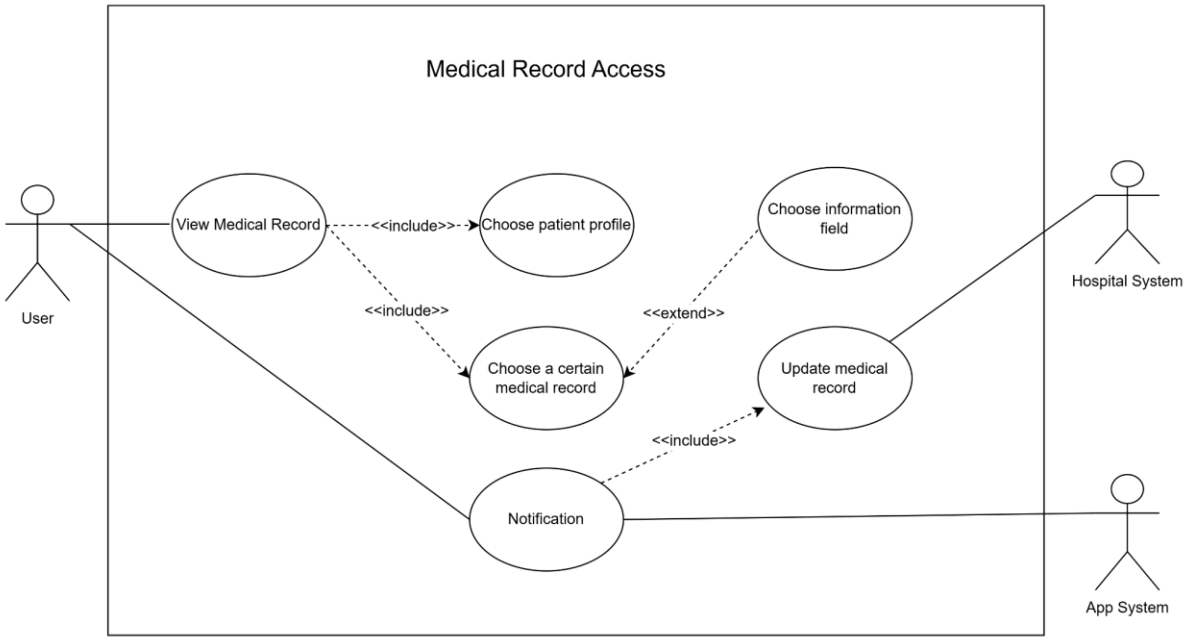


Figure 3. 40. Use case diagram of Medical record access (Source: Author)

The **Use Case Diagram for Medical Record Access** shows how a user, the hospital system, and the app system interact to access and update medical records:

View Medical Record: The user initiates the process to view a medical record, which includes selecting a patient profile.

Choose Specific Medical Record: The user can choose a specific medical record and view detailed information fields.

Update Medical Record: The hospital system allows for updates to the medical record, ensuring information is current.

Notification: Notifications are included to keep the user informed of any updates or necessary actions.

Table 3. 33. Use case description of Medical record access process (Source: Author)

Use Case Name	Medical record access

Use Case No.	06	
Primary Actor	User	
Secondary Actor	Hospital system, app system	
Use Case Story	This use case allows the patient to access their medical records through the system. The user can view their complete profile, select specific medical records, and view details of certain fields within the record.	
Trigger	The user initiates a request to view their medical records.	
Pre-Condition	The user is authenticated in the application. Internet connection is active. The medical records are stored in the system database.	
Post-Condition	The user can view their selected medical records. Any updates made to the records are stored in the database. The system logs all access and updates actions for audit purposes.	
Primary Flow (PF)	**Title:** User Accesses Medical Record	
	Actor Action	**System Response**
	1. The user initiates the "Xem hồ sơ bệnh án" process.	2. The app system provides a list of profiles.
	3. The user selects a specific profile to view.	4. The app system provides a list of available medical records.
	5. The user selects a specific medical record to view.	6. The system displays the selected medical record,

		allowing the user to choose detailed information fields, such as diagnosis, treatment history, or lab results.
Alternate Flow 1 (AF1)	**Title:** Choosing Information Field	
	Actor Action	**System Response**
	1. The user selects specific information fields within the record.	2. The app system displays the selected field details for in-depth review.
Alternate Flow 2 (AF2)	**Title:** Updating Medical Record (Hospital system only)	
	Actor Action	**System Response**
		1. The hospital system updates new medical records.
		2. The app system displays an notification message to the user: "Bệnh án của bạn đã được cập nhật"
	7. Go back to step 1 of PF.	

e) User story

Table 3. 34. User Story of View Medical Record process

View medical record

As a	user of the Mobile Appointment Scheduling System.
I want	to easily access their medical records through the application.
So that	I can review my complete profile and specific medical details at their convenience.
Acceptance Criteria	1. When the user opens the app, they see a prominent "Xem hồ sơ tế" button on the home screen. Clicking this button, the user anticipates the system to load quickly and display a list of their available medical records, allowing for easy navigation. 2. After reviewing the list, the user selects a specific record by clicking on the desired record. The user expects the system to display the selected medical record, including details like diagnosis, treatment history, and lab results, in a clear and organized format. 3. The user wants the option to expand specific sections of the record by clicking on tabs labeled "Lịch Sử Khám Bệnh," "Kết Quả Xét Nghiệm," and "Thuốc Đang Sử Dụng." Upon selecting any tab, the user expects the system to present the relevant information without delay. 4. After any updates to their medical records by the hospital system, the user anticipates seeing an alert pop-up or banner message within the app that states, "Bệnh án của bạn đã được cập nhật," ensuring they are promptly informed about any changes. 5. The user expects that all actions taken in the app, such as viewing or updating records, are securely logged. They would like to see a confirmation message after each access, such as "Thông tin truy cập

	đã được ghi lại," to provide reassurance about the security of their personal health information.

f) Sequence Diagram

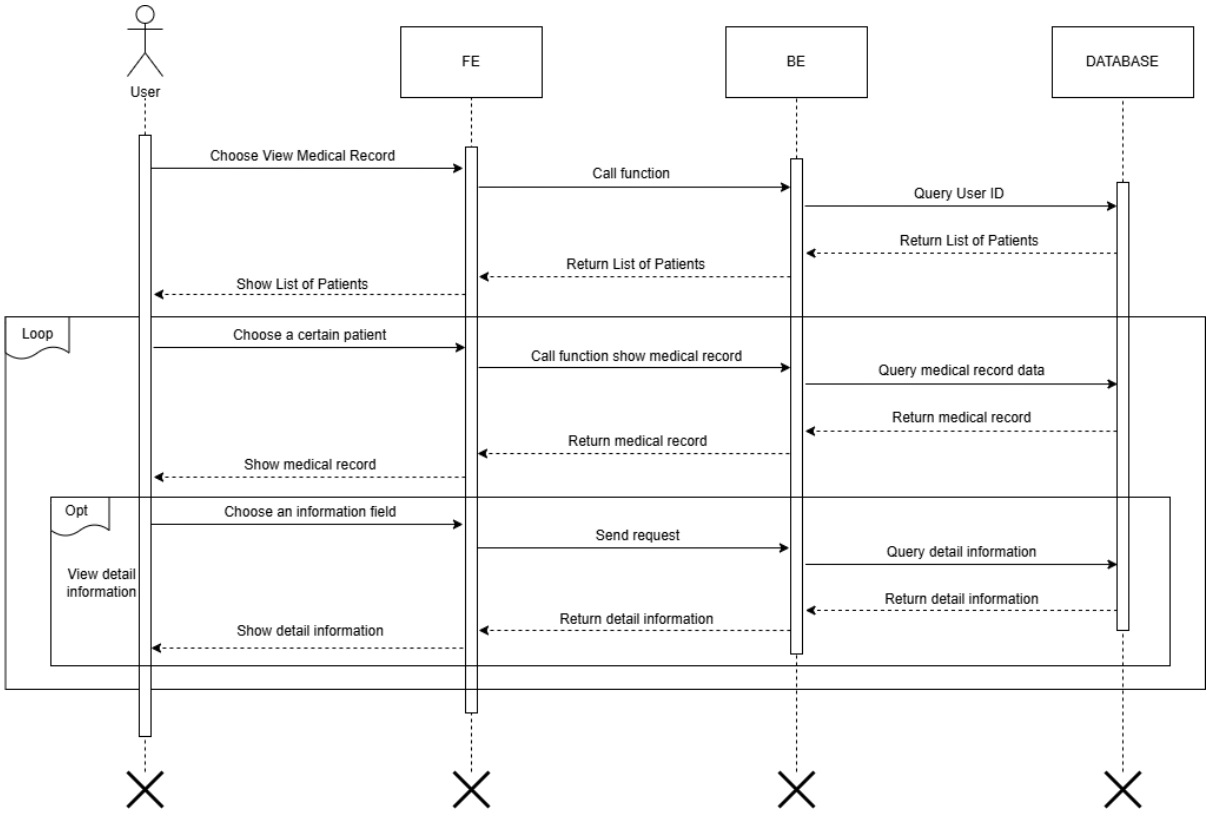


Figure 3. 41. Sequence Diagram of View Medical Record (Source: Author)

The **Sequence Diagram for Viewing a Medical Record** outlines the interactions between the user, front-end (FE), back-end (BE), and database systems in the process of retrieving a medical record.

Initiate View Request: The user selects the "View Medical Record" option, prompting the front-end to call a function that retrieves a list of patients.

Select Patient: The user chooses a specific patient from the list. The front-end then requests the medical record data from the back-end.

Retrieve Medical Record: The back-end queries the database for the medical record data, which is then returned to the user through the front-end.

View Detailed Information: The user can choose specific information fields within the medical record for further details, which are retrieved from the database and displayed to the user.

g) Activity Diagram

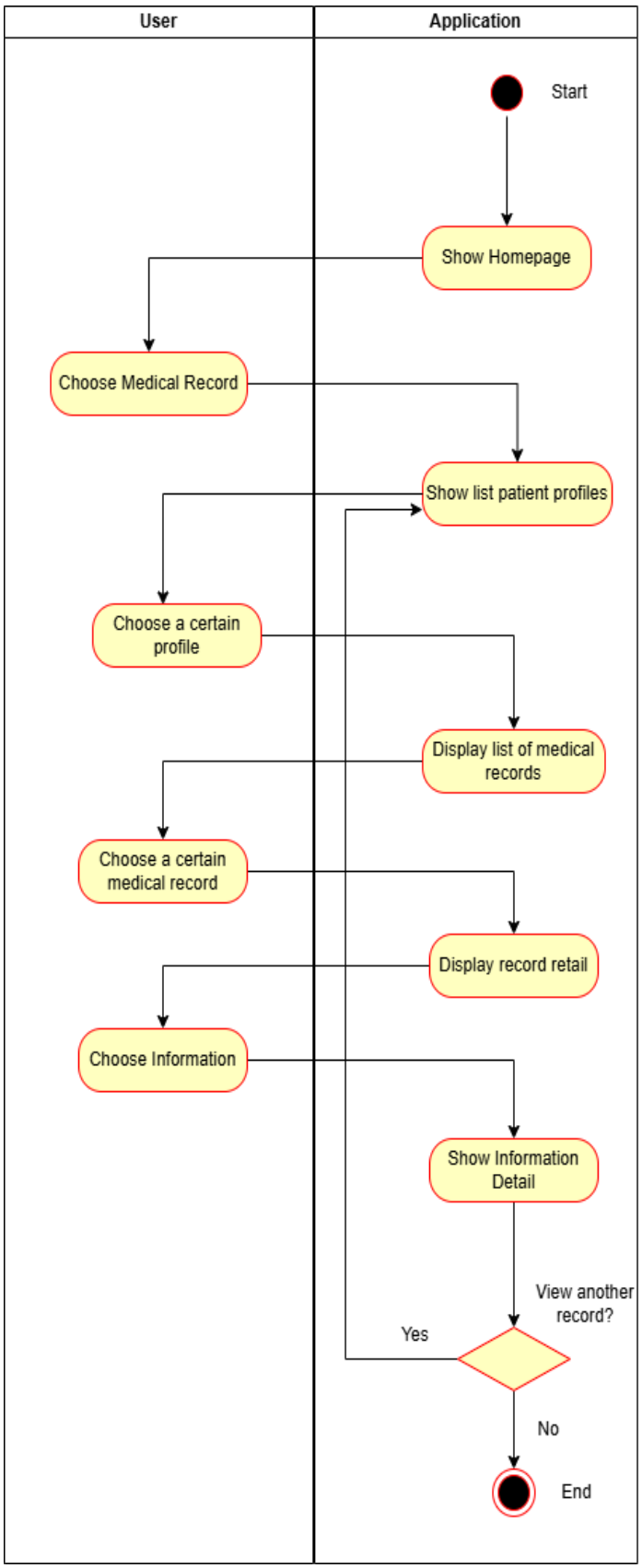


Figure 3. 42. Activity Diagram of View Medical Record (Source: Author)

The **Activity Diagram of View Medical Record** outlines the steps for a user to access and view a medical record in the system:

1. **Choose Medical Record**: The user starts by selecting the option to view medical records.
2. **Select Profile**: The user chooses a specific patient profile.
3. **Choose Medical Record**: The system displays a list of medical records for the selected profile, and the user selects a specific record.
4. **View Information**: The user can view detailed information within the record.
5. **Loop Option**: The user is prompted to view another record, and the process repeats if desired; otherwise, the session ends.

h) Data Dictionary

Table 3. 35. Data element dictionary of function Medical Record (Source: Author)

Label	Alias	Type & Length	Default Value	Acceptable values	Source	Security	Responsible User(s)	Description/Comments
Medical Record ID		String (12 characters)	None	12-character string start with MR__	Hospital database	Hospital Information System	Hospital Information System Hospital Database Aministrat or istrators	Unique identifier for the medical record
Chronic Conditions	Overview of patient' health condition	String (5000 characters)	None	Any text about health conditions that are specified in hospital database	Hospital database	Hospital Information System	Hospital Information System Hospital Database Aministrat or istrators	Details about ongoing health issues

Disease	Medical Condition	String (255 characters)	None	Specific disease names	User Input	Sensitive Data (encrypted)	Doctor, Medical Staff	Name of the disease diagnosed for the patient.
Test Result	Diagnostic Outcome	Text (5000 characters)	None	Descriptive results	Lab Reports	Sensitive Data (encrypted)	Doctor, Medical Staff	Results from medical tests conducted on the patient.
Medical Record Summary	Health Overview	Text (10000 characters)	None	Detailed summary	Medical Record	Sensitive Data (encrypted)	Doctor, Medical Staff	Summary of the patient's medical history, diagnosis, and key information.
Treatment Progress	Patient Status	Text (500 characters)	None	Progress details	Doctor's Notes	Sensitive Data (encrypted)	Doctor, Medical Staff	Description of the patient's response to treatment and their recovery progress.
Doctor's Advice	Medical Recommendations	Text (5000 characters)	None	Recommendations	Doctor's Notes	Sensitive Data (encrypted)	Doctor	Recommendations or instructions provided by the doctor for further care or medication.

Past Medical Events	Patient' medical history	String (5000 characters)	None	Any text about patient's medical history	Hospital database	Hospital Information System	Hospital Information System Hospital Database Aministrator istrators	Significant illnesses, surgeries, or treatments the patient has undergone in the past.
Prescription ID		String (20)	Auto-generated	Alphanumeric; unique identifier	Pharmacy System	Public	Doctors, Pharmacists	Unique identifier for each prescription pharmacy records.

Table 3. 36. Data flow dictionary of function Medical Record (Source: Author)

Label	**Description**	**Alternate Name(s)**	**Origin**	**Destination**	**Record**	**Volume and Frequency**
Check in QR Code	Pateint provide Check in QR Code to scan		User	Scan QR Code process	QR Code, Booking ID	Verifies patient's presence at the facility.
Appointment Information	Provide Appointment Information to update status		Booking record	Update status Process	Booking ID, Doctor, Time, Patient ID, Name	Updates appointment information after booking is confirmed.
Payment Status to update process	Status of payment was sent to update checkup status		Payment record data store	Update status Process	Payment record status	Used to verify if payment has been completed for check-up.
Payment Status to hospital	Status of payment was sent back to the hospital system to		Payment record data store	Update status Process	Payment record status	Confirms payment before scheduling next check-up.

	confirm the next check-up					
Validation checkup status	Update validation check up status to data store		Update status Process	Checkup list data store	Check-up validation status	Validates patient's eligibility for the next check-up.
Updated checkup status	Receive updated checkup status		Checkup list data store	Update status Process	Updated check-up information	Confirms that the patient's check-up status is current.
Unpaid checkup list	Provide unpaid checkup list to notify user		Checkup list data store	Payment notification process	List of unpaid Booking ID	Identifies checkups pending payment.
Payment Notification	Notification sent to the user to notify about unpaid checkup list		Payment notification process	User	Booking ID, Payment amount	Reminds user of pending payments for checkups.
Information to queue number creation	Sent information to create queue number		User	Queue number notification Process	Patient ID, Prescription ID	Provides queue position for patient based on prescription ID.

Table 3. 37. Data store dictionary of function Medical Record (Source: Author)

Label	Description	Alternate Name(s)	Attributes	Volume and Frequency
Medical Record	Stores patient's medical history, treatments, notes		Patient Profile, Medical History	100,000 - 1,000,000 records (hospital-wide), updated per patient cycle

Table 3. 38. Data process of function Medical Record (Source: Author)

Process Name	Description	Process Number	Input Data Flows	Output Data Flows	Process Description
Display Medical Record	Retrieves and displays the patient's medical record for review	5.1	Patient profile, Booking ID, Validation Information	Medical Record Detail, Response	Allows users to view the details of their medical record after validating the necessary credentials and inputs.

Table 3. 39. Data entity of function View Medical Record (Source: Author)

Entity Name	Description	Input Data Flows	Output Data Flows
User	Represents the patient or user interacting with the system. They can view medical record	- Select view medical record	- Medical record information.
Hospital	Manager Medical Record	- User account information. - Appointment information (patient ID, patient profile).	- Medical checkup status.

i) User Interface

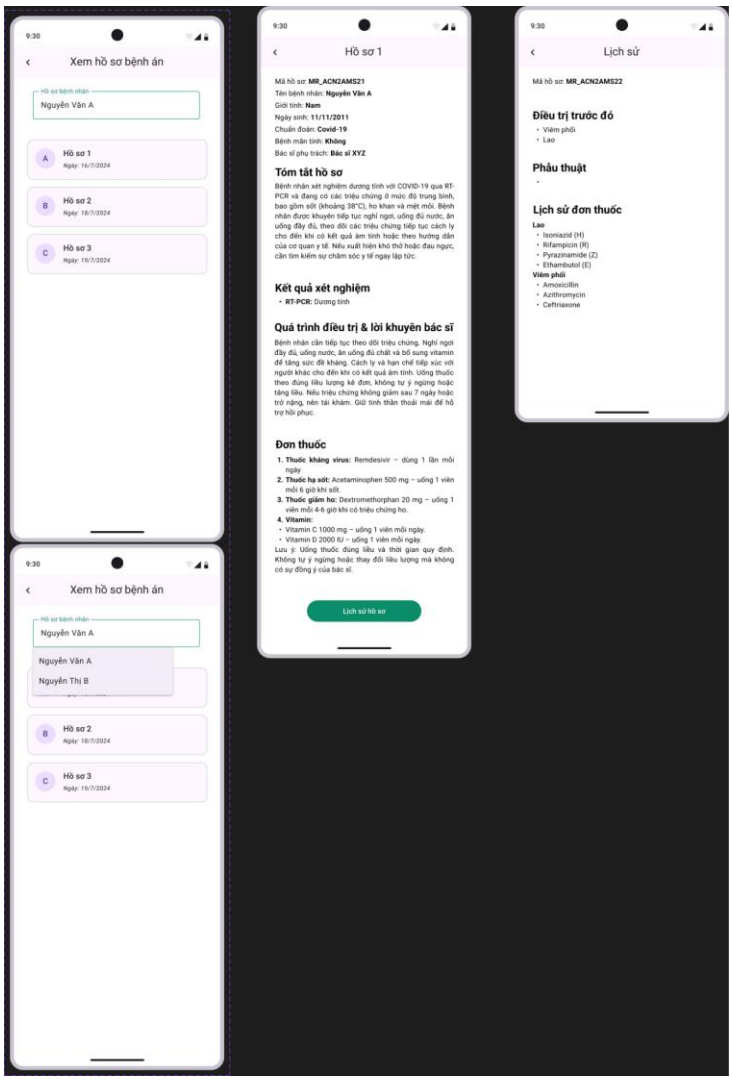


Figure 3. 43. User interface of viewing medical record (Source: Author)

The interface in this image demonstrates the process of viewing and selecting medical records within a healthcare application. Here's a concise description:

Medical Record Selection Screen:

- Users can view and select from a list of medical records associated with a patient, each labeled with a unique identifier and visit date.
- A dropdown menu allows switching between profiles to access records for different patients.

Medical Record Details Screen:

- Displays detailed patient information, including name, gender, date of birth, and diagnosis.
- Sections include:
  - **Record Summary**: Overview of the diagnosis and symptoms.
  - **Test Results**: Specific test findings.
  - **Treatment Process**: Description of the prescribed treatment plan.
  - **Medication List**: Detailed prescription with dosages and instructions.

j) Validation

Table 3. 40. Data validation of View medical record function (Source: Author)

Failed Validation	**Error Message Displayed**
Patient has an existing profile and has prior visit history	User won't be able to view medical record because system doesn't display

3.4.6. Tracking

a) Function introduce

Function meaning

The function allows patients to monitor the progress of their medical examination at the hospital in real time. If the examination process requires additional tests, blood work, prescriptions, or payments, patients can follow all these steps through this feature.

This function ensures patients stay informed about each stage of their examination, including the next steps and the departments they need to visit, reducing wait times and enhancing the overall quality of medical services

Business rules

The business rule for this function is users can update their progress by scanning the qr at the hospital. When need to have additional payment, the application will guide the user in the payment process.

b) BPMN

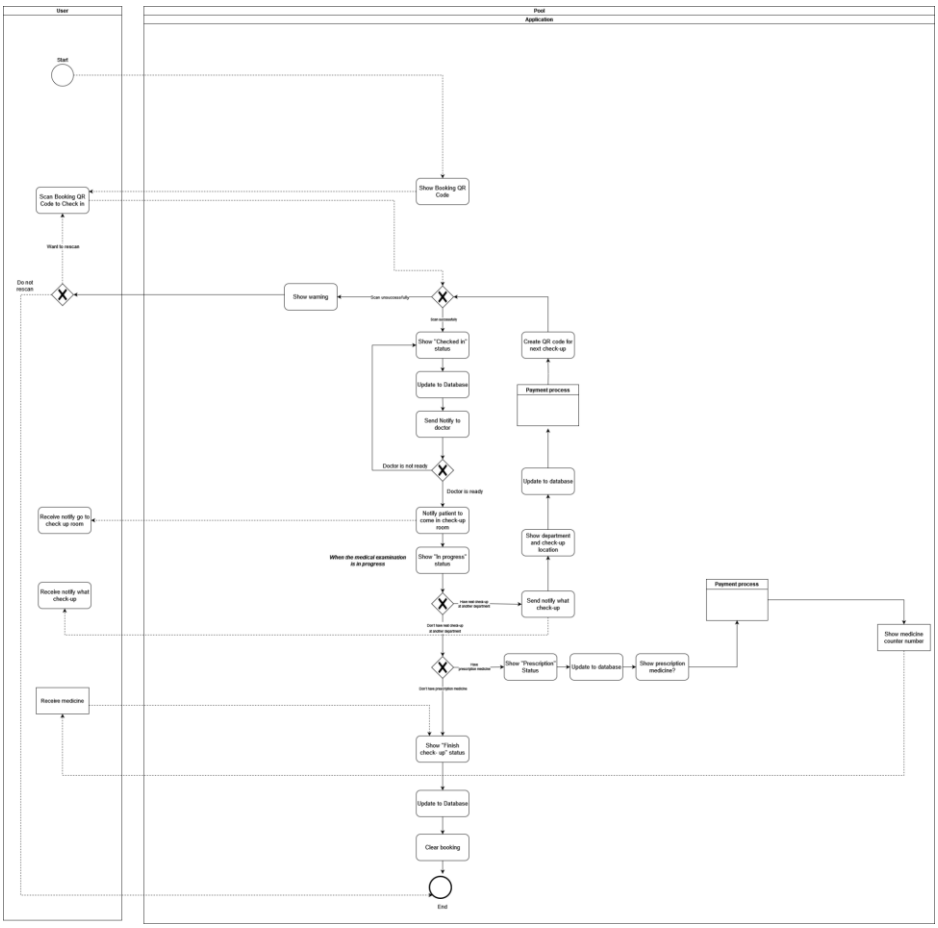


Figure 3. 44. BPMN of Check-up and Hospital Interaction process (Source: Author)

The **BPMN of the Check-up and Hospital Interaction Process** reveals the steps involved in a patient's check-up process at the hospital:

Check-in: The user scans a QR code to check in for the appointment.

Status Update: If the doctor is not ready, the system shows a waiting notification. Once the doctor is ready, the user is notified.

Check-up Progress: The patient is notified of their progress through the check-up stages.

Payment Process: If additional payments are required, a QR code is generated, and payment is processed.

Completion: After the check-up, the system updates the status, shows the prescription, and provides medicine collection details.

c) DFD

- Users check in using a QR code at the hospital.

- The system updates the appointment status, provides the queue number, and sends notifications regarding the next steps in the examination process.

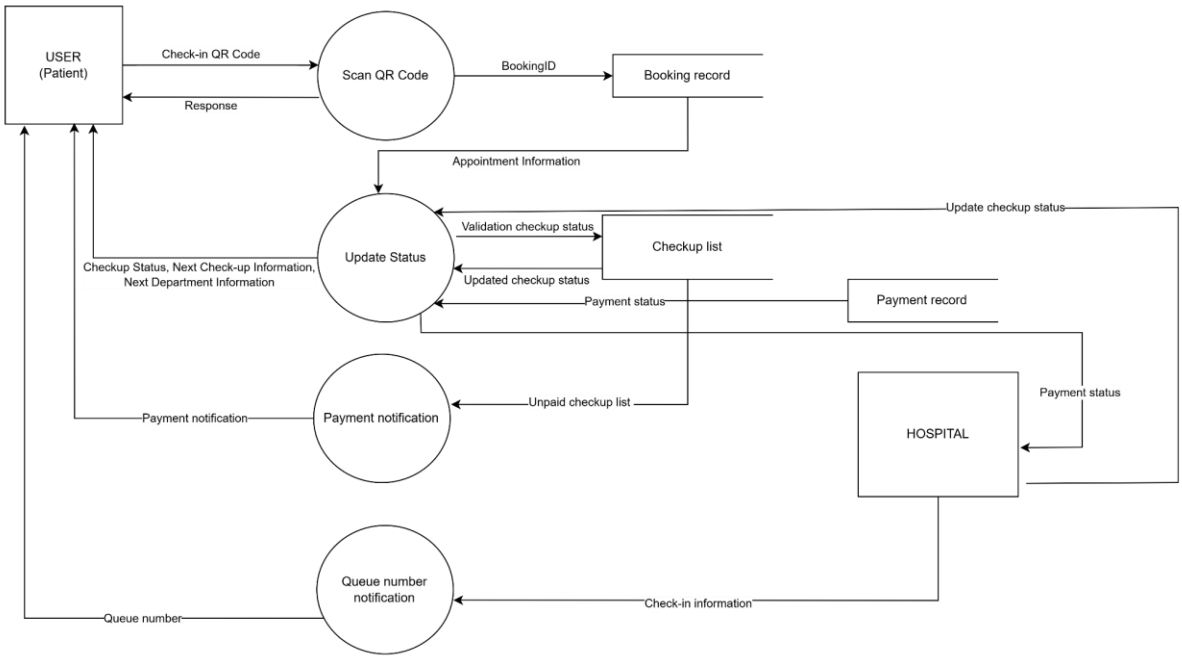


Figure 3. 45. Check-up and Hospital Interaction process DFD level 1 (Source: Author)

The **DFD Level 1 of the Check-up and Hospital Interaction Process** shows the flow of information between the patient, hospital system, and related entities:

1. **Scan QR Code**: The patient scans a QR code to check in, sending the booking ID to retrieve booking records and appointment information.
2. **Update Status**: The system validates the check-up status and updates it accordingly, providing the patient with current check-up status, next steps, or department information.
3. **Payment Notification**: If there are unpaid fees, the system notifies the patient of payment requirements and updates payment status upon completion.
4. **Queue Number Notification**: The patient receives a queue number and check-in confirmation to manage the flow within the hospital.

5. **Interaction with Hospital System**: The hospital system receives updated check-up and payment statuses, ensuring seamless coordination and record-keeping across departments.

d) Use case

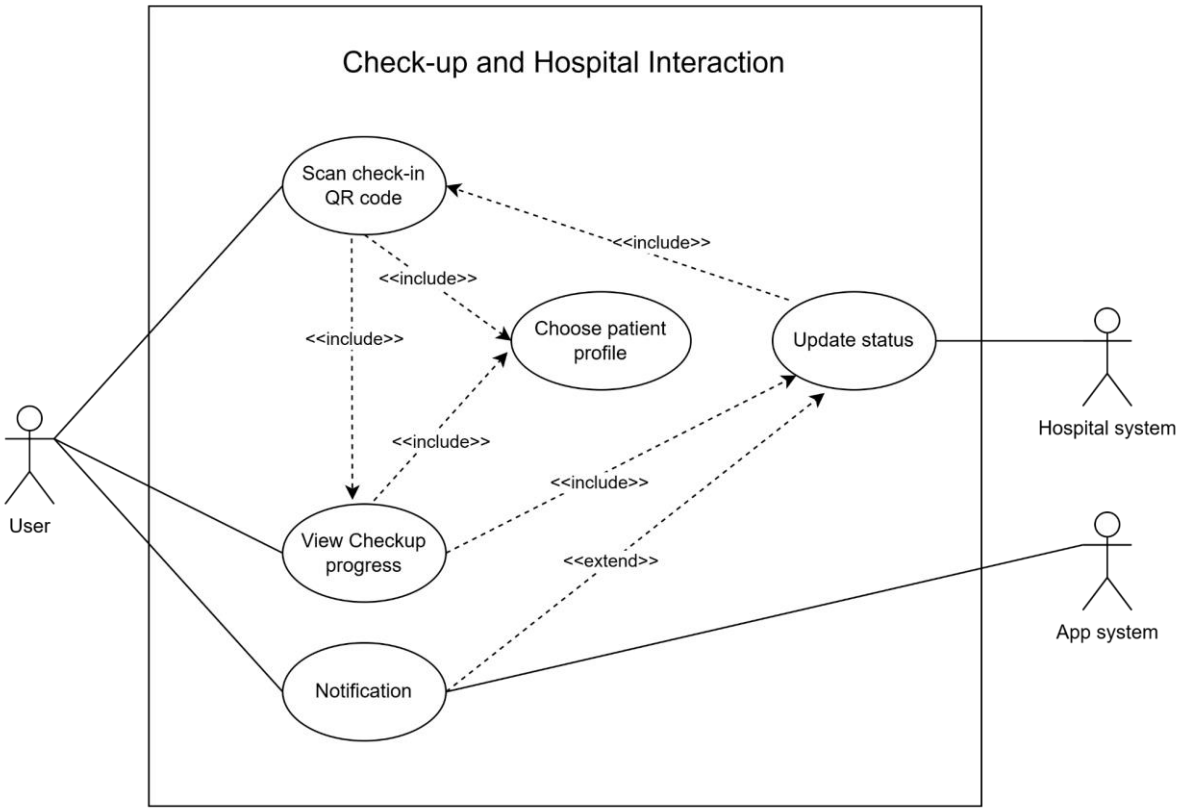


Figure 3. 46. Use case diagram of Check-up and Hospital interaction (Source: Author)

The **Use Case Diagram for Check-up and Hospital Interaction** outlines the main interactions between a user (patient) and the hospital and app systems during a check-up process:

Scan Check-in QR Code: The patient scans a QR code to initiate the check-in process.

Choose Patient Profile: The user selects the relevant patient profile for the check-up.

Update Status: The system updates the check-up status, interacting with the hospital system.

View Checkup Progress: The patient can view the current progress of their check-up.

Notification: Notifications are sent to the user, keeping them informed about their check-up status and any required actions.

Table 3. 41. Use case description of Check-up and Hospital interaction process (Source: Author)

Use Case Name	Check-up and Hospital interaction
Use Case No.	07
Primary Actor	User
Secondary Actor	Hospital system, app system
Use Case Story	This use case describes the process of a patient checking in for a hospital appointment, waiting in a queue, consulting with a doctor, making a payment, and checking out.
Trigger	Patient arrives at the hospital for their scheduled appointment and scans the first check-in QR code.
Pre-Condition	User has installed the app and logged in successfully. User’s device has a stable internet connection. User has previously scheduled an appointment. Hospital database is connected and operational. User’s device meets app requirements. Device's camera is functional for scanning QR codes.
Post-Condition	System updates appointment status to "Completed." Notifications about examination results and follow-up instructions are sent to the user.

<table>
<tr><td></td><td colspan="2">User activity is logged for security and analysis.
Users can track their medical history and upcoming appointments in the app.</td></tr>
<tr><td rowspan="8">Primary Flow (PF)</td><td colspan="2">Title: User does check-up process</td></tr>
<tr><td>Actor Action</td><td>System Response</td></tr>
<tr><td>1. User access check-up and hospital interaction process</td><td>2. The system provides a list of profiles.</td></tr>
<tr><td>3. The user selects a specific profile to view.</td><td>4. System shows a check-up process.</td></tr>
<tr><td>5. User scans the QR code in the examination room for tracking purposes.</td><td>6. The system logs the QR code scan to confirm the patient’s presence in the examination room.</td></tr>
<tr><td rowspan="2">7. Users finish the current examination.</td><td>8. The hospital system updates new check-up steps for the check-up lists.</td></tr>
<tr><td>9. System displays an notification message to the user: "Quy trình khám sức khỏe của bạn đã cập nhật"</td></tr>
<tr><td>10. Users receive notifications about the next steps in the</td><td></td></tr>
</table>

	check-up process.	
Alternate Flow 1 (AF1)	**Title:** Invalid QR Code	
	Actor Action	**System Response**
	1. The patient scans the QR code, but it is invalid.	2. The system displays an error message: "Mã không hợp lệ. Vui lòng thử lại hoặc tạo mã mới."
	3. The patient requests a new QR code.	4. The system generates a new QR code and sends it to the patient.

e) User story

Table 3. 42. User Story of Check-up and Hospital interaction (Source: Author)

Check-up and Hospital interaction	
As a	user of the Mobile Appointment Scheduling System.
I want	to complete the check-up process smoothly within the application.
So that	I can efficiently manage my hospital visit and receive timely notifications about my health status.

Acceptance Criteria	1. When the user arrives at the hospital and opens the app, they choose “Quản lí khám bệnh” and choose the specific profile 2. User see a "Bắt đầu quy trình Khám" button prominently displayed on the home screen. Upon clicking this button, the system should initiate the check-up process and display the relevant steps clearly. 3. After scanning the check-in QR code, the user expects a confirmation message indicating their presence in the examination room. The system should log this action seamlessly without any noticeable delay. 4. Once the examination is completed, the user should receive a notification that states, "Quy trình khám sức khỏe của bạn đã cập nhật," confirming that the next steps are now available. 5. If the user encounters an invalid QR code, they should see an error message saying, "Mã không hợp lệ. Vui lòng thử lại hoặc tạo mã mới," and the option to request a new QR code should be straightforward and efficient. 6. All user actions within the check-up process, including QR code scans and appointment updates, should be securely logged by the system. The user should receive a confirmation message after each significant action, such as "Hoạt động của bạn đã được ghi lại," to ensure the security of their health information.

f) Sequence Diagram

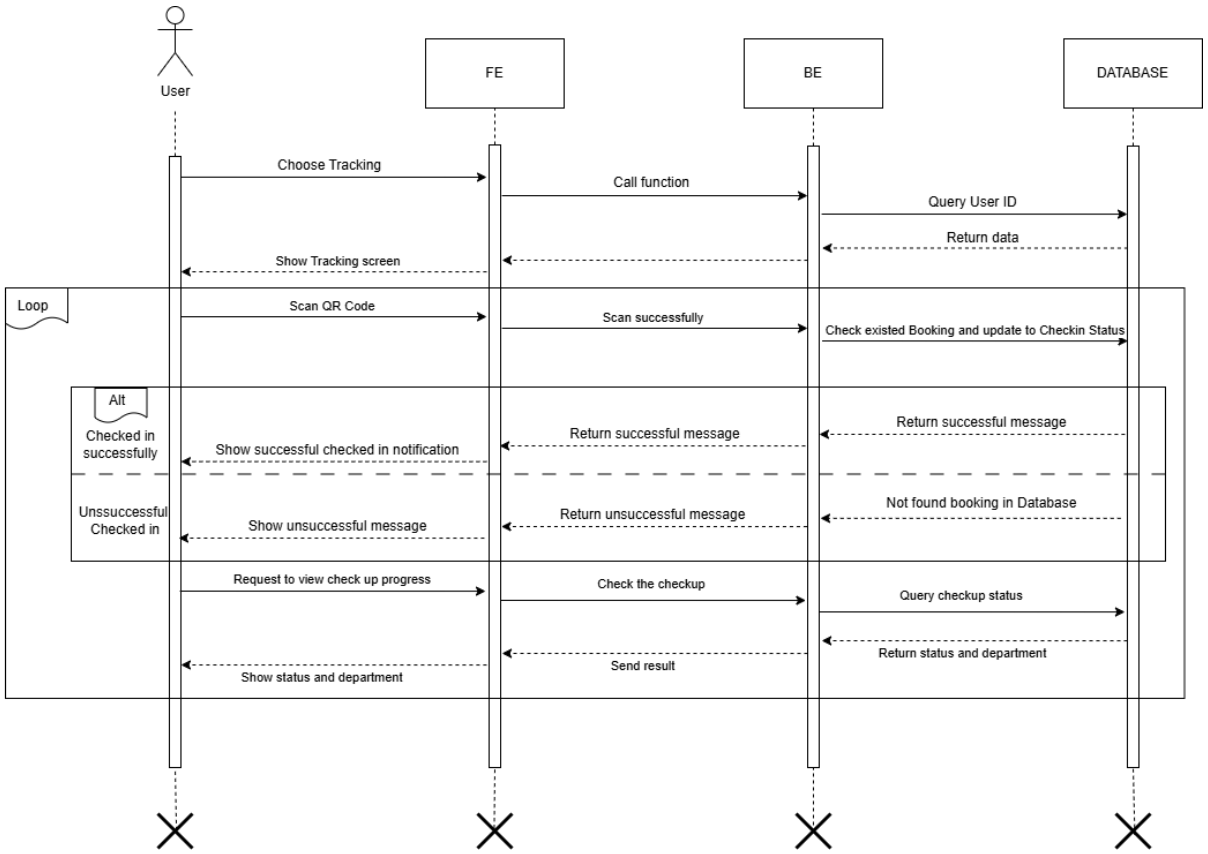


Figure 3. 47. Sequence Diagram of function Tracking (Source: Author)

The **Sequence Diagram of the Tracking Function** mentions the user interaction with the tracking feature for check-in at a hospital:

Choose Tracking: The user selects the tracking option, and the front-end (FE) displays the tracking screen.

Scan QR Code: The user scans a QR code, triggering the system to verify and update the booking's check-in status.

Check-In Status: Based on the verification, the system sends either a successful or unsuccessful check-in notification.

View Checkup Progress: The user can request to view the check-up progress, with the system querying the check-up status and returning the relevant information, including status and department.

g) Activity Diagram

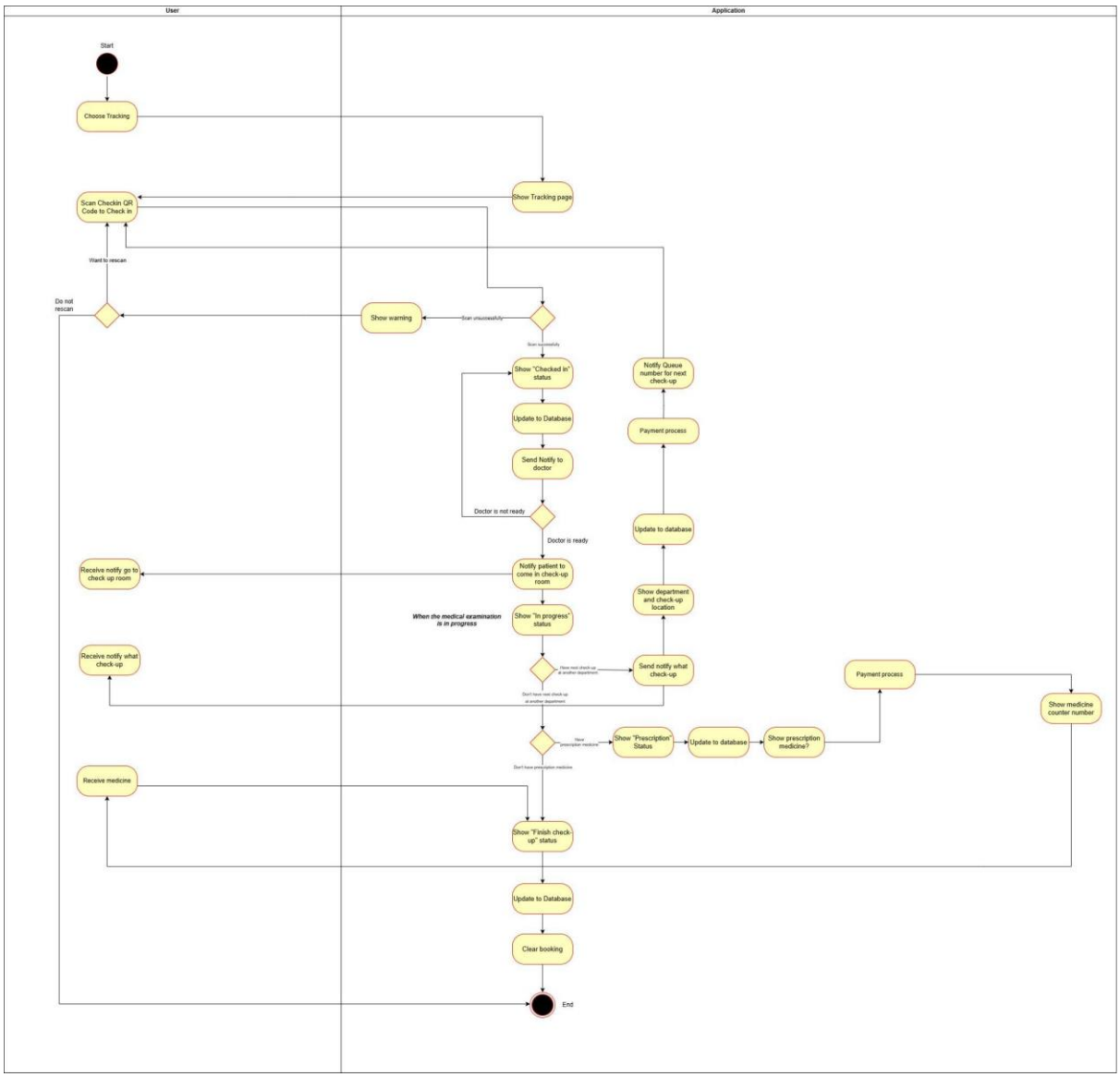


Figure 3. 48. Activity Diagram of Tracking (Source: Author)

The **Activity Diagram of Tracking** presents the steps a user follows for tracking a hospital check-up:

Choose Tracking: The user initiates tracking.

Scan QR Code: The user scans a check-in QR code, and the system verifies it.

Check-In and Notifications: Depending on availability, the system updates the check-in status, notifies the user of their queue, and shows relevant updates as the check-up progresses.

Doctor Readiness and Status Updates: The user receives notifications as the doctor becomes ready, and check-up stages are completed.

Prescription and Medicine Pickup: The system displays prescription details and provides directions for medicine collection, completing the tracking process.

h) Data Dictionary

Table 3. 43. Data element dictionary of function Tracking (Source: Author)

Label	Alias	Type & Length	Default Value	Acceptable values	Source	Security	Responsible User(s)	Description/Comments
Check-up ID	CheckUpID	String (20)	Auto-generated	Alphanumeric; unique identifier	Health System	Confidential; View access for authorized medical staff	Doctors, Nurses	Unique identifier for each check-up.
Check-up Name	CheckUpName	String (50)	None	Standard names for check-up types	Medical Staff Input	Public; View access for authorized medical staff	Doctors, Nurses	Descriptive name of the check-up, such as "Annual Physical" or "Blood Pressure Screening,"
Check-in QR Code	Check-in Code	QR code	None	Valid QR code format	System-generated	Only user and Hospital Database Aministrator can view	Reception Department	QR code for check-in during the visit
Checkup Status	Status	String (15)	"Pending”	Pending, Completed,	User Input	Masked	Medical Staff, Reception	Represents the status of a patient's medical checkup.
Queue Number	Queue ID	Numeric (5 digits)	None	Valid queue number	System-generated	Only user and Hospital Database Aministrator can view	Reception Department	User's queue number for the medical visit

Table 3. 44. Data flow dictionary of function Tracking (Source: Author)

Label	Description	Alternate Name(s)	Origin	Destination	Record	Volume and Frequency
Check-in information	Provide check-in information to create queue number		Hospital	Queue number notification Process	Patient ID, Booking ID, doctor, time	Per appointment, when patient arrives
Queue number	Create queue number notification for patient		Queue number notification Process	User	Patient ID, Booking ID, doctor, time	Per check-in, updated in real-time

Table 3. 45. Data store dictionary of function Tracking (Source: Author)

Label	Description	Alternate Name(s)	Attributes	Volume and Frequency
Booking Record	Stores the record of the booking, including booking details, payment status, and refund history.	Appointment Data Store	Booking ID, Booking Date, Time Slot, Payment Status	50,000 - 500,000 records, updated every time a booking is created, modified, or payment status changes.
Payment Record	Stores all payment-related details		Payment status, Invoice ID, Total amount	5,000 - 50,000 records, updated per completed transaction
Medical Record	Stores patient's medical history, treatments, notes		Patient Profile, Medical History	100,000 - 1,000,000 records (hospital-wide), updated per patient cycle
Checkup list	Contains the list of checkups scheduled for the patients	Appointment List	Checkup ID, Patient ID, Doctor ID, Checkup Date, Status	100,000 - 1,000,000 records Updated daily, accessed frequently

Table 3. 46. Data process of dictionary of Tracking (Source: Author)

Process Name	Description	Process Number	Input Data Flows	Output Data Flows	Process Description
Scan QR Code	Verifies the user's credentials by scanning a QR code.	6.1	QR Code Data	Verification Result	The system reads the QR code and verifies the information for identification or access purposes.
Update Status	Updates the status of a medical appointment or payment.	6.2	Status Update Request, Payment Confirmation, Appointment Status	Updated Status	Updates the status of either a payment or appointment after an action is performed.
Payment Notification	Sends a notification to the user about the payment need to pay.	6.3	Payment Confirmation	Payment Notification	Notifies the user about successful or failed payment transactions.
Queue Number Notification	Sends a notification with the user's queue number for their appointment.	6.4	Appointment Details, Queue Number	Queue Number Notification	The system informs the user of their queue number, allowing them to track their appointment time.

Table 3. 47. Data entity dictionary of Tracking (Source: Author)

Entity Name	Description	Input Data Flows	Output Data Flows

User	Represents the patient or user interacting with the system. They can sign up for an account, log in, schedule appointments, and make payments.	- Personal information for Sign Up (name, phone number, email, citizen ID). - Phone number and OTP for Login. - Appointment information (patient profile, symptoms, time, doctor). - Payment method.	- Sign Up result (successful or failed). - Login result (successful or failed). - Invoice. - Appointment result (QR code, appointment information). - Medical record information. - Payment status.
Hospital	Manages user accounts, medical records, and appointment statuses. The hospital responds to appointment bookings and payments.	- User account information. - Appointment information (patient ID, patient profile).	- Update user account. - Update medical records. - Appointment result, appointment status. - Medical checkup status.

i) User Interface

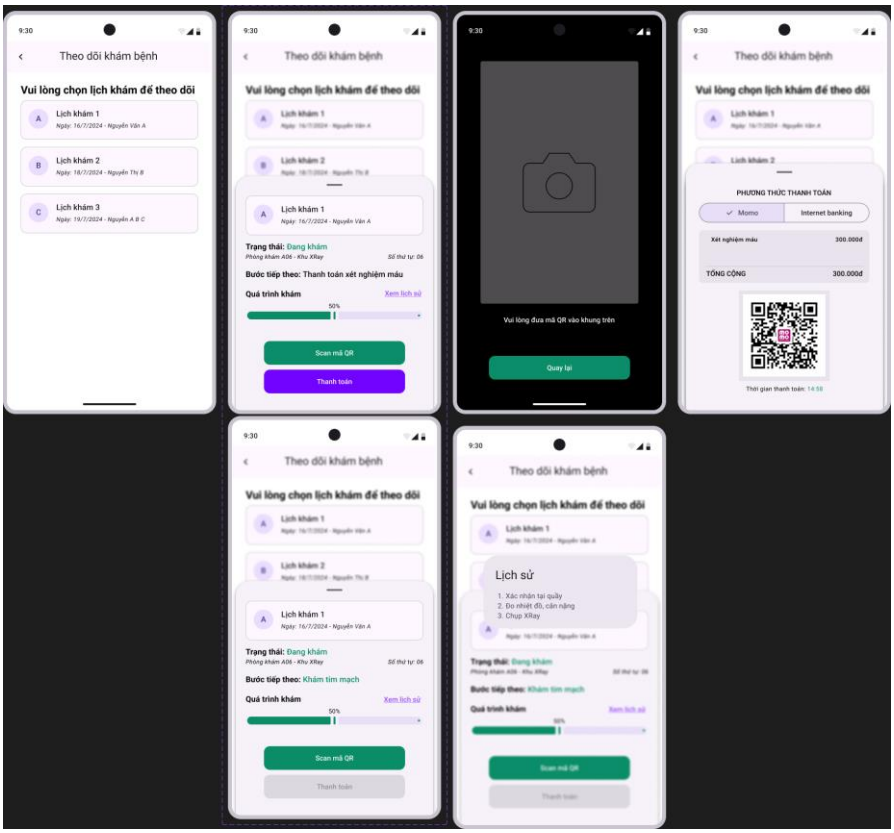


Figure 3. 49. User interface of Tracking examination (Source: Author)

The interface in this figure illustrates the process of tracking medical examination progress within a healthcare application, along with payment options. Here's a description of each screen:

Select Profile for Tracking Screen:

- Users are prompted to select a profile from a list of available patient profiles.
- Each profile displays information such as the name, date, and specific examination or service details, making it easy to choose the relevant profile for tracking.

Examination Tracking Screen:

- After selecting a profile, users can view the **Status** of the examination (e.g., "In Progress").
- Detailed information about the **current step** in the medical process (e.g., "Blood Test") is provided.
- A progress bar indicates the completion level of the examination process.
- Buttons are available to **Scan QR Code** for additional actions or proceed to **Payment** for pending fees.

QR Code Scanning Screen:

- This screen allows users to scan a QR code, possibly for verification or payment.
- A prompt instructs users to align the QR code within the camera frame.
- A button to "Retry" is available in case the scan is unsuccessful.

Payment Method Screen:

- Similar to previous payment screens, users can select **MoMo** or **Internet Banking** for payment.
- The total amount due for the service (e.g., "Blood Test") is displayed along with a QR code for payment.

- The payment timestamp is shown at the bottom, indicating when the transaction was initiated.

Further Examination Tracking:

- Additional tracking details are shown, with a progress bar indicating the examination's progress.
- The screen updates with the next step in the examination process (e.g., “Blood Pressure Check”) as the user advances.

j) Validation

Table 3. 48. Data validation of Tracking (Source: Author)

Failed Validation	**Error Message Displayed**
The payment must be successful to go to next step and queue number to be created	User won’t be able to proceed to next step and receive a queue number
User must scan Check-in QR code for the Check-up status to be updated	System will not update Check-up status until user scan Check-in QR code

Part 4: Database design

4.1 Overview and purpose

This section focuses on the database design process essential for supporting the hospital's Mobile appointment scheduling and medical records management system. The purpose of this section is to establish a structured and efficient database that meets patient's needs, ensuring the system can handle complex data requirements related to patient information, appointments, medical records, and payments. By defining entities, relationships, attributes, and applying data normalization principles, this section aims to create a stable and scalable foundation for the hospital's information system. The outcome will be a well-designed database that supports accurate data storage, retrieval, and processing, thus enhancing operational efficiency and data integrity.

4.2 Deliverables

This chapter presents the key deliverables essential to constructing a robust database for the system, each playing a critical role in ensuring an effective design and structure for the hospital's operational needs. These deliverables collectively form the database backbone, aligning with the hospital's requirements for accuracy, efficiency, and scalability.

Entity Table

This table includes detailed information on entities, relationships, attributes, data types, and constraints identified during the database design phase.

The purpose of this deliverable is to outline the core components of the database, specifying how data will be organized and restricted to ensure data integrity and consistency. By defining entities and relationships, this table forms the blueprint for structuring data across different parts of the system.

Entity Relationship Diagram (ERD)

The ERD visually represents the entities and relationships within the hospital's database. This deliverable provides a clear and structured view of how different data elements are connected, which aids in understanding the flow of information within the system.

The purpose of the ERD is to serve as a communication tool between technical and non-technical stakeholders, ensuring a shared understanding of the database structure.

Normalized Tables

The3NF normalized tables ensure that data redundancy is minimized and data integrity is maintained. This deliverable presents the database in a streamlined, organized format by breaking down tables into their simplest form following normalization rules.

The purpose of normalization is to optimize the database for efficient data processing and storage, reducing unnecessary duplication and potential inconsistencies.

Physical Database in SQL Server

The physical database is implemented in SQL Server, providing a concrete, functioning database model that can be tested and deployed. This deliverable translates the logical design into a real database system, allowing for actual data storage and retrieval.

The purpose of this database is to enable the hospital's application to interact with stored data effectively, supporting the functionalities of the mobile app and ensuring smooth data management for appointments, medical records, and patient information.

These deliverables together form the foundation of a well-structured database system, ensuring that all hospital data is organized, accessible, and reliable for various application functions.

4.3 Identify entities, relationship, attributes, data types, and constraints

4.3.1 Entities

Table 4. 1. Data type of Entities Account (Source: Author)

Attribute Name	Data type	Key	Description	Constraint
User ID	Varchar(10)	Primary Key (PK)	Uniquely represents the user's account in the system	UNIQUE, NOT NULL
User Name	Nvarchar(40)		User’s full name	NOT NULL
User Email	Varchar(30)		User's email address	UNIQUE, NOT NULL, must follow email

				format
User Phone Number	Varchar(10)		Phone number for authentication and OTP	UNIQUE, NOT NULL, 10-digit phone format
OTP Code	Varchar(6)		One-time authentication code	NOT NULL, must be 6 digits
Login Result	Boolean		Authentication result after OTP	Default: 0 (False)
Signup Result	Boolean		Result of the registration process	Default: 0 (False)

Table 4. 2. Data type of Entities Profile (Source: Author)

Attribute Name	**Data type**	**Key**	**Description**	**Constraint**
Patient ID	Varchar(10)	Primary Key (PK)	Uniquely represents the patient's profile	UNIQUE, NOT NULL
Patient Name	Nvarchar(40)		Patient’s full name	NOT NULL
Patient Citizen Identification	Varchar(12)		Patient’s Citizen ID	UNIQUE, 12-digit numeric format
Patient Insurance Number	Varchar(15)		Patient’s health insurance policy	UNIQUE
Patient Phone Number	Varchar(10)		Phone number for patient communication	UNIQUE, NOT NULL, 10-digit phone format
Patient Email	Varchar(30)		Patient's email address	UNIQUE, NOT NULL, must follow email format
Patient Gender	Nvarchar(4)		Patient's gender	CHECK (Value in [‘Nam’, ‘Nữ’, ‘Khác’])
Patient Age	Int		Patient's age	CHECK (Age >

				0)
Patient Date of Birth	Date		Patient's date of birth	NOT NULL
Patient Address	Nvarchar(50)		Patient's address	NOT NULL
Patient Nationality	Nvarchar(20)		Patient's country	NOT NULL
Patient Allergy	Nvarchar(50)		Allergies	Optional
Patient Blood Type	Varchar(3)		Patient's blood type	Optional

Table 4. 3. Data type of Entities Appointment (Source: Author)

Attribute Name	**Data type**	**Key**	**Description**	**Constraint**
Appointment ID	Varchar(10)	Primary Key (PK)	Unique identifier for appointment	UNIQUE, NOT NULL
Time Slots	Varchar(10)		Appointment time slots	NOT NULL
Department Name	Nvarchar(20)		Department associated with appointment	NOT NULL
Patient ID	Varchar(10)	Foreign Key (FK)	Links to Profile Table's Patient ID	REFERENCES Profile(Patient ID), NOT NULL
Invoice Number	Varchar(10)		Invoice identifier for appointment	UNIQUE

Table 4. 4. Data type of Entities Review (Source: Author)

Attribute Name	**Data type**	**Key**	**Description**	**Constraint**
Review ID	Varchar(10)	Primary Key (PK)	Unique identifier for review	UNIQUE, NOT NULL
Rating	Int		User-provided rating	CHECK (Rating

				BETWEEN 1 AND 5), NOT NULL
Comment	Nvarchar(500)		User feedback	Optional
Review Timestamp	Datetime		Timestamp of review submission	NOT NULL, Auto-generated

Table 4. 5. Data type of Entities Payment (Source: Author)

Attribute Name	**Data type**	**Key**	**Description**	**Constraint**
Payment ID	Varchar(10)	Primary Key (PK)	Unique identifier for payment	UNIQUE, NOT NULL
Timestamp of Payment	Datetime		Time of payment	NOT NULL
Payment Method	Varchar(10)		Method of payment	CHECK (Value in [‘Banking’, ‘Momo’]), NOT NULL
Payment Amount	Int		Total payment amount	CHECK (Amount > 0), NOT NULL
Payment Status	Boolean		Status of payment	CHECK (Status IN (0,1)), Default: 0 (Pending)
Invoice Number	Varchar(10)		Related invoice identifier	REFERENCES Appointment(Invoice Number), UNIQUE, NOT NULL
Invoice Timestamp	Datetime		Timestamp of invoice	NOT NULL
Payment QR Code	Binary large object		QR code for payment	Must be an image file format, NOT

	(BLOB)			NULL

Table 4. 6. Data type of Entities Refund (Source: Author)

Attribute Name	**Data type**	**Key**	**Description**	**Constraint**
Refund ID	Varchar(10)	Primary Key (PK)	Unique identifier for refund	UNIQUE, NOT NULL
Refund Amount	Int		Total amount of refund	CHECK (Amount > 0), NOT NULL
Refund Timestamp	Datetime		Timestamp of refund request	NOT NULL, Auto-generated
Refund Status	Boolean		Status of refund	CHECK (Status IN (0,1)), Default: 0 (Pending)
Refund Reason	Nvarchar(50)		Reason for refund	NOT NULL

Table 4. 7. Data type of Entity Invoice (Source: Author)

Attribute Name	**Data type**	**Key**	**Description**	**Constraint**
Invoice number	Varchar(10)	Primary Key (PK)	Unique identifier for each invoice	UNIQUE, NOT NULL
Invoice Timestamp	Datetime		The timestamp when the invoice was issued	NOT NULL, Auto-generated
Invoice details	Nvarchar(50)		Details of invoice include services name and price	NOT NULL

Table 4. 8. Data type of Entity Check-up (Source: Author)

Attribute Name	**Data type**	**Key**	**Description**	**Constraint**

Check-up ID	Varchar(10)	Primary Key (PK)	Unique identifier for each check-up	UNIQUE, NOT NULL
Room ID	Varchar(10)		Unique identifier for each room in the hospital	UNIQUE, NOT NULL
Check-up Name	Varchar(10)		Descriptive name of check-up	Must be in the hospital's check-up categories
Check-in QR Code	BLOB		QR code for check-in	Must be an image file format, NOT NULL
Check-up Status	Boolean		Check-up status	CHECK (Status IN (0,1)), Default: 0 (Pending)
Queue Number	Int		Queue number for medical room visit order	NOT NULL

4.3.2 Relationships between entities

Table 4. 9. Relationships between application's entities (Source: Author)

Entities	**Entities**	**Relationship**	**Name of relationship**
Account	Profile	1:n	Have/Belong to
Profile	Appointment	1:n	Have/Belong to
Profile	Payment	1:n	Pay/Paid by
Profile	Refund	1:n	Have/Belong to
Appointment	Review	1:1	Have/Belong to
Appointment	Payment	1:1	Contain/Belong to
Appointment	Check-up	1:n	Have/Belong to
Payment	Refund	1:1	Requested/Linked
Payment	Invoice	1:1	Have/Belong to
Check-up	Payment	1:1	Contain/Belong to

4.4 Entity Relationship Diagram

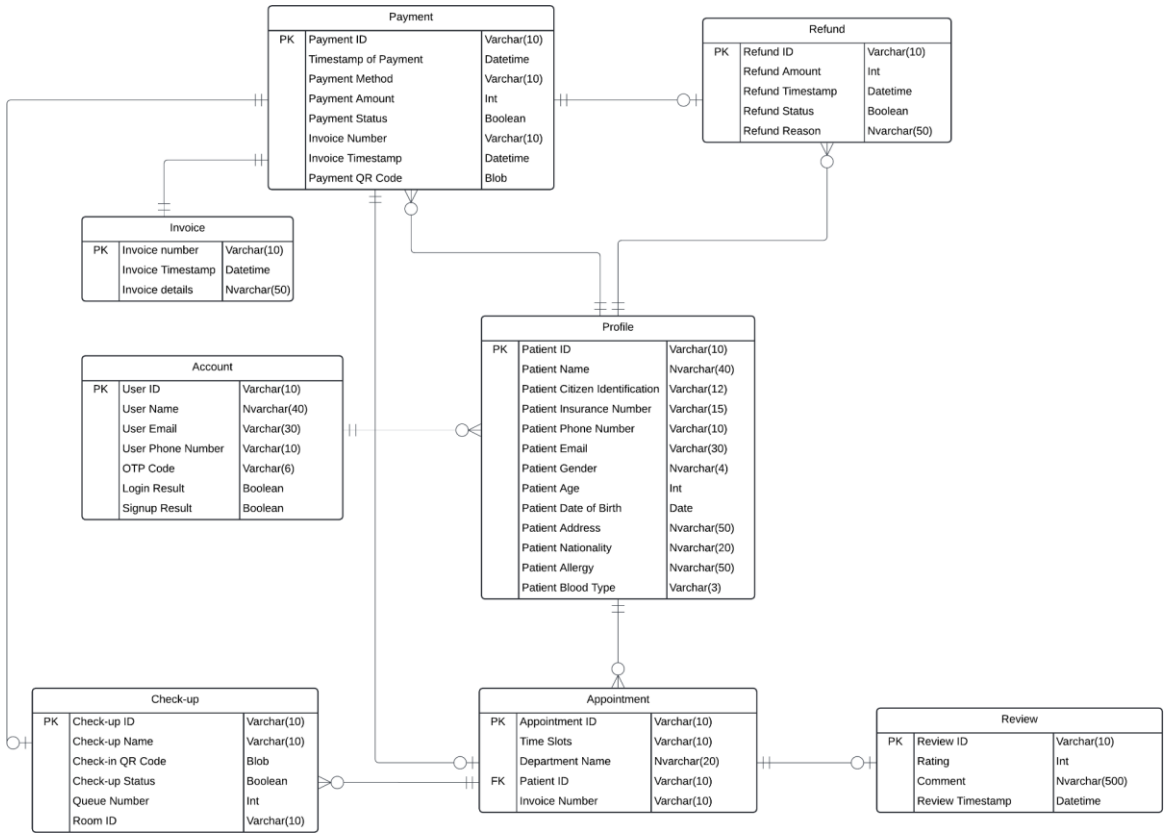


Figure 4. 1. ER diagram of the application (Source:Author)

4.5. Data Normalization

Database Normalization:

Normalization is a process of organizing data in a relational database to minimize redundancy and dependency. The primary goal is to ensure data consistency and improve database efficiency. It involves decomposing a database into smaller, related tables to avoid issues such as data anomalies and ensure that each piece of information is stored in only one location.

Normalization Levels:

There are several normal forms (NF), but the most commonly used are First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF). Each form builds upon the preceding one.

4.5.1. Current Data Analysis

The system consists of eight entities: Account, Profile, Appointment, Review, Payment, Refund, Medical Record, and Check-up. Each entity is represented by a table with corresponding attributes and constraints.

4.5.2. Data Normalization

a) 1NF (First Normal Form):
All tables comply with 1NF because:

- There are no repeating groups of attributes.

- Each column contains atomic values.

- A clearly defined primary key is present.

b) 2NF (Second Normal Form):
2NF requirements:

- Must satisfy all 1NF criteria.

- Each non-key attribute must depend entirely on the entire primary key, meaning there cannot be partial dependencies on any part of the primary key.

Medical Record Table: If a Medical Record ID corresponds to multiple pieces of health information (ongoing health issues, diseases, test results, etc.), a separate table should be created for each type of information and linked to the Medical Record table using the Medical Record ID.

c) 3NF (Third Normal Form):
3NF requirements:

- The table must meet 2NF conditions.

- Transitive dependencies (non-key attributes depending on other non-key attributes) must be removed.

Appointment Table:

- Department Name might depend on Patient ID.

- Create a Department table with attributes Department ID (PK) and Department Name.

- Replace Department Name with Department ID (FK) in the Appointment table.

Payment Table:

-Invoice Timestamp depends on Invoice Number, creating a transitive dependency.

- Move Invoice Timestamp to the Appointment table

4.5.3. Data Structure after Normalization

This section will not repeat tables that remain unchanged or normalized.

Table 4. 10. Table appointment (Source: Author)

Attribute Name	**Data type**	**Key**	**Description**	**Constraint**
Appointment ID	Varchar(10)	Primary Key (PK)	Unique identifier for appointment	UNIQUE, NOT NULL
Time Slots	Varchar(10)		Appointment time slots	NOT NULL
Department ID	Varchar(10)	Foreign Key (FK)	Unique identifier for for department associated with appointment	NOT NULL
Patient ID	Varchar(10)	Foreign Key (FK)	Links to Profile Table's Patient ID	REFERENCES Profile(Patient ID), NOT NULL
Invoice Number	Varchar(10)		Invoice identifier for appointment	UNIQUE

Table 4. 11. Table department (Source: Author)

Attribute Name	Data type	Key	Description	Constraint
Department ID	Varchar(10)	Primary Key (PK)	Unique identifier for department	UNIQUE, NOT NULL
Department Name	Nvarchar(20)		Department Name	NOT NULL

Table 4. 12. Table Department Appointment (Source: Author)

Attribute Name	**Data Type**	**Key**	**Description**	**Constraint**
DepartmentID	VARCHAR(10)	Foreign Key (FK)	Links to Department table	REFERENCES Department(DepartmentID), NOT NULL
AppointmentID	VARCHAR(10)	Foreign Key (FK)	Links to Appointment table	REFERENCES Appointment(AppointmentID), NOT NULL
Primary Key		Primary Key (PK)	Composite key between Department and Appointment	PRIMARY KEY (DepartmentID, AppointmentID)

Table 4. 13. Table payment (Source: Author)

Attribute Name	**Data type**	**Key**	**Description**	**Constraint**

Payment ID	Varchar(10)	Primary Key (PK)	Unique identifier for payment	UNIQUE, NOT NULL
Timestamp of Payment	Datetime		Time of payment	NOT NULL
Payment Method	Varchar(10)		Method of payment	CHECK (Value in [‘Banking’, ‘Momo’]), NOT NULL
Payment Amount	Int		Total payment amount	CHECK (Amount > 0), NOT NULL
Payment Status	Boolean		Status of payment	CHECK (Status IN (0,1)), Default: 0 (Pending)
Invoice Number	Varchar(10)	Foreign Key (FK)	Related invoice identifier	REFERENCES Invoice(Invoice Number), UNIQUE, NOT NULL
Payment QR Code	Binary large object (BLOB)		QR code for payment	Must be an image file format, NOT NULL

Table 4. 14. Table Check-up (Source: Author)

Attribute Name	**Data type**	**Key**	**Description**	**Constraint**
Check-up ID	Varchar(10)	Primary Key (PK)	Unique identifier for each check-up	UNIQUE, NOT NULL
Check-up Name	Varchar(10)		Descriptive name of check-up	Must be in the hospital’s check-up categories
Check-in QR Code	BLOB		QR code for check-in	Must be an image file format, NOT NULL

Check-up Status	Boolean		Check-up status	CHECK (Status IN (0,1)), Default: 0 (Pending)
Queue Number	Int		Queue number for medical room visit order	NOT NULL
Room ID	Varchar (20)	Foreign Key (FK)	Examination Room ID	REFERENCE Examination Room (Room ID), NOT NULL

Table 4. 15. Table Examination Room (Source: Author)

Attribute Name	**Data type**	**Key**	**Description**	**Constraint**
Room ID	Varchar(10)	Primary Key (PK)	Unique identifier for each room	UNIQUE, NOT NULL
Room Name	Nvarchar(20)		Descriptive name of check-up	NOT NULL

4.6 Physical database in any DBMS (SQL server, SQlite or others)

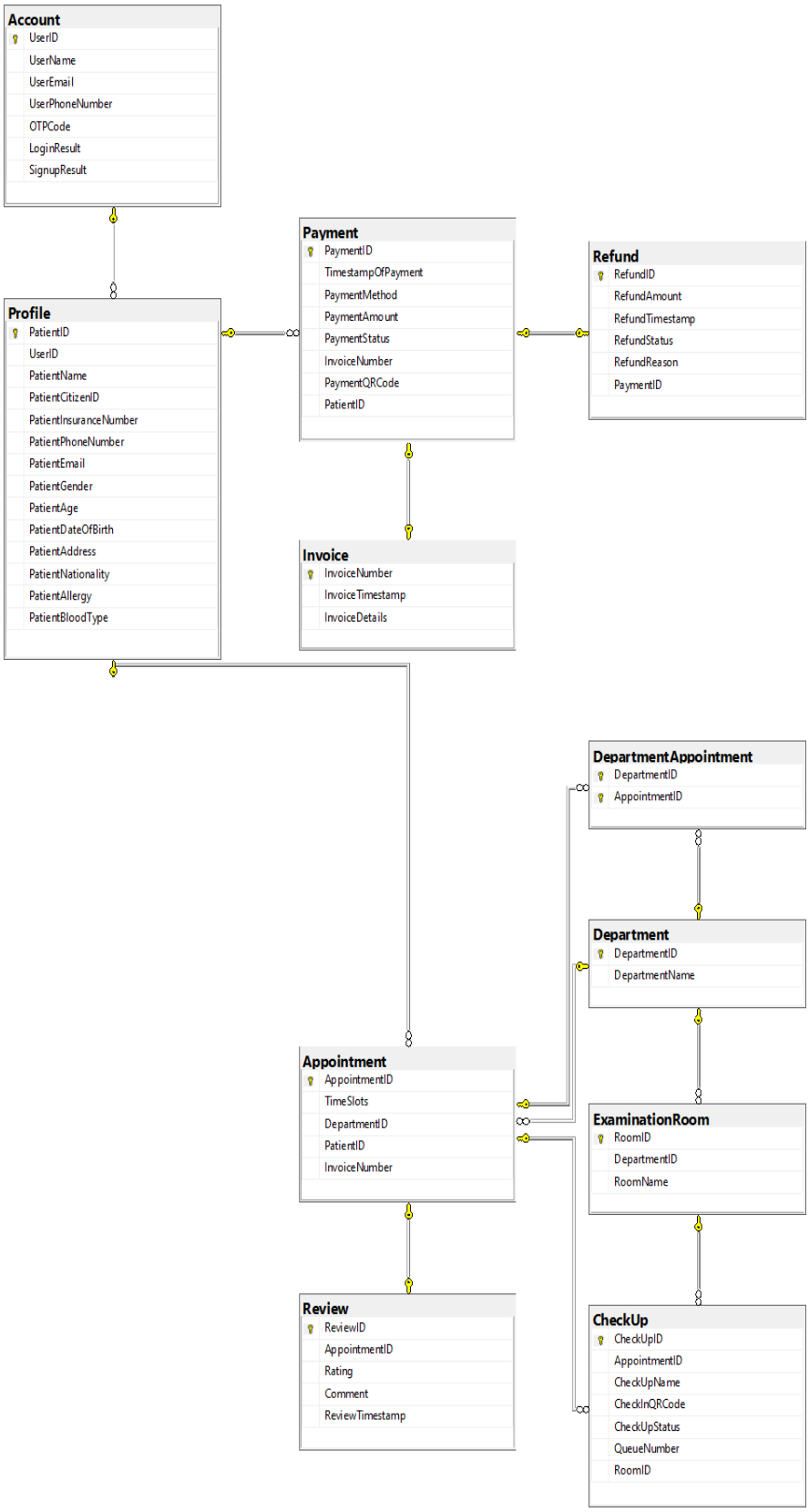


Figure 4. 2. Designing physical tables (Source: Author)

Part 5: Project Plan

5.1. Work Breakdown Structure

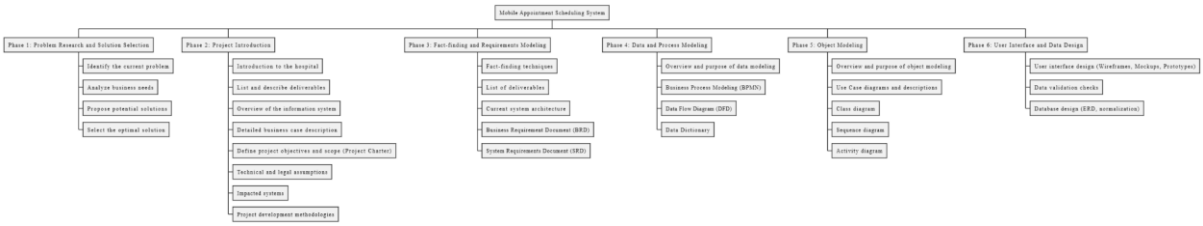


Figure 5. 1. Work Breakdown Structure of the project (Source: Author)

5.2. Project Milestones



Figure 5. 2. Project Milestone (Source: Author)

Part 6: Lessons learnt

Table 6. 1. Table of issue - lessons learned (Source: Author)

No.	Issue / what did not work well	What the team should have done instead?	Lessons Learned
1	Inefficient communication leading to misunderstandings	Set up regular check-in meetings and used a centralized communication tool	Effective communication prevents misunderstandings
2	Delays in task completion due to poor time management	Created a timeline with clearly deadlines for each task	Proper time management helps the team stay on track
3	Inconsistent work quality among team members	Established quality standards and regular peer reviews to maintain consistency	Setting standards and reviewing work improves quality and reduces errors
4	Low motivation and engagement from some members over time	Scheduled team-building activities and acknowledged contributions regularly	Keeping team morale high improves productivity and keeps everyone engaged

Table 6. 2. Table of worked well – lesson learned (Source: Author)

No.	What worked Well	Lesson Learned (Best Practice)
1	Regular team meetings to update progress	Regular meetings help the team stay updated on progress and resolve issues promptly
2	Clear task assignments for each member	Clear assignments help each member understand their responsibilities and reduce

		task duplication
3	Everyone contributed ideas in discussions	Encouraging contributions helps create well-rounded decisions that have team support
4	Established a shared folder for project documents	A shared folder allows easy access and saves time searching for documents

References

Beck, K., Beedle, M., Van Bennekum, A., Cockburn, A., Cunningham, W., Fowler, M., . . . Thomas, D. (2001). *Manifesto for Agile software development*. https://agilemanifesto.org/

Bệnh viện Đa khoa Khu vực Thủ Đức. (n.d.). *Giới thiệu*. Bệnh viện Đa khoa Khu vực Thủ Đức. http://www.benhvienkhuvucthuduc.vn/gioi-thieu/

Bệnh viện Đa khoa Khu vực Thủ Đức. (2020, July 16). *TP.HCM phê duyệt dự án BV Đa khoa Khu vực Thủ Đức 1.915 tỉ đồng*. http://www.benhvienkhuvucthuduc.vn/article/tp.hcm-phe-duyet-du-an-bv-da-khoa-khu-vuc-thu-duc-1.915-ti-dong-3/

Government of Vietnam. (2023, April). *Nghị định số 13/2023/NĐ-CP của Chính phủ về việc bảo vệ dữ liệu cá nhân*. Vietnam Government Portal. https://vanban.chinhphu.vn

Martin, James (1991). *Rapid Application Development*. Macmillan. ISBN 0-02-376775-8

Hồ Chí Minh City Party Committee. (2021, December 26). *Khởi công xây dựng mới bệnh viện đa khoa khu vực Thủ Đức*. https://hcmcpv.org.vn/tin-tuc/khoi-cong-xay-dung-moi-benh-vien-da-khoa-khu-vuc-thu-duc-1491887232

Huy, Q. (2021, November 15). *Khởi công xây dựng bệnh viện đa khoa khu vực Thủ Đức quy mô 1.000 giường*. Sài Gòn Giải Phóng. https://www.sggp.org.vn/khoi-cong-xay-dung-benh-vien-da-khoa-khu-vuc-thu-duc-quy-mo-1000-giuong-post617100.html

Loc, H. (2021, November 15). *Khởi công xây mới Bệnh viện Đa khoa khu vực Thủ Đức.* Tuổi trẻ. https://tuoitre.vn/khoi-cong-xay-moi-benh-vien-da-khoa-khu-vuc-thu-duc-2021111513042823.htm

The Ministry of Health. (2017, December 29). Thông tư 54/2017/TT-BYT ban hành bộ tiêu chí ứng dụng công nghệ thông tin tại các cơ sở khám bệnh, chữa bệnh. National Database of Legal Documents. https://vbpl.vn/boyte/Pages/vbpq-van-ban-goc.aspx?ItemID=128249

People's Committee of Ho Chi Minh City. (1994). *Quyết định số 725/QĐ-UB-NC về việc thành lập Trung tâm Y tế huyện Thủ Đức*. Vietnam Government Portal. https://vanban.chinhphu.vn

People's Committee of Ho Chi Minh City. (1999). *Quyết định số 6784/QĐ-UB về việc nâng cấp Trung tâm Y tế huyện Thủ Đức thành Bệnh viện Đa khoa Khu vực Thủ Đức*. Vietnam Government Portal. https://vanban.chinhphu.vn

Lien, V. (2024, October 3). *Hai bệnh viện Đa khoa khu vực Thủ Đức, Củ Chi sẽ nghiệm thu vào đầu năm 2025*. Khoa Học Phổ Thông. https://khoahocphothong.vn/hai-benh-vien-da-khoa-khu-vuc-thu-duc-cu-chi-se-nghiem-thu-vao-dau-nam-2025-257254.html

Appendix